

The quest for conserved RNAs in viral genomes

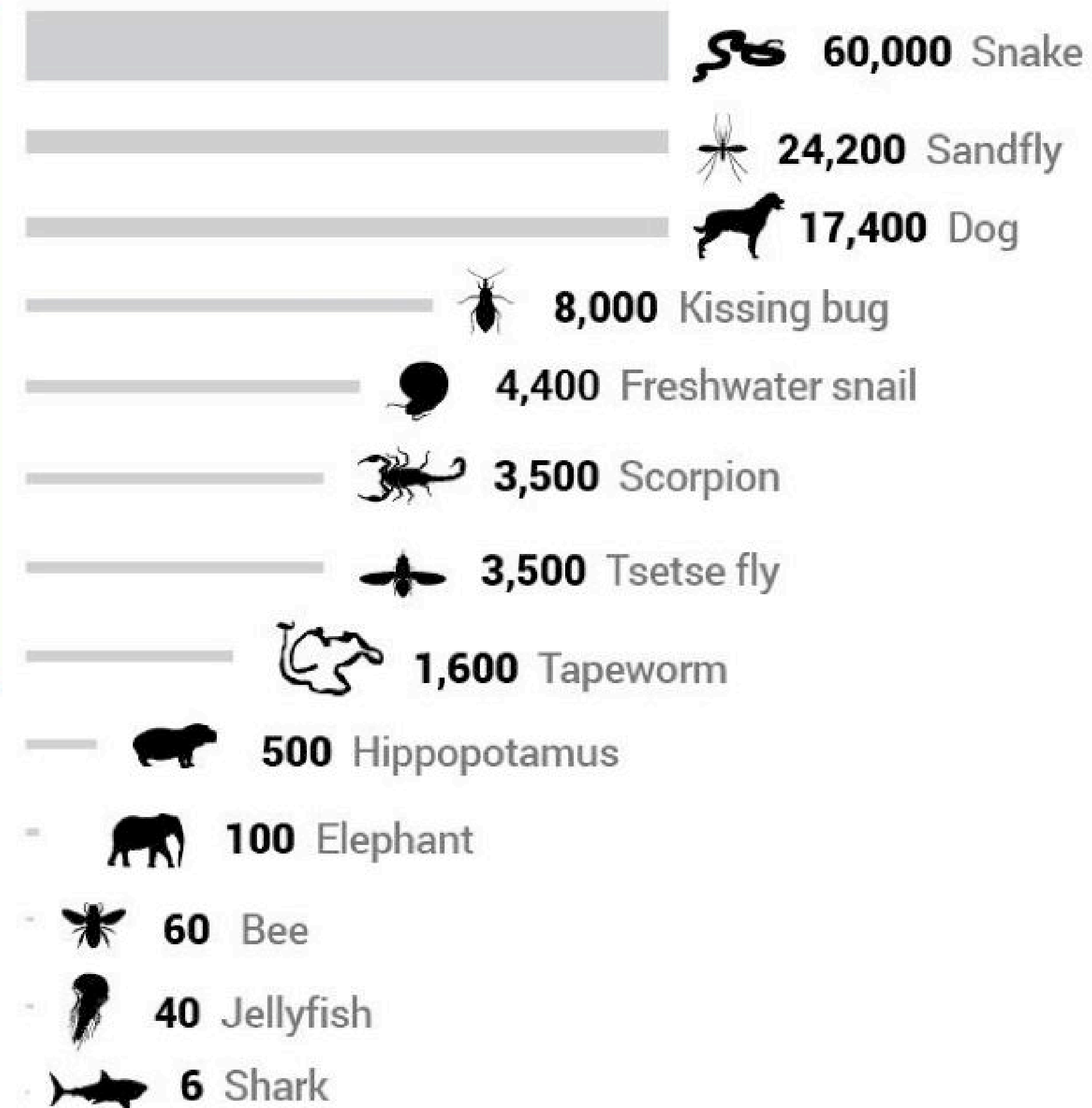
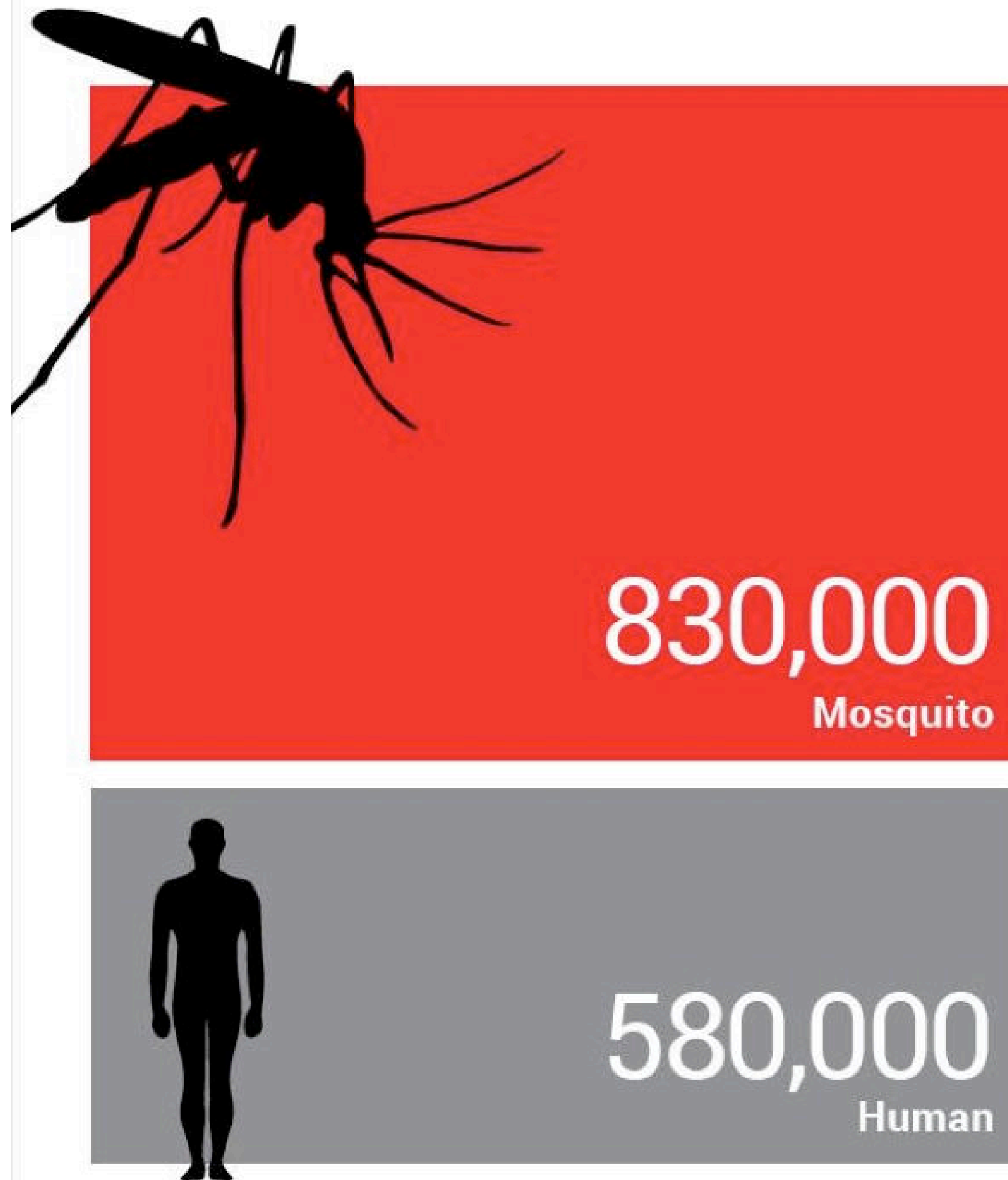
Michael T. Wolfinger

Research Group Bioinformatics and Computational Biology
& Department of Theoretical Chemistry
University of Vienna
Austria

The World's Deadliest Animals

Number of people killed by animals, 2015

gates
notes

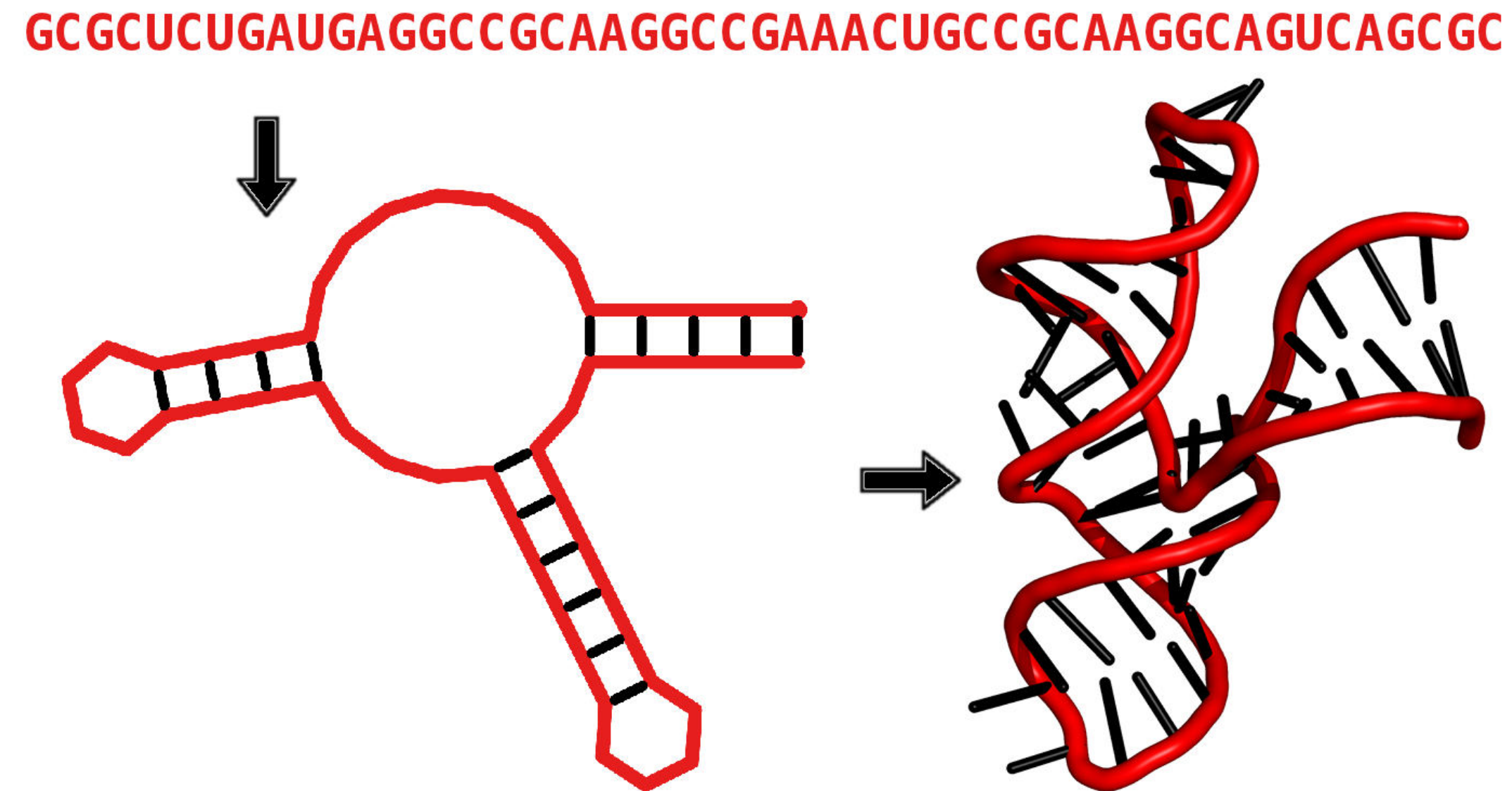


Sources: IHME, WHO, CrocBITE, FAO, Norwegian Institute for Nature Research, International Shark Attack File, National Geographic, PBS, National Science Foundation, CDC, WWF, Wilderness & Environmental Medicine, Nature, French Institute of Research for Development. All calculations have wide error margins.

Part I:

RNA structure prediction

The RNA Folding Problem

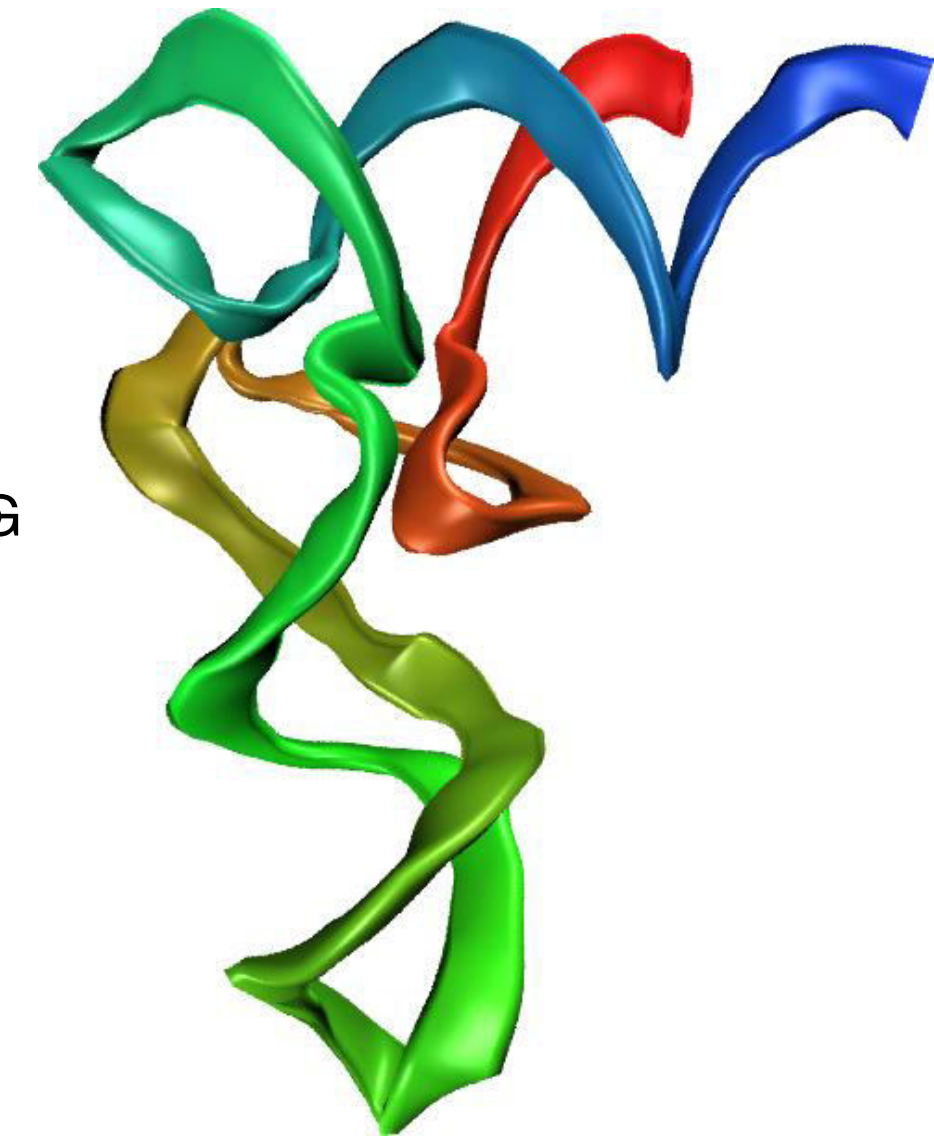
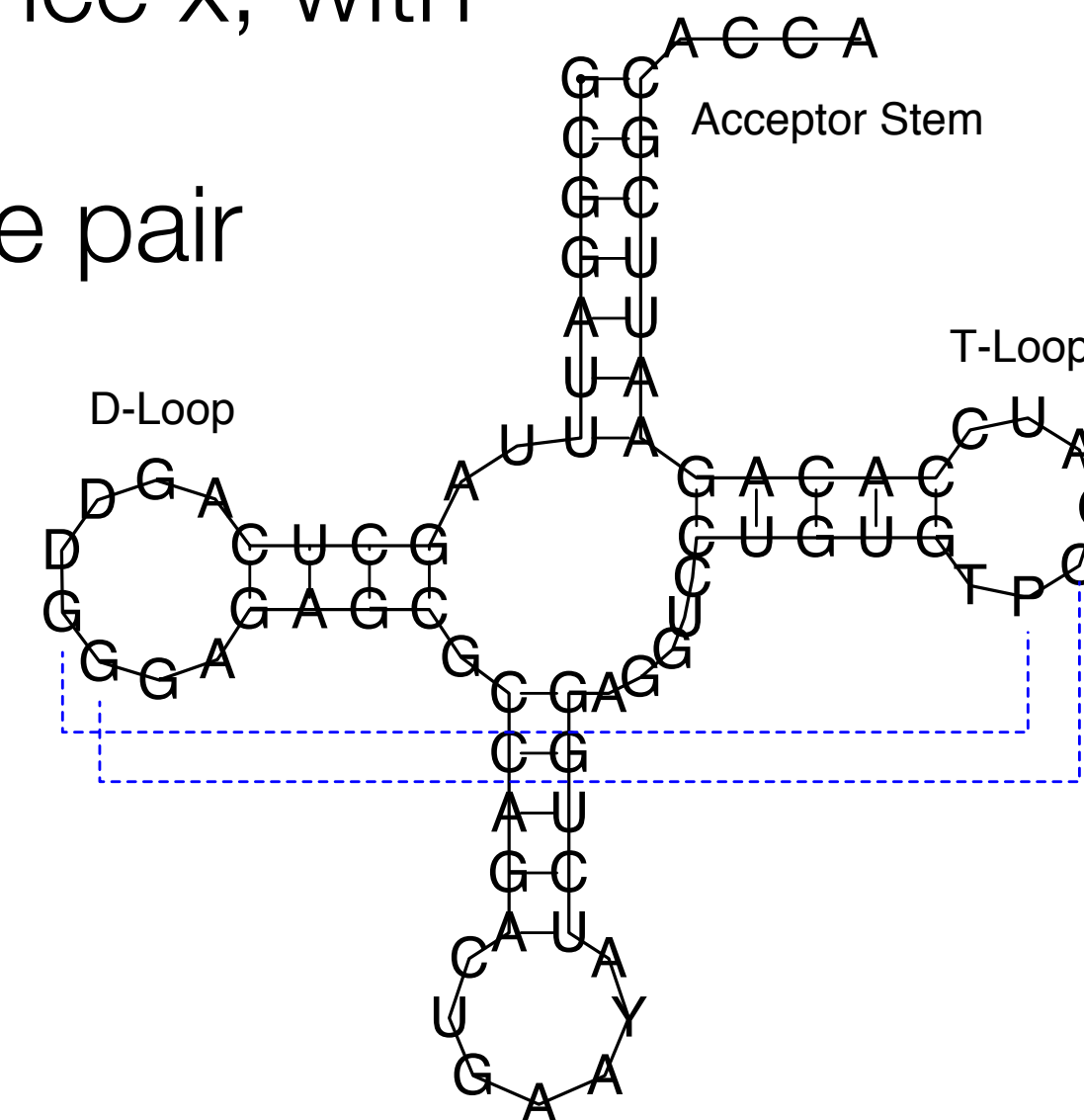


- Hierarchical folding: Secondary structure forms first then helices arrange to form tertiary structure
- Secondary structures cover most most of the folding energy
- Convenient and biologically useful description
- Computationally easy to handle
- Tertiary structure prediction needs knowledge of secondary structure

Secondary Structures

A **secondary structure** is a list of base pairs (i,j) on a sequence x , with

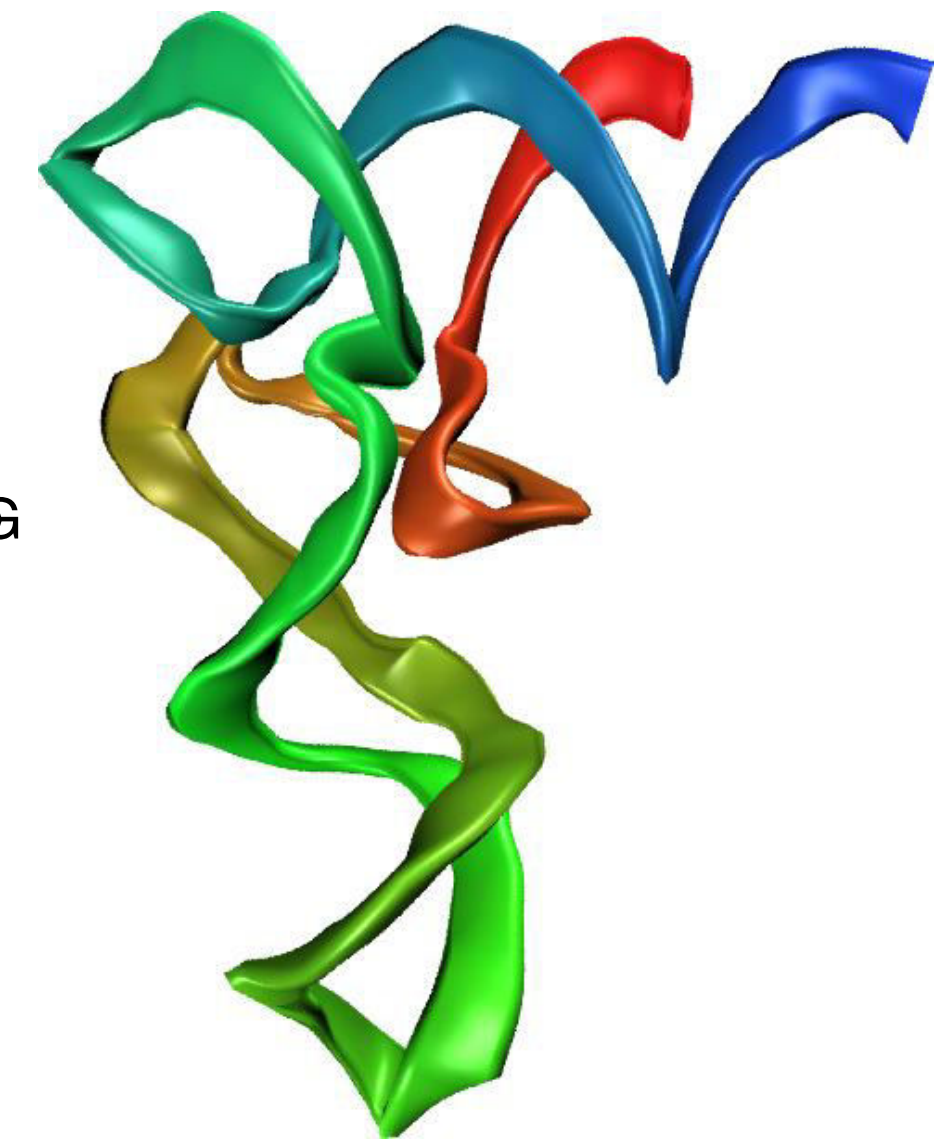
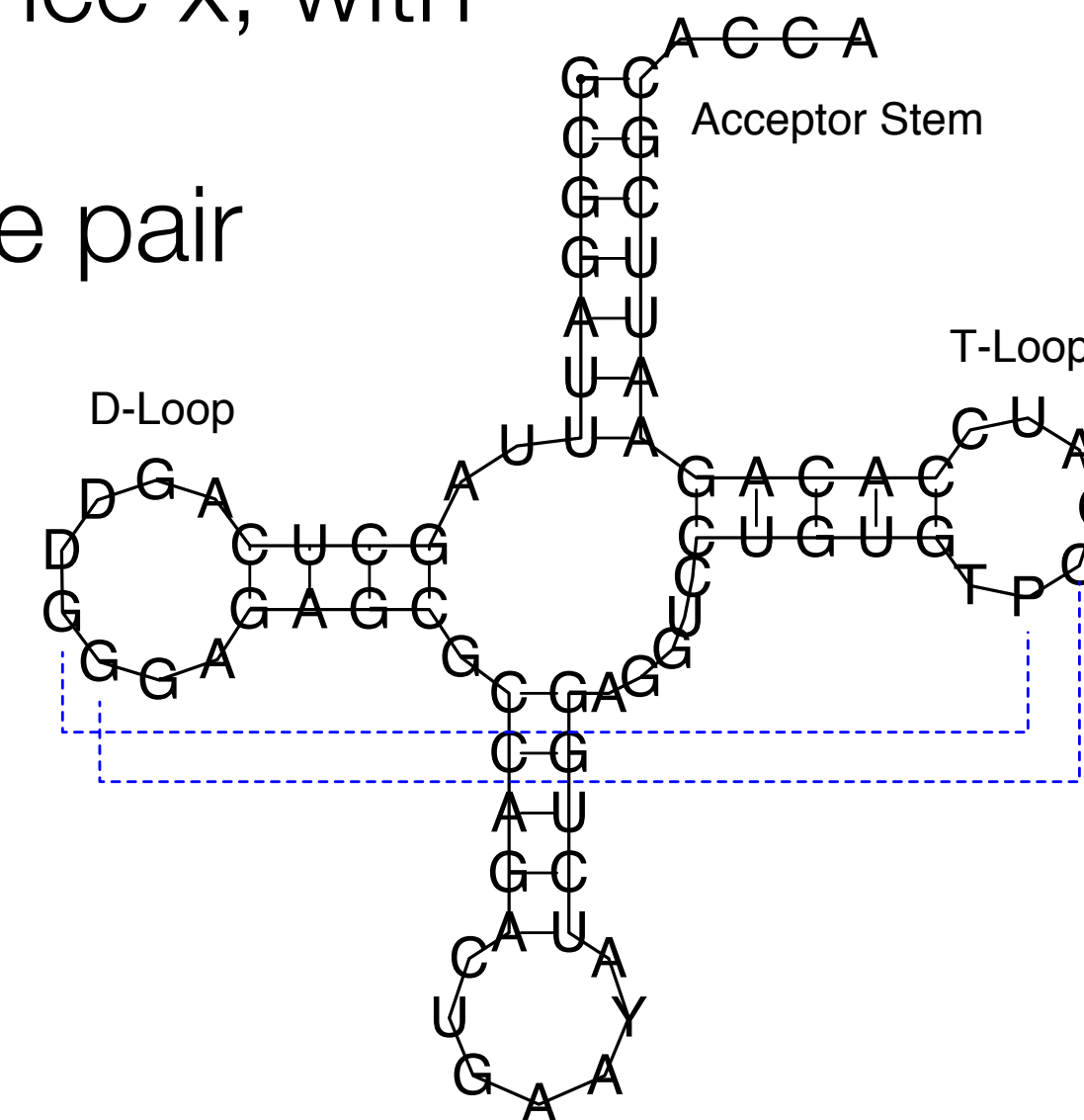
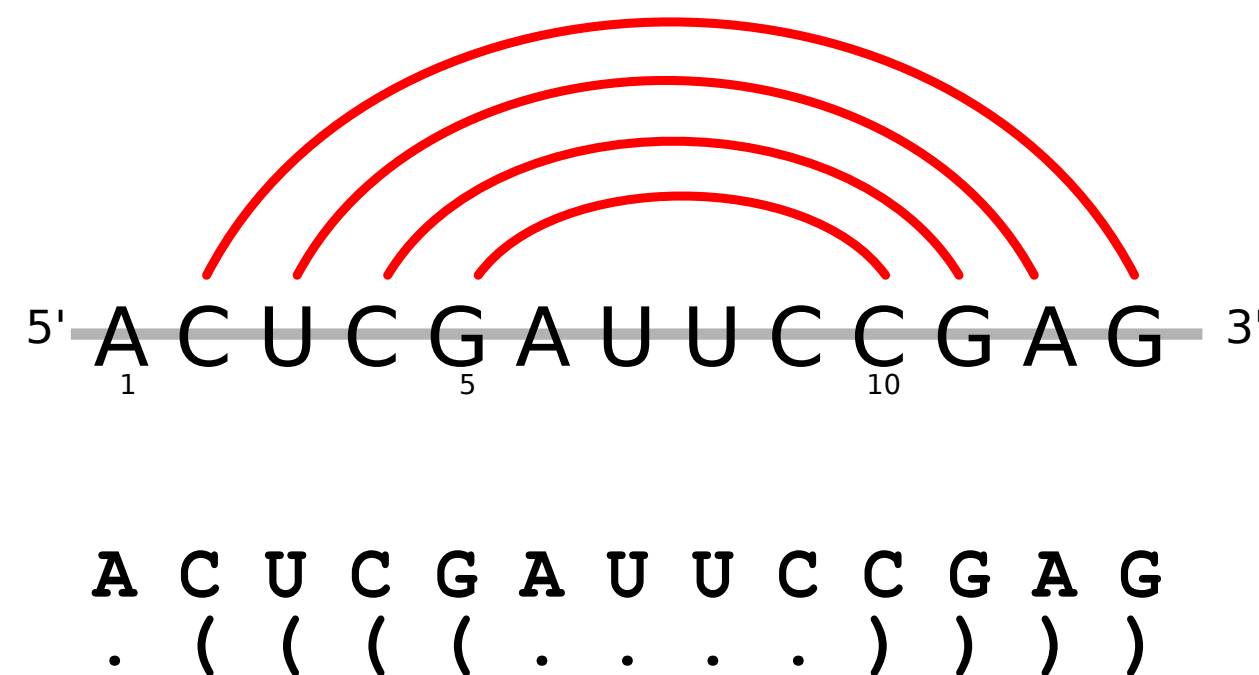
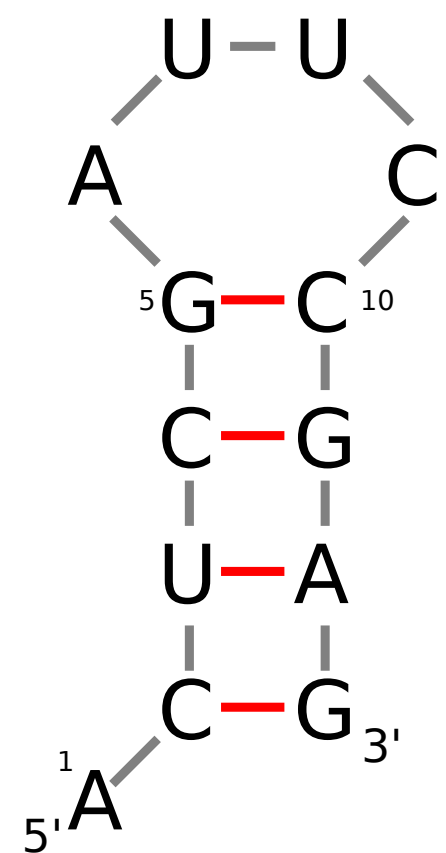
- Any nucleotide (sequence position) can form at most one pair
- No pseudo-knots: No pairs (i,j) and (k,l) with $i < k < j < l$
- If (i,j) is a pair then $x_i x_j \in \{GC, CG, AU, UA, GU, UG\}$
- If (i,j) is a base pair, then $j - i > 3$



Secondary Structures

A **secondary structure** is a list of base pairs (i,j) on a sequence x , with

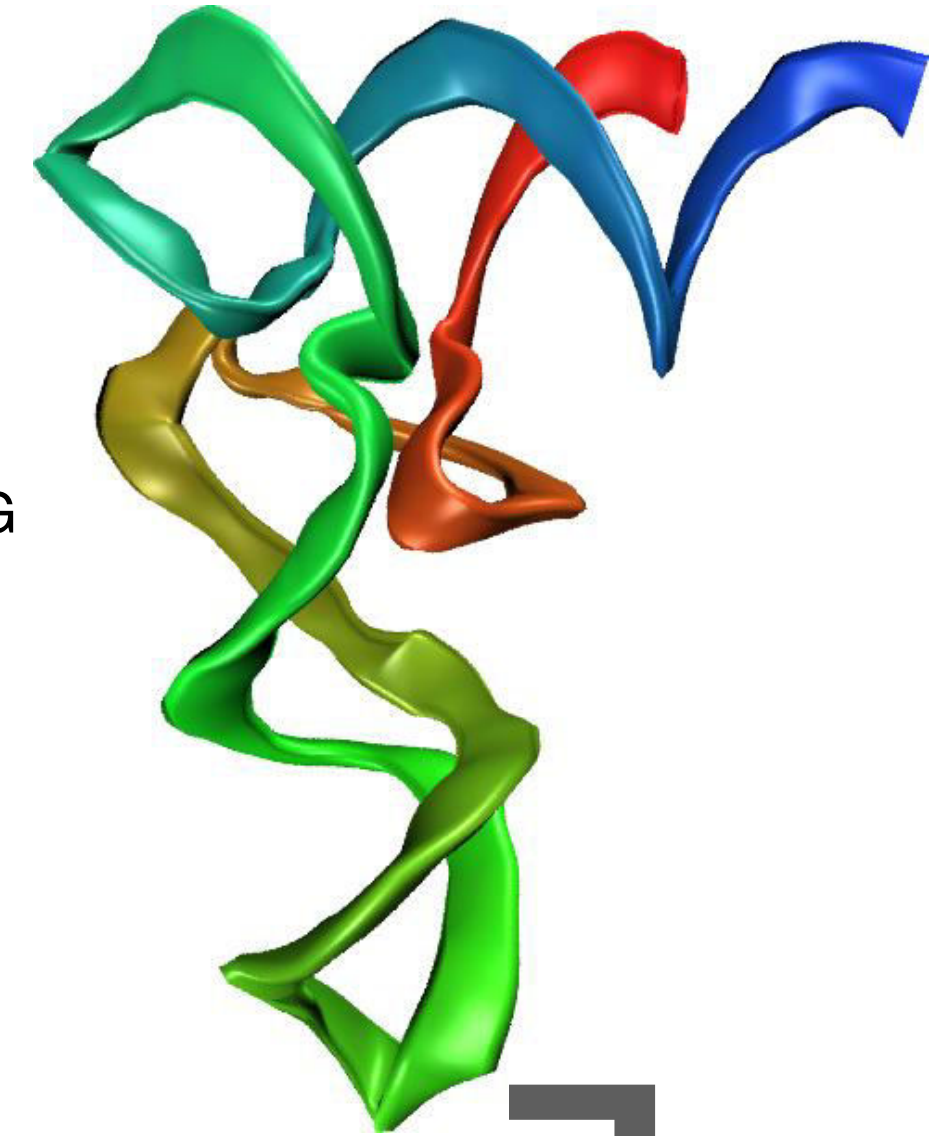
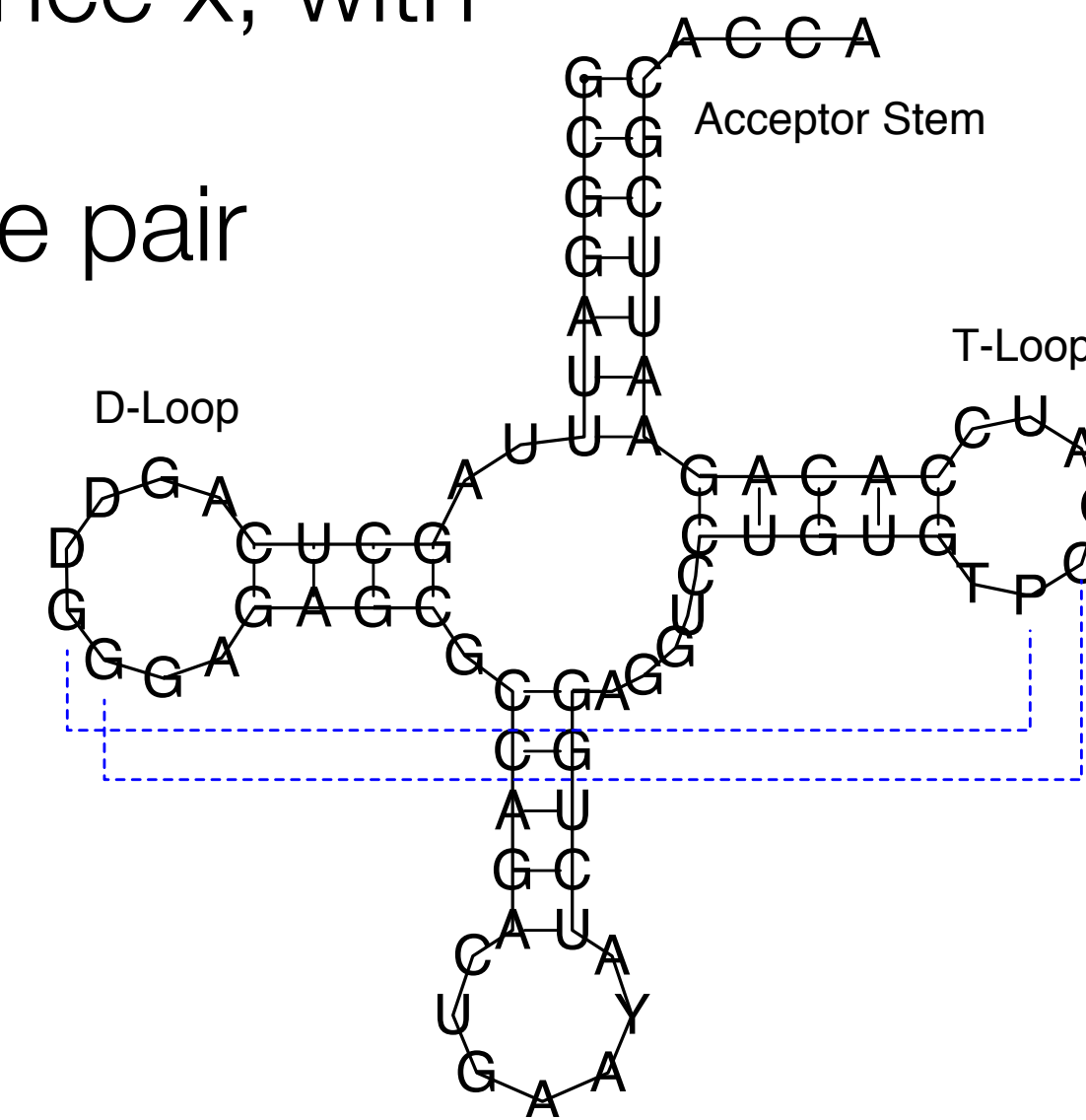
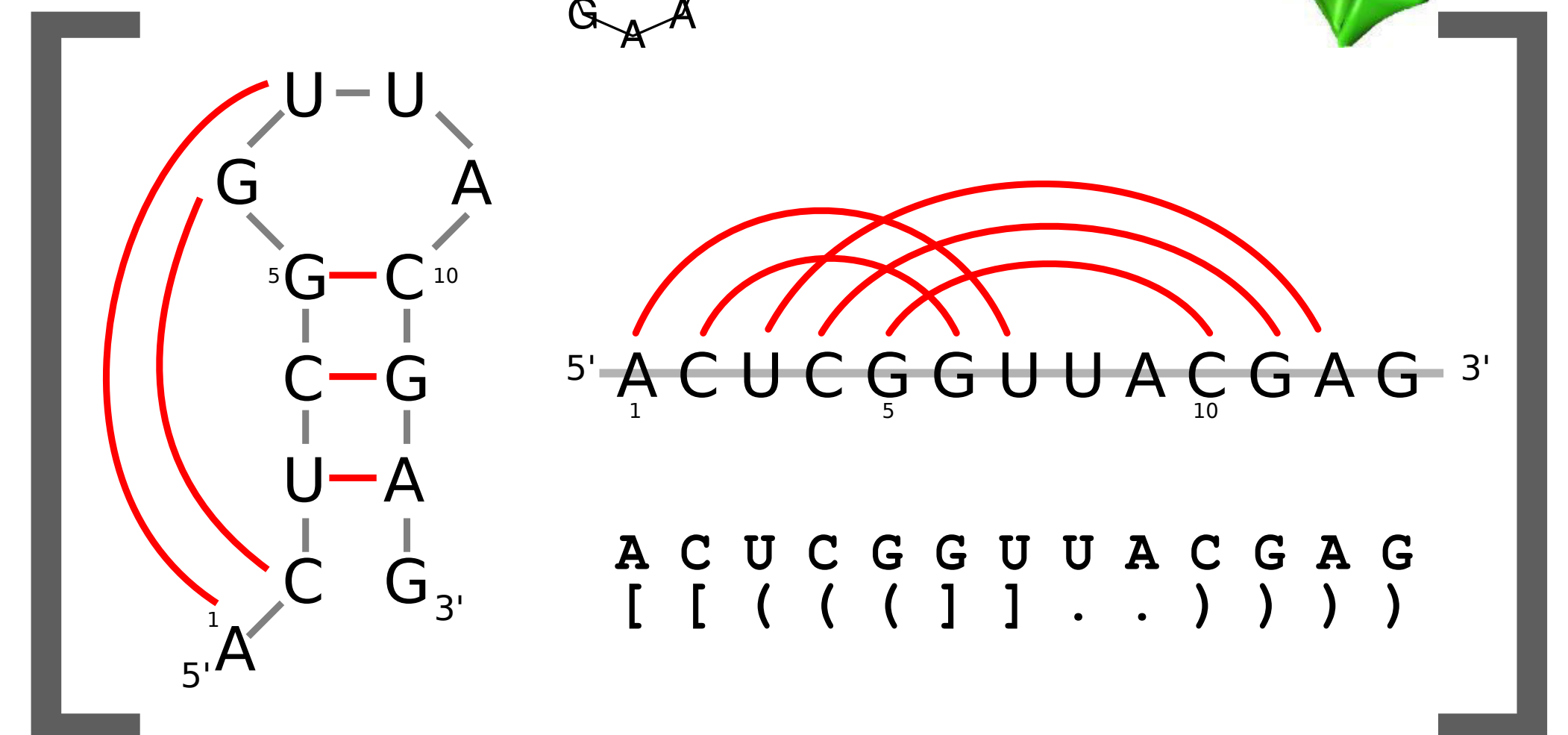
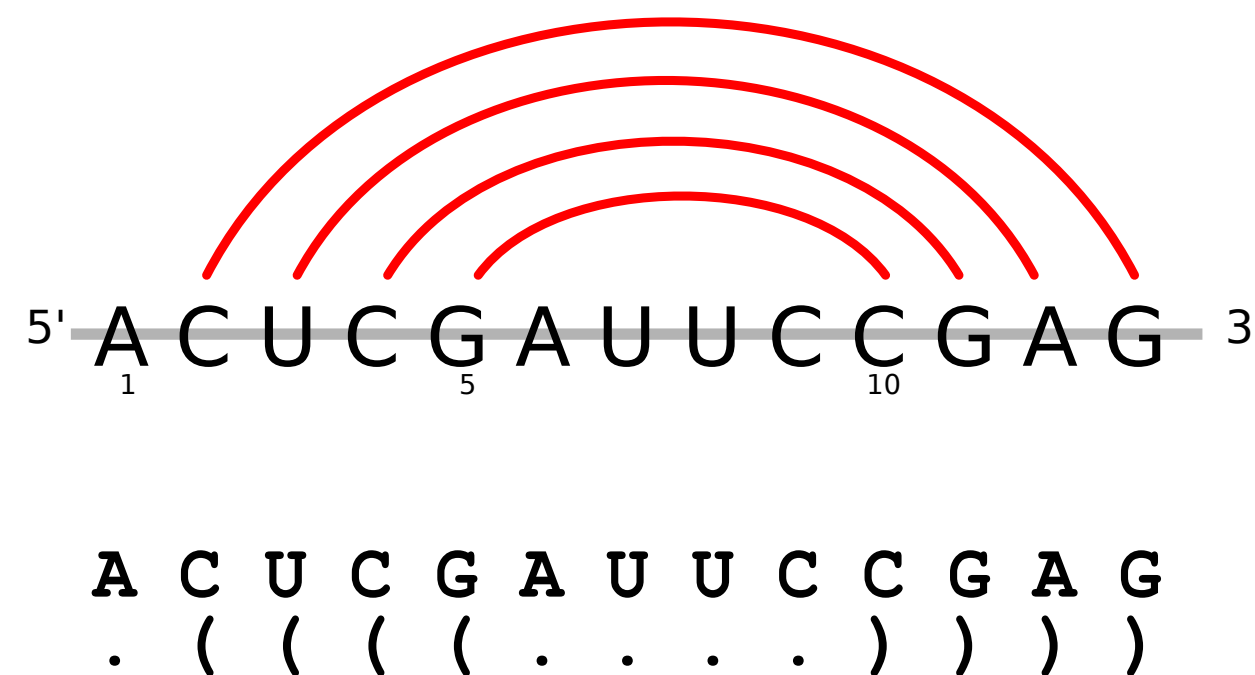
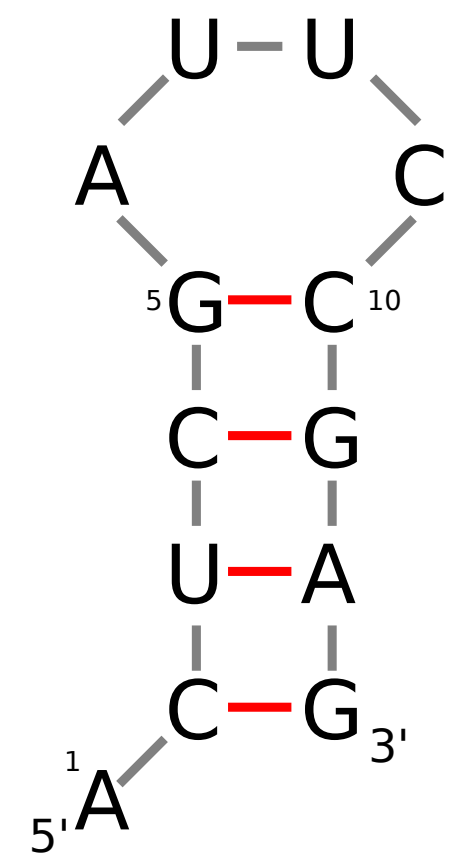
- Any nucleotide (sequence position) can form at most one pair
- No pseudo-knots: No pairs (i,j) and (k,l) with $i < k < j < l$
- If (i,j) is a pair then $x_i x_j \in \{GC, CG, AU, UA, GU, UG\}$
- If (i,j) is a base pair, then $j - i > 3$



Secondary Structures

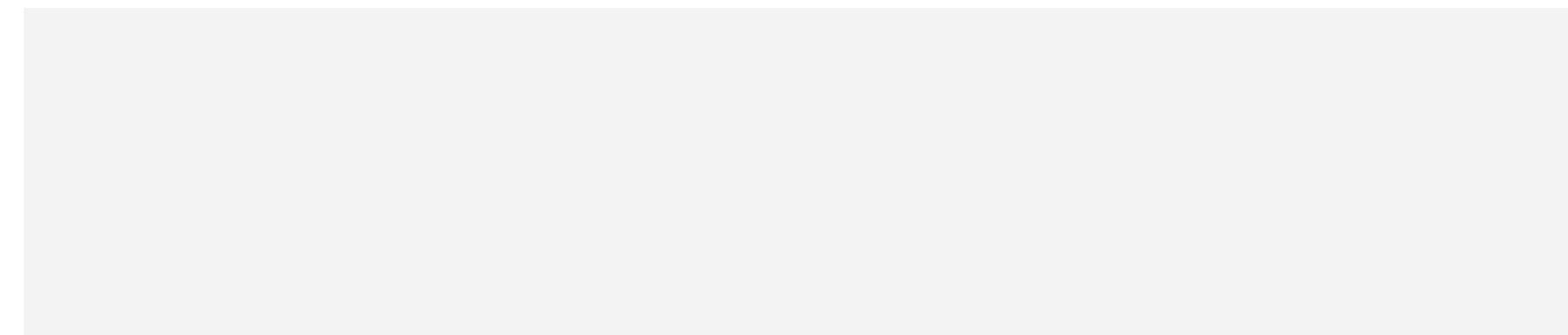
A **secondary structure** is a list of base pairs (i,j) on a sequence x , with

- Any nucleotide (sequence position) can form at most one pair
- No pseudo-knots: No pairs (i,j) and (k,l) with $i < k < j < l$
- If (i,j) is a pair then $x_i x_j \in \{GC, CG, AU, UA, GU, UG\}$
- If (i,j) is a base pair, then $j - i > 3$



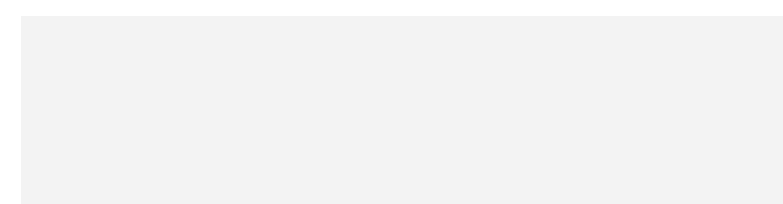
Conformation Space

The number of secondary structures for a sequence $x = x_1 \dots x_n$ can be computed recursively



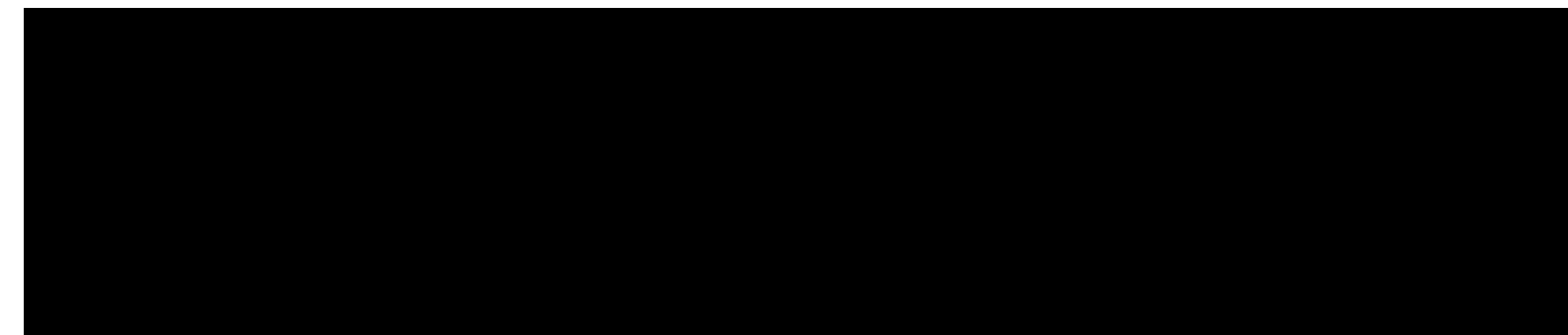
if $x_i x_k \in \{GC, CG, AU, UA, GU, UG\}$, otherwise

For sequences with equal $\{A, U, G, C\}$ content, the number of conformations grows asymptotically with sequence length



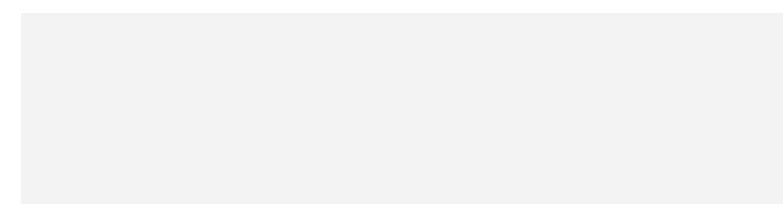
Conformation Space

The number of secondary structures for a sequence $x = x_1 \dots x_n$ can be computed recursively



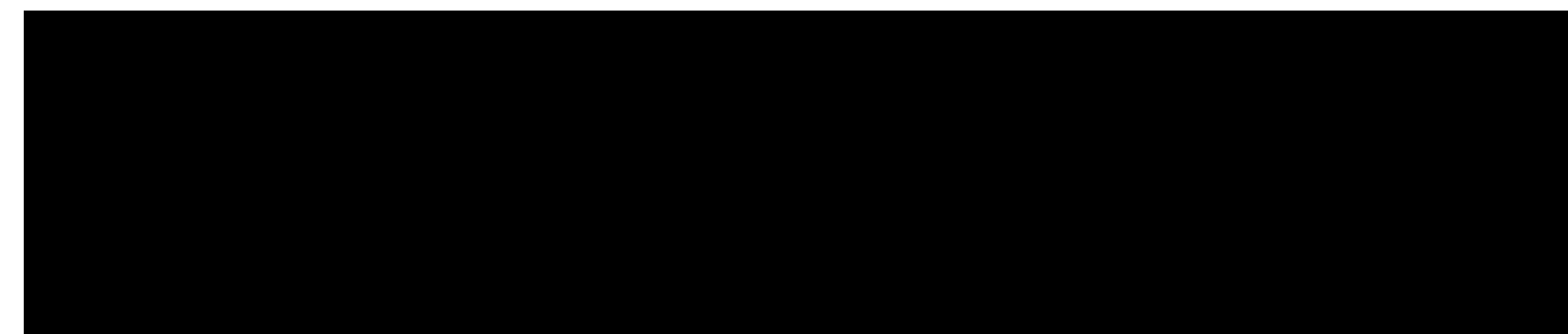
if $x_i x_k \in \{GC, CG, AU, UA, GU, UG\}$, otherwise

For sequences with equal $\{A, U, G, C\}$ content, the number of conformations grows asymptotically with sequence length



Conformation Space

The number of secondary structures for a sequence $x = x_1 \dots x_n$ can be computed recursively



if $x_i x_k \in \{GC, CG, AU, UA, GU, UG\}$, otherwise

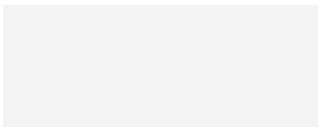
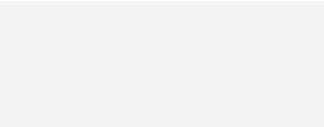
For sequences with equal $\{A, U, G, C\}$ content, the number of conformations grows asymptotically with sequence length



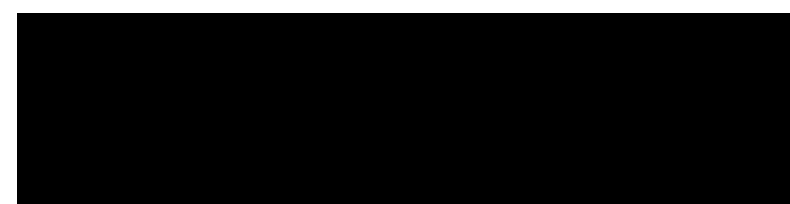
Conformation Space

The number of secondary structures for a sequence $x = x_1 \dots x_n$ can be computed recursively



 if $x_i x_k \in \{GC, CG, AU, UA, GU, UG\}$, otherwise 

For sequences with equal $\{A, U, G, C\}$ content, the number of conformations grows asymptotically with sequence length



Conformation Space

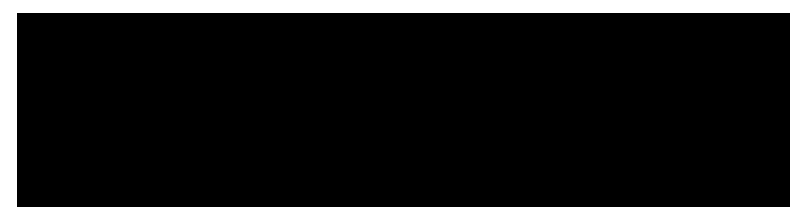
The number of secondary structures for a sequence $x = x_1 \dots x_n$ can be computed recursively



Nature 'exploits' this property

if $x_i x_k \in \{GC, CG, AU, UA, GU, UG\}$, otherwise

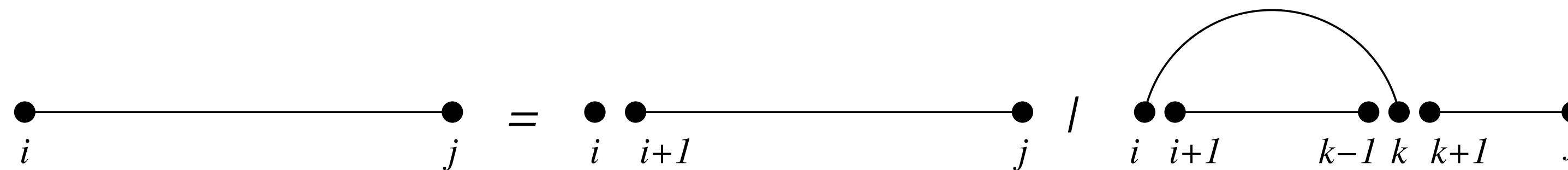
For sequences with equal $\{A, U, G, C\}$ content, the number of conformations grows asymptotically with sequence length



Solving the RNA Folding Problem

Toy model for RNA folding: assign energies to base pairs $\varepsilon(x, y)$

Easily solved by **Dynamic Programming**: recursive computation with tabulation of intermediate results



$$E_{ij} = \min_{i < k \leq j} \left\{ E_{i+1,j} ; \left(E_{i+1,k-1} + E_{k+1,j} + \varepsilon(x_i, x_k) \right) \right\}$$

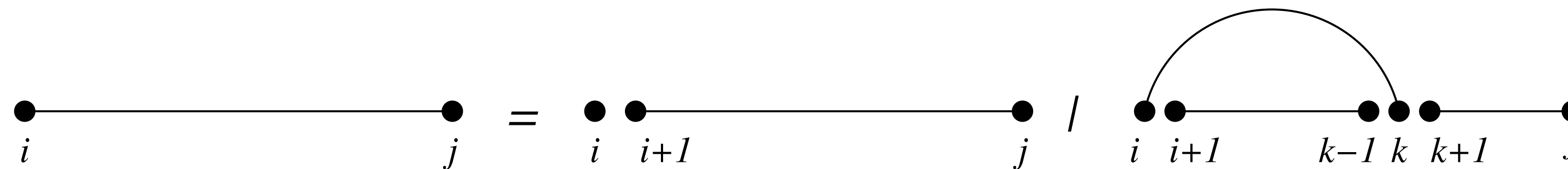
- E_{1n} is the best possible energy for our sequence
- Backtracing through the E table yields the corresponding structure
- The algorithm requires $\mathcal{O}(n^2)$ memory and $\mathcal{O}(n^3)$ CPU time

In practice this toy model is not good enough !
We need loop-dependent energies for serious predictions

Solving the RNA Folding Problem

Toy model for RNA folding: assign energies to base pairs $\varepsilon(x, y)$

Easily solved by **Dynamic Programming**: recursive computation with tabulation of intermediate results

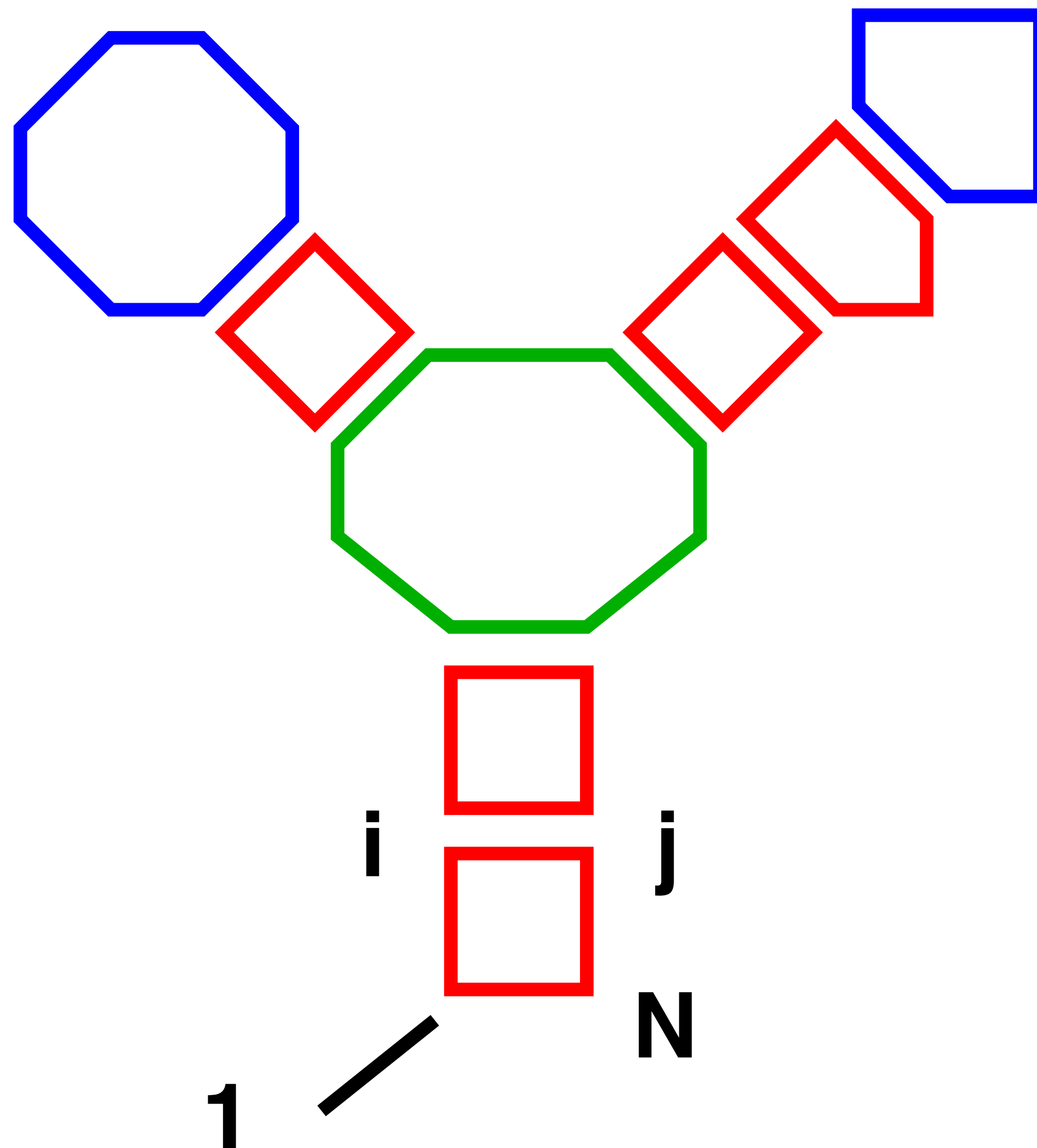


$$E_{ij} = \min_{i < k \leq j} \left\{ E_{i+1,j} ; \left(E_{i+1,k-1} + E_{k+1,j} + \varepsilon(x_i, x_k) \right) \right\}$$

- E_{1n} is the best possible energy for our sequence
- Backtracing through the E table yields the corresponding structure
- The algorithm requires $\mathcal{O}(n^2)$ memory and $\mathcal{O}(n^3)$ CPU time

In practice this toy model is not good enough !
We need loop-dependent energies for serious predictions

Loop Decomposition

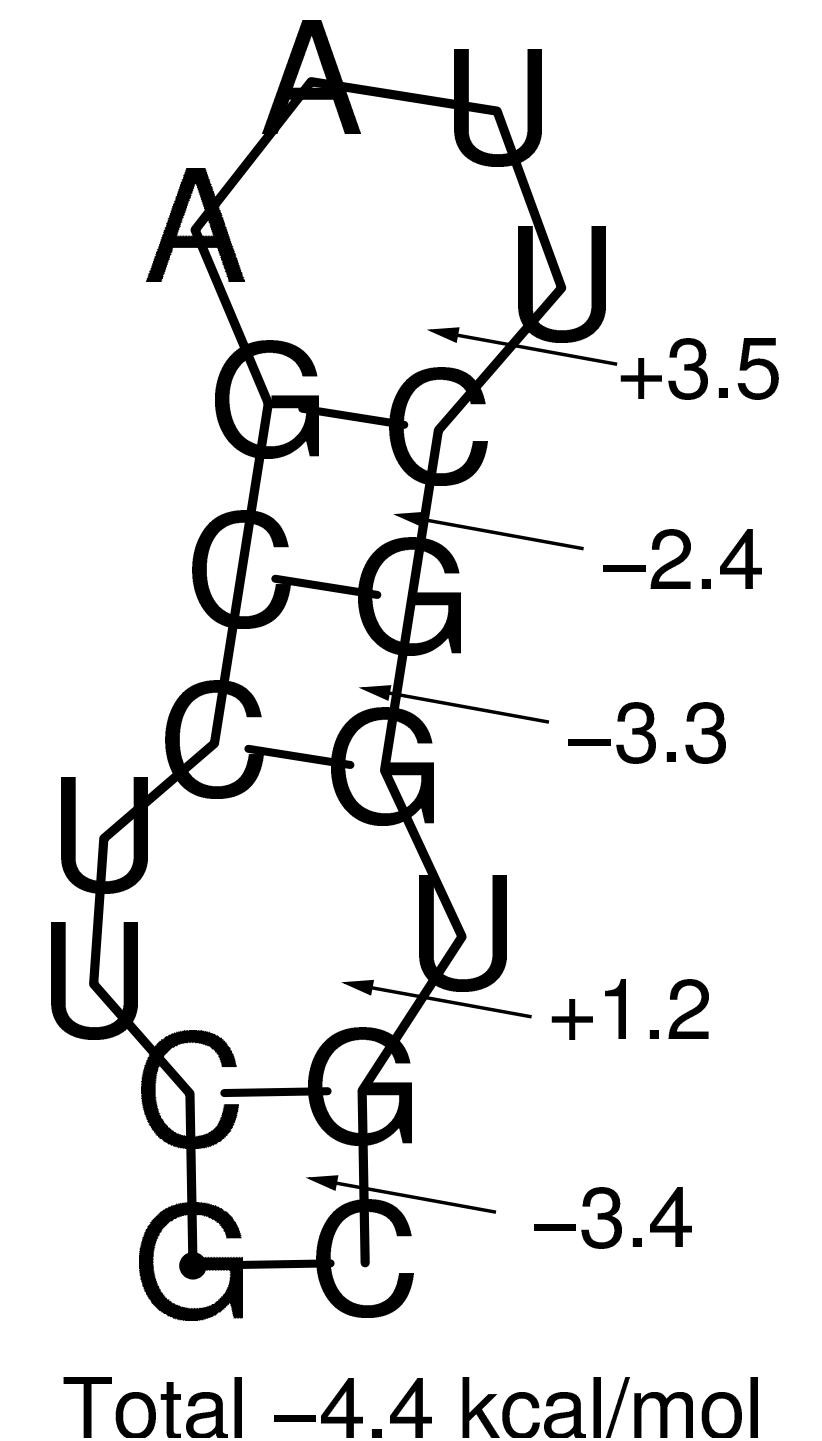


Nearest Neighbour Model

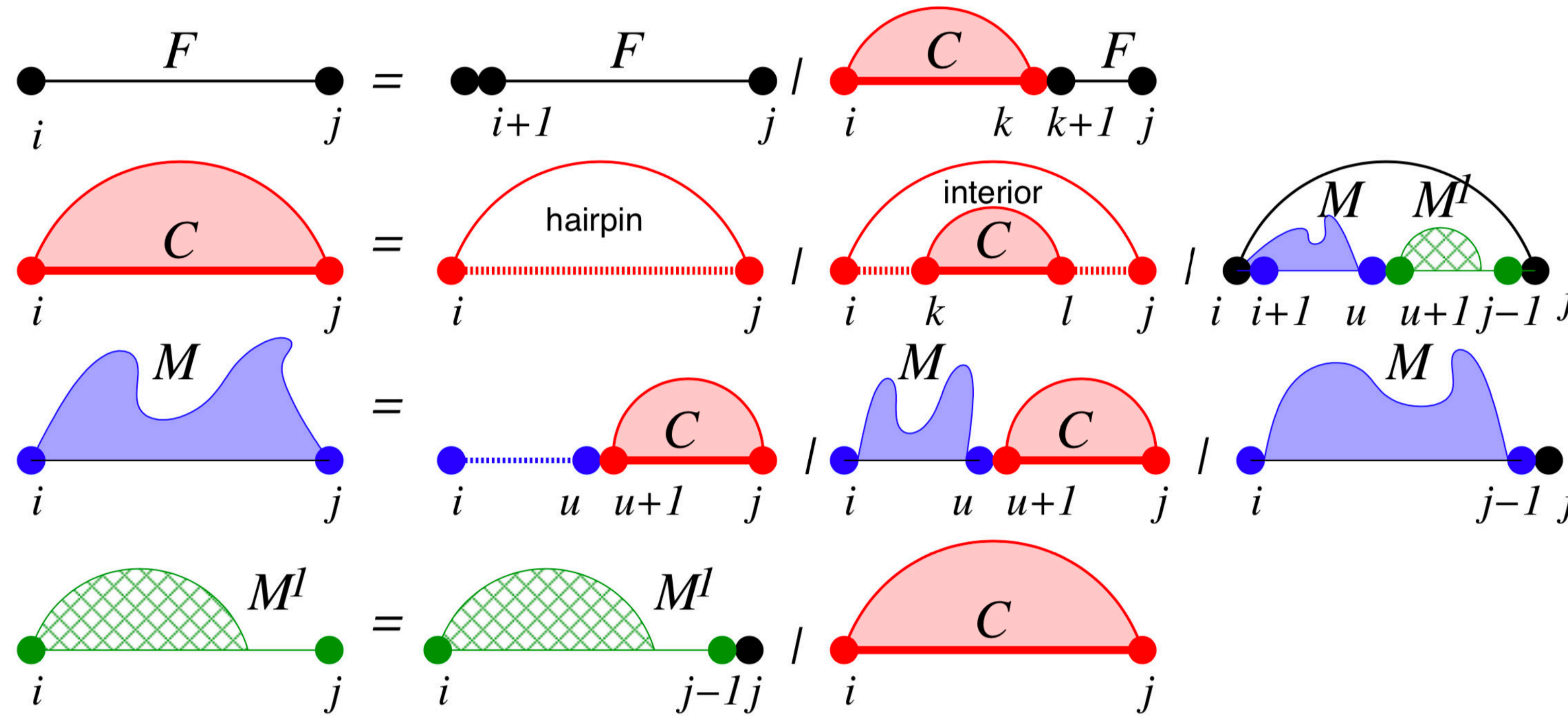
The standard energy model expresses the free energy of a structure as the sum over its loop energies

$$E(\mathcal{S}) = \sum_{l \in \mathcal{S}} E(l)$$

- Good approximation for most oligonucleotides
- Loop energies depend on loop type/size and some sequence dependence
- Most relevant parameters are experimentally measured; some still guesswork
- Secondary structures are macro-states, hence energies are **temperature-dependent free energies**
- Training parameters is becoming a viable alternative to experiment



Folding with Loop Based Energies



F_{ij} free energy of the optimal substructure on the subsequence $x[i..j]$.

C_{ij} optimal free energy on $x[i..j]$, where (i, j) pair.

M_{ij} $x[i..j]$ is part of a multiloop and contains at least one pair.

M^1_{ij} same as M_{ij} but contains exactly one component closed by (i, h) .

Partition Function

Recall:



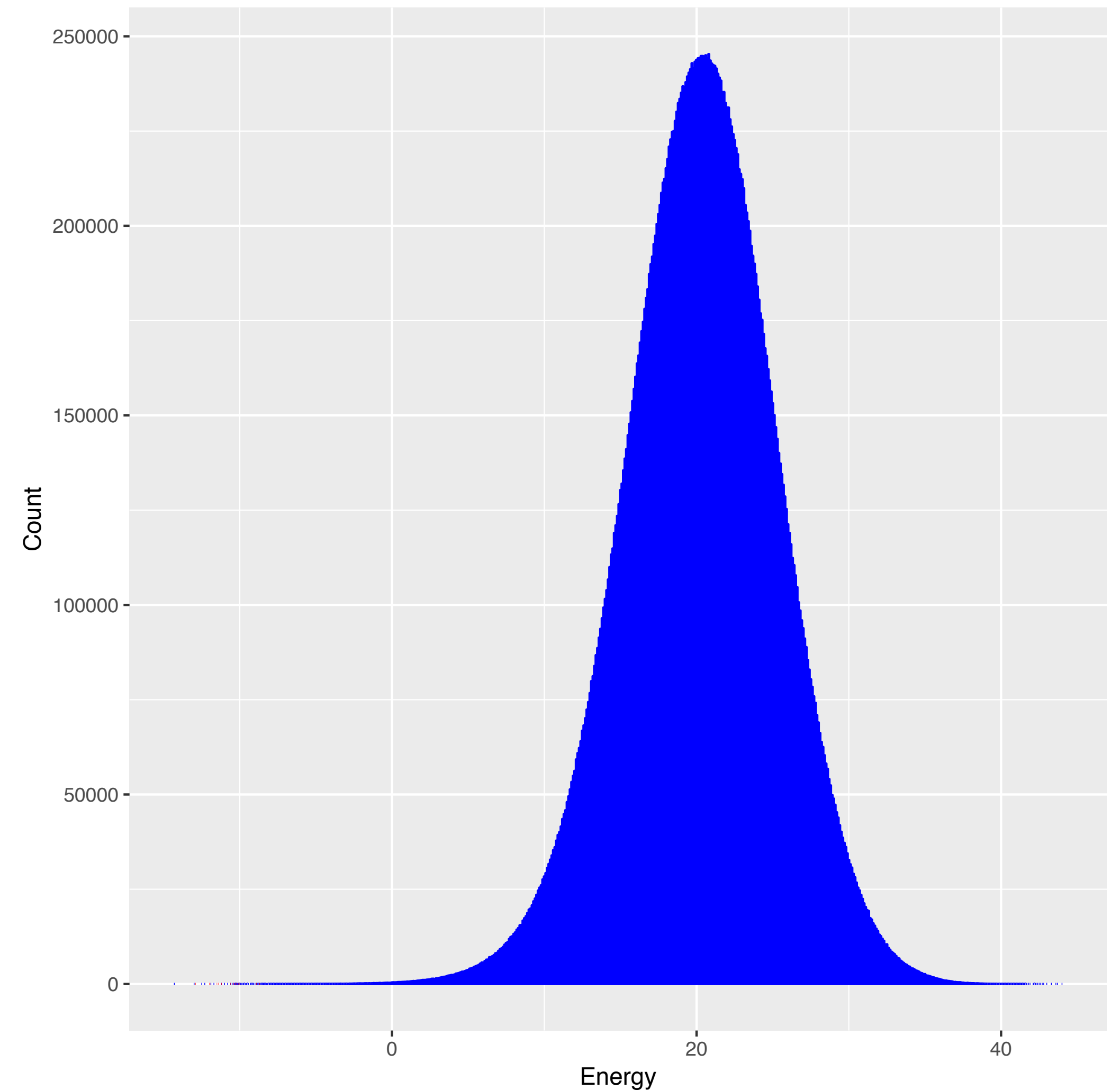
RNA is a biopolymer and ruled by thermodynamics

The **partition function** is the fundamental quantity of statistical mechanics and all thermodynamic properties can be derived from it

$$Z = \sum_{\Psi} \exp\left(-\frac{E(\Psi)}{RT}\right)$$

E.g. the free energy of formation is given by

$$\Delta G = -RT \ln Z$$



Partition Function

Recall:



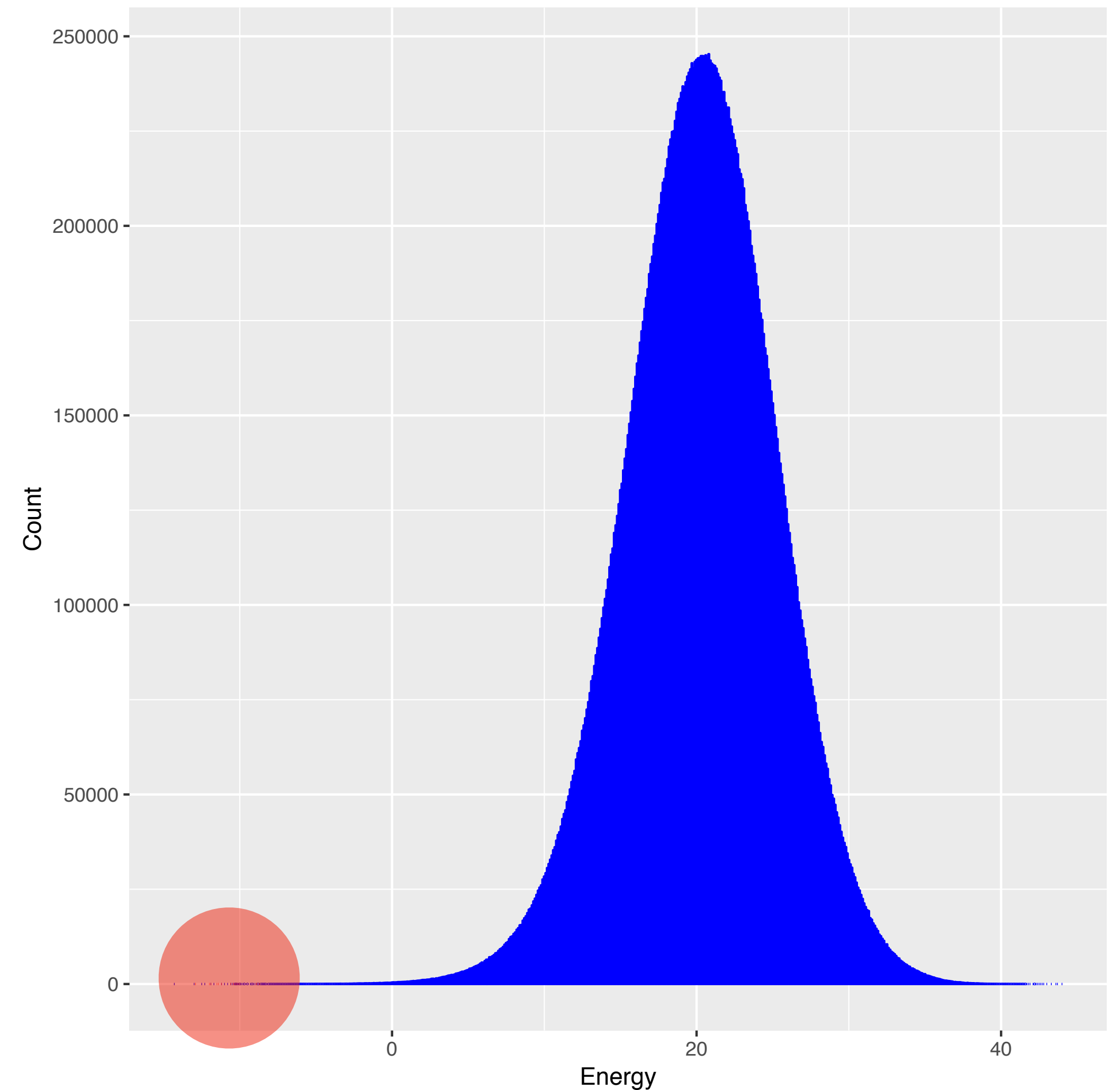
RNA is a biopolymer and ruled by thermodynamics

The **partition function** is the fundamental quantity of statistical mechanics and all thermodynamic properties can be derived from it

$$Z = \sum_{\Psi} \exp\left(-\frac{E(\Psi)}{RT}\right)$$

E.g. the free energy of formation is given by

$$\Delta G = -RT \ln Z$$



Computing the Partition Function

The recursion has the same structure as for energy minimisation, with two differences

- replace minimum operation by sums
- addition of energies by products of partition functions

$$E_{ij} = \min_{i < k \leq j} \left\{ E_{i+1,j} ; \left(E_{i+1,k-1} + E_{k+1,j} + \varepsilon(x_i, x_k) \right) \right\}$$

$$Z_{ij} = Z_{i+1,j} + \sum_{k, (i,k) \text{ pairs}} Z_{i+1,k-1} Z_{k+1,j} \exp(-\varepsilon(x_i, x_k)/RT)$$

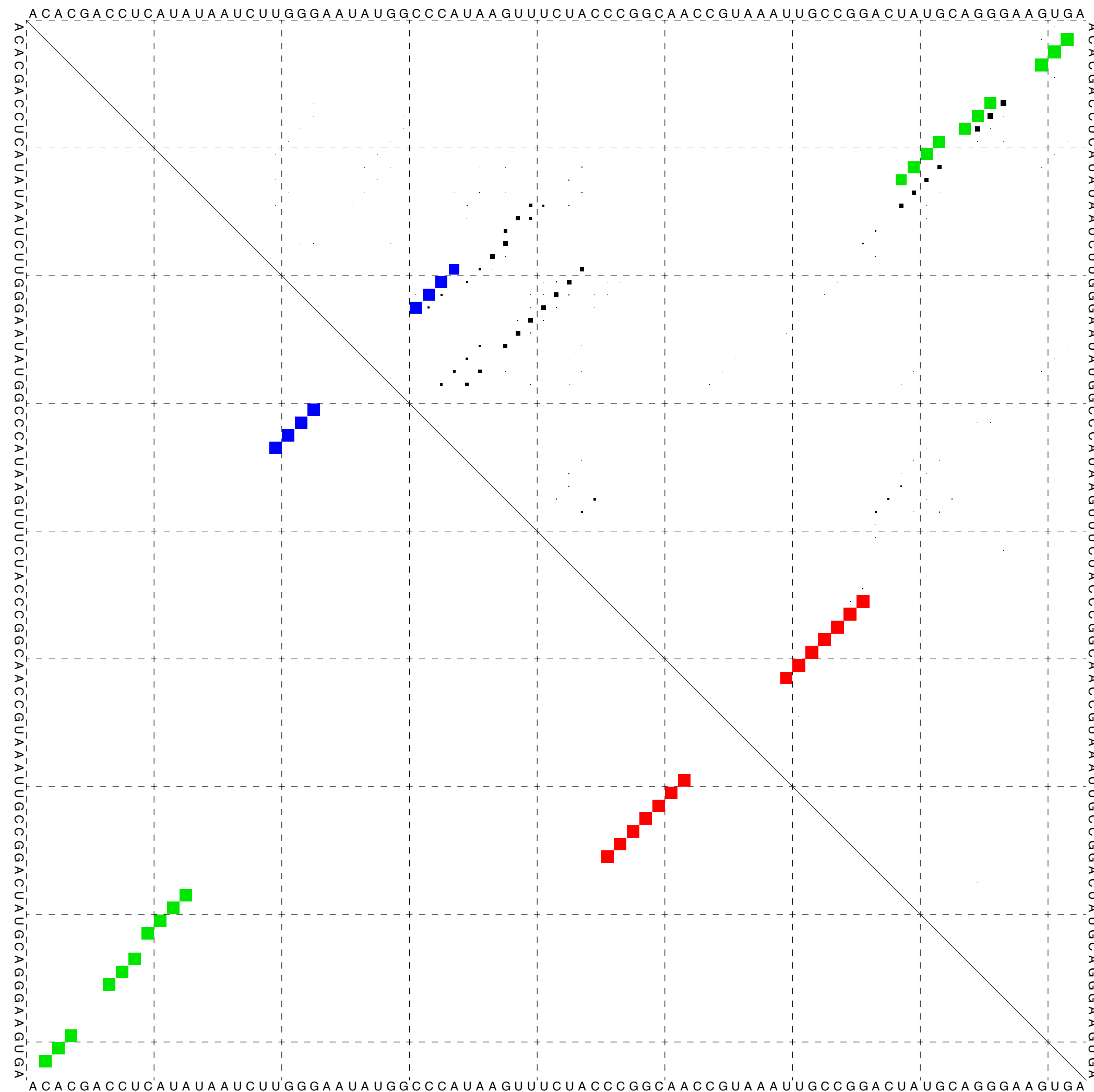
The probability of structure features can be computed from Z, e.g. the probability that a pair is formed

$$p_{ij} = \sum_{\Psi, (i,j) \in \Psi} p(S)$$

For efficient computation define the partition function $\hat{Z}_{ij}$ for structures outside the subsequence $x[i..j]$

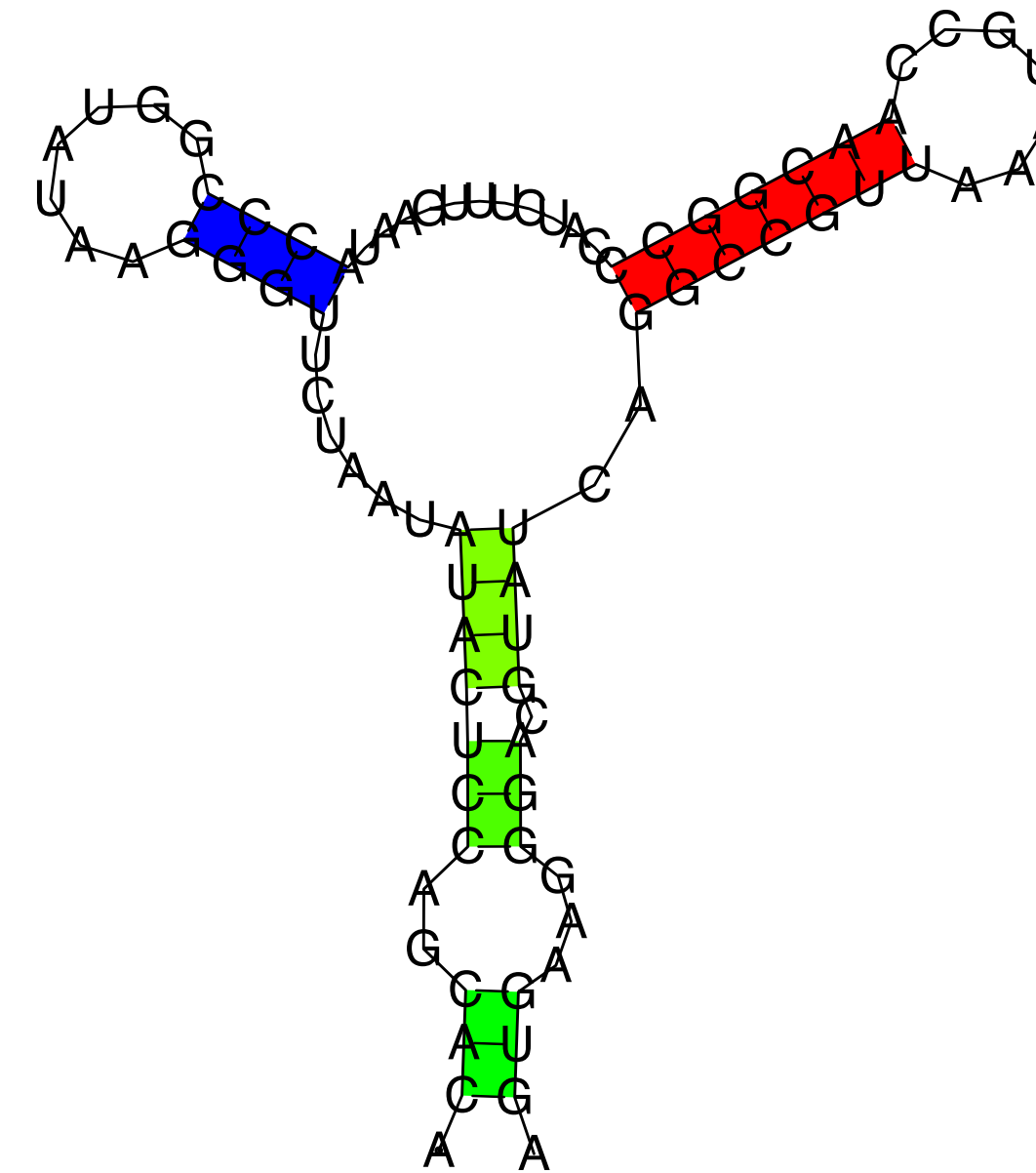
$$p_{ij} = \hat{Z}_{ij} Z_{i+1,j-1} \exp(-\varepsilon_{ij}/RT) / Z$$

Representing Ensembles of RNA Structures



Ensembles of structures (thermodynamic equilibrium) are best represented by base pair probabilities.

A pair (i, j) with probability p is represented by a square in row i and column j with area p .



The Vienna RNA Package

- Minimum free energy and partition function folding
- Complete suboptimal folding
- Inverse folding / RNA design
- Comparison of secondary structures
- Specific heat curves
- Inclusion of structure probing data
- Analysis of folding kinetics / co-transcriptional folding
- Utilities for plotting and annotation structures
- 2.5D prediction: G-quadruplexes and pseudo-knots
- Prediction of consensus structures

For the programmer:

- A C library to link against your programs
- Python/Perl scripting language interface

The Vienna RNA Package

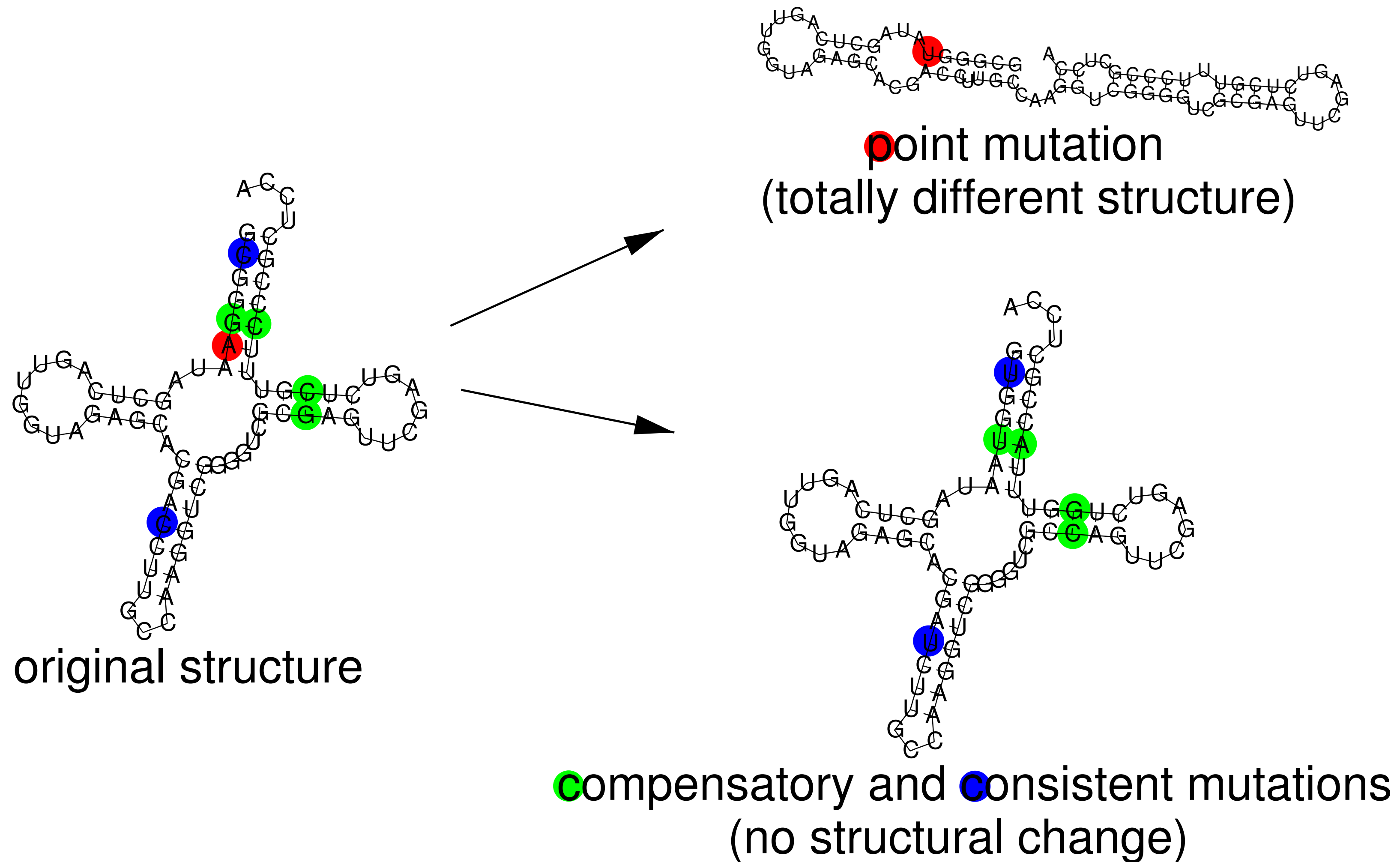
- Minimum free energy and partition function folding
- Complete suboptimal folding
- Inverse folding / RNA design

Free software, C source code and fold server available at

<http://www.tbi.univie.ac.at/RNA/>

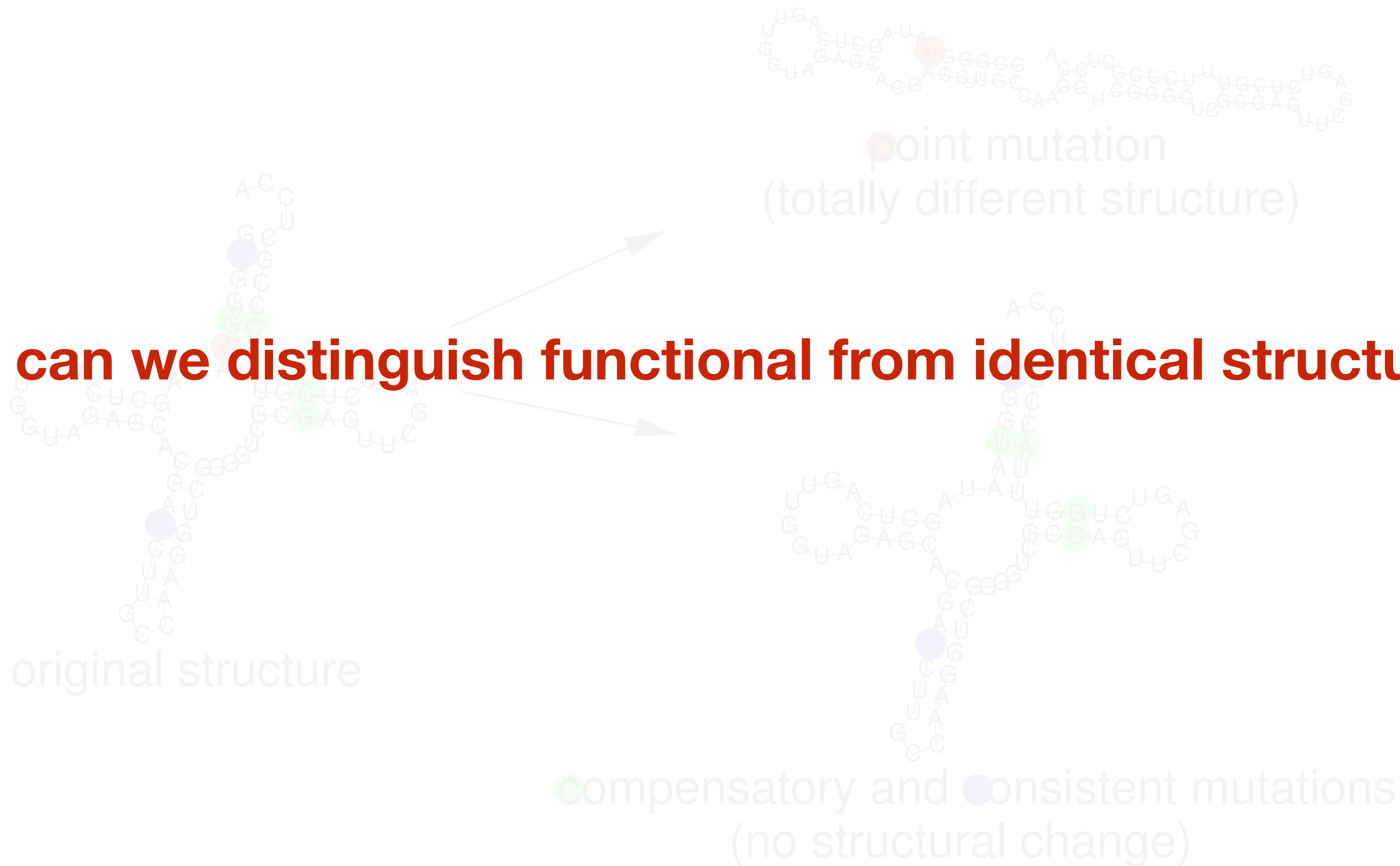
- Comparison of secondary structures
 - Specific heat curves
 - Inclusion of structure probing data
 - Analysis of folding kinetics / co-transcriptional folding
 - Utilities for plotting and annotation structures
 - Prediction of consensus structures
- For the programmer:
- A C library to link against your programs
 - Python/Perl scripting language interface

Functional Structures: Point Mutations



Functional Structures: Point Mutations

How can we distinguish functional from identical structures?



Consensus Structures: Alignment Folding

Combine covariance analysis and folding into one DP algorithm

- Apply conventional folding algorithm to alignment
- Use a modified energy function that includes covariance score

$$E_c(A, \Psi) = \sum_k E(A_k, \Psi) + cv \cdot \sum_{(i,j) \in \Psi} B_{ij}$$

- Can be used for all variants: MFE, partition function, ...
- Efficient: $\mathcal{O}(N \cdot n^2 + n^3)$ CPU and $\mathcal{O}(n^2)$ memory for alignment length n and N sequences
- Same results as RNAfold for single sequences

```
GGGCCGGUAGCUCAUUAAGGCAGAGCGUCUGACUCUUAACAGACGGUCGCGUGUUCGAAUCGCGUCCGGCCCA
GCCGAAAUAGCUCAGUU-GGGAGAGCGUUAGACUGAAGACUAAAGGUCCCCGGUUCAAUCCCGGGUUUCGCCA
GCGGAUUUAGCUCAGUU-GGGAGAGCGCCAGACUGAAGAUUUGGAGGUCCUGUGUUCGAUCCACAGAAUUCGCA
CAGAGUGUAGCUUA---ACACAAAGCACCCAACUACACUAGGAGAU-UUCAACU-UAACUUGACCGCUCUGA
(((((((..((((.....))))).((((.....))))).(((.....)))))))).((((.....))))).
mfe: -29 kcal/mol = (-21.3 + -7.7) kcal/mol
```

Consensus Structures: Alignment Folding

Combine covariance analysis and folding into one DP algorithm

RNAalifold

- Apply conventional folding algorithm to alignment
- Use a modified energy function that includes covariance score

$$E_c(A, \Psi) = \sum_k E(A_k, \Psi) + cv \cdot \sum_{(i,j) \in \Psi} B_{ij}$$

- Can be used for all variants: MFE, partition function, ...
- Efficient: $\mathcal{O}(N \cdot n^2 + n^3)$ CPU and $\mathcal{O}(n^2)$ memory for alignment length n and N sequences
- Same results as RNAfold for single sequences

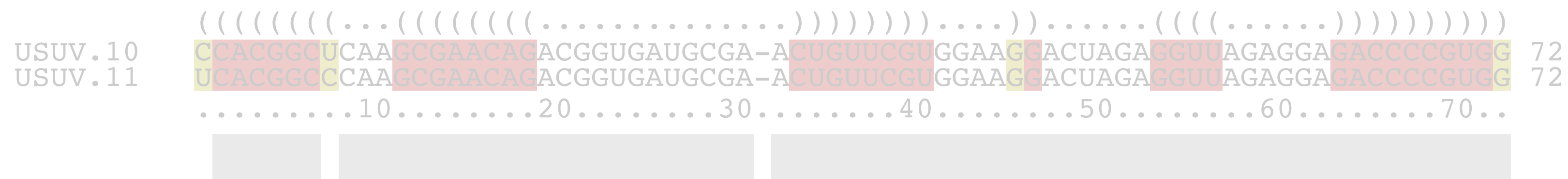
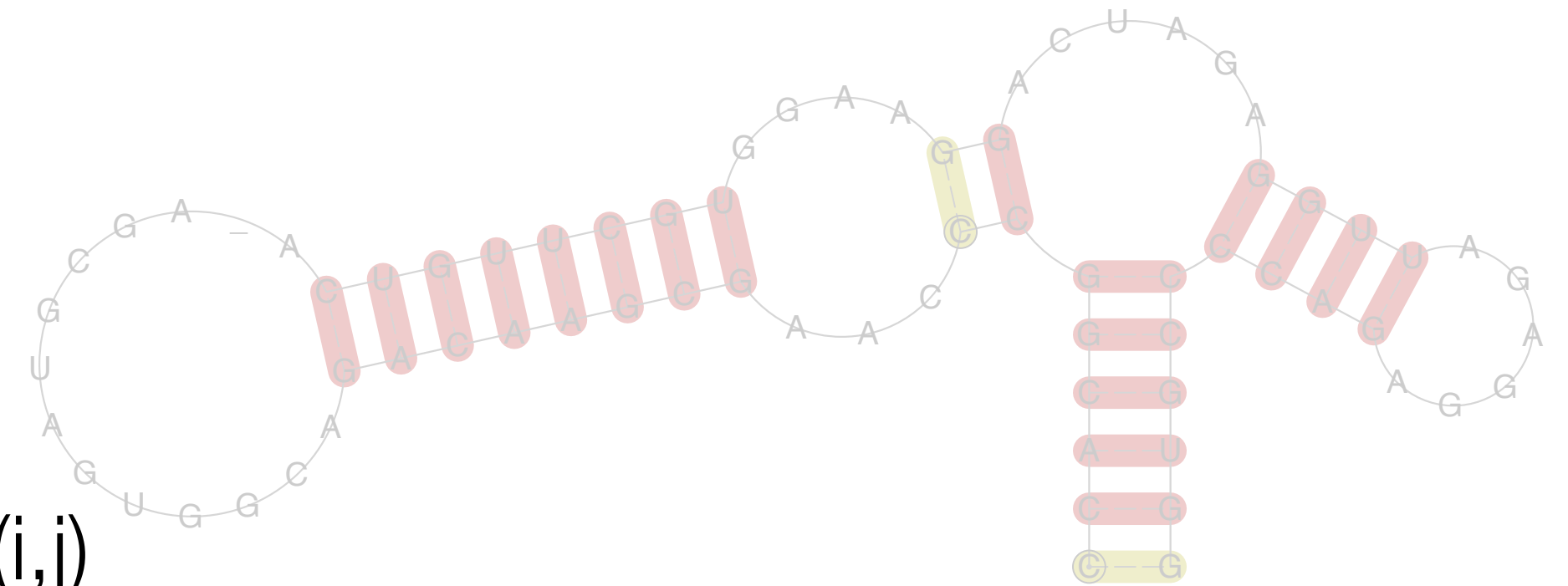
```
GGGCCGGUAAGCUCAUUUAGGCAAGAGCGUCUGACUCUUAUUCAGACGGUCGCGUGUUCGAAUCGCGUCCGGCCCA
GCCGAAAUAGCUCAGUU-GGGAGAGCGUUAGACUGAAGAUCUAAAGGUCCCCGGUUCAAUCCCGGGUUUCGGCA
GCGGAUUUAAGCUCAGUU-GGGAGAGCGCCAGACUGAAGAUUUGGAGGUCUGUGUUCGAUCCACAGAAUUCGCA
CAGAGUGUAAGCUUA---ACACAAAGCACCCAAACUUACACUUAGGAGAU-UUCAACU-UAACUUGACC GCUCUGA
```

((((((((..((((.....))))).((((.....))))).(((.....))))).(((.....))))).(((.....))))).

mfe: -29 kcal/mol = (-21.3 + -7.7) kcal/mol

RNA Covariation as Evolutionary Trait

- High mutation rate in RNA viruses due to error-prone RdRP
- For base pair (i,j): GC/CG/AU/UA/GU/UG
- Consistent mutation: different standard combinations
- Compensatory mutation: both positions are mutated
- Presence of both strongly supports predicted base pair (i,j)



types of pairs:

	1	2	3	4	5	6
0						
1						
2						

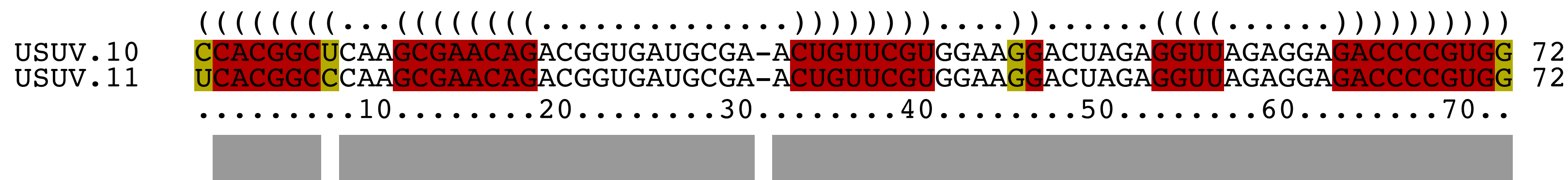
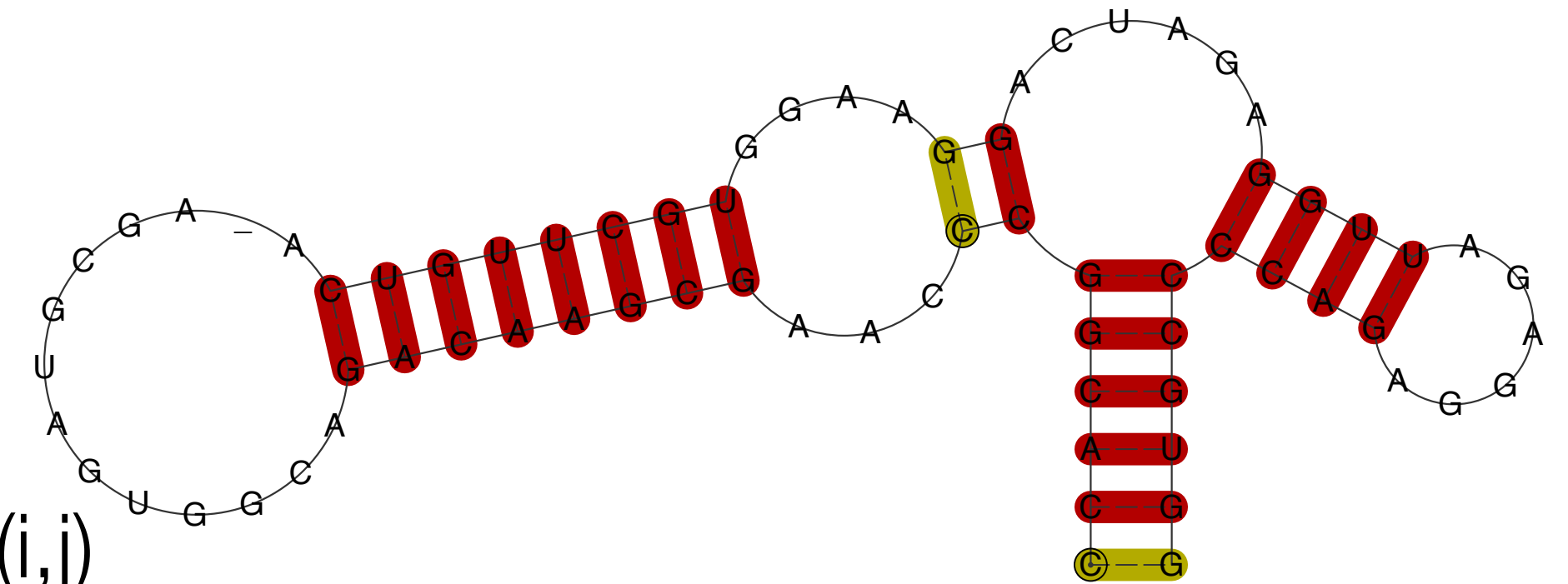
incompatible pairs in column

allowed pairs:

A-U	G-C	G-U
U-A	C-G	U-G

RNA Covariation as Evolutionary Trait

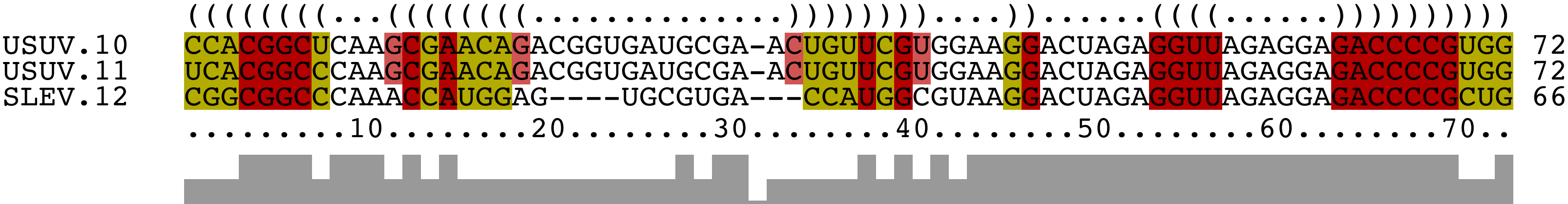
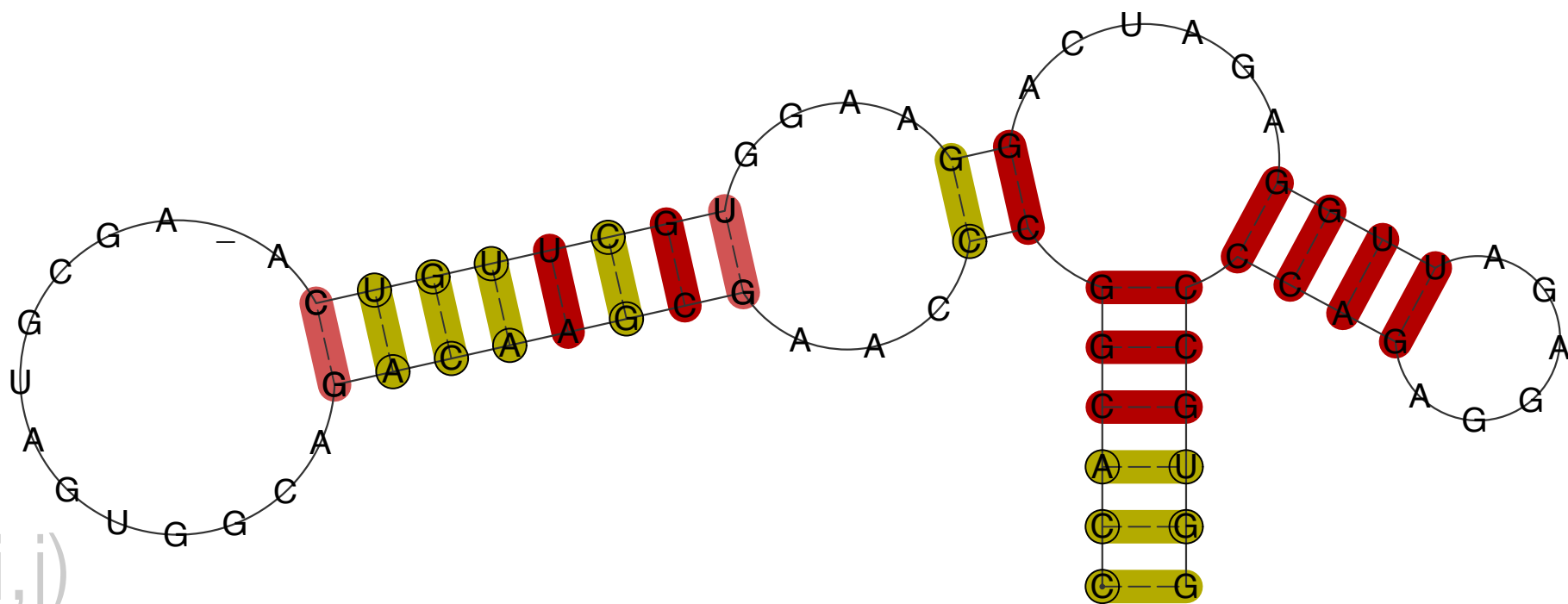
- High mutation rate in RNA viruses due to error-prone RdRP
- For base pair (i,j): GC/CG/AU/UA/GU/UG
- Consistent mutation: different standard combinations
- Compensatory mutation: both positions are mutated
- Presence of both strongly supports predicted base pair (i,j)



		types of pairs:					
		1	2	3	4	5	6
incompatible pairs in column	0						
	1						
	2						
allowed pairs:		A-U U-A		G-C C-G		G-U U-G	

RNA Covariation as Evolutionary Trait

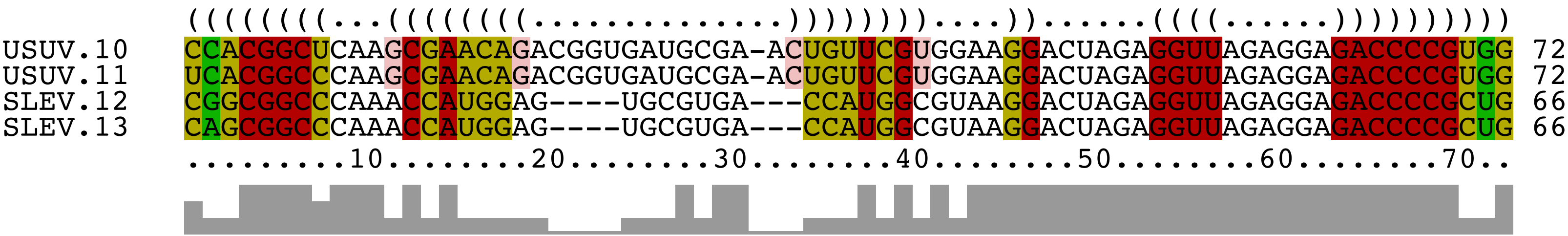
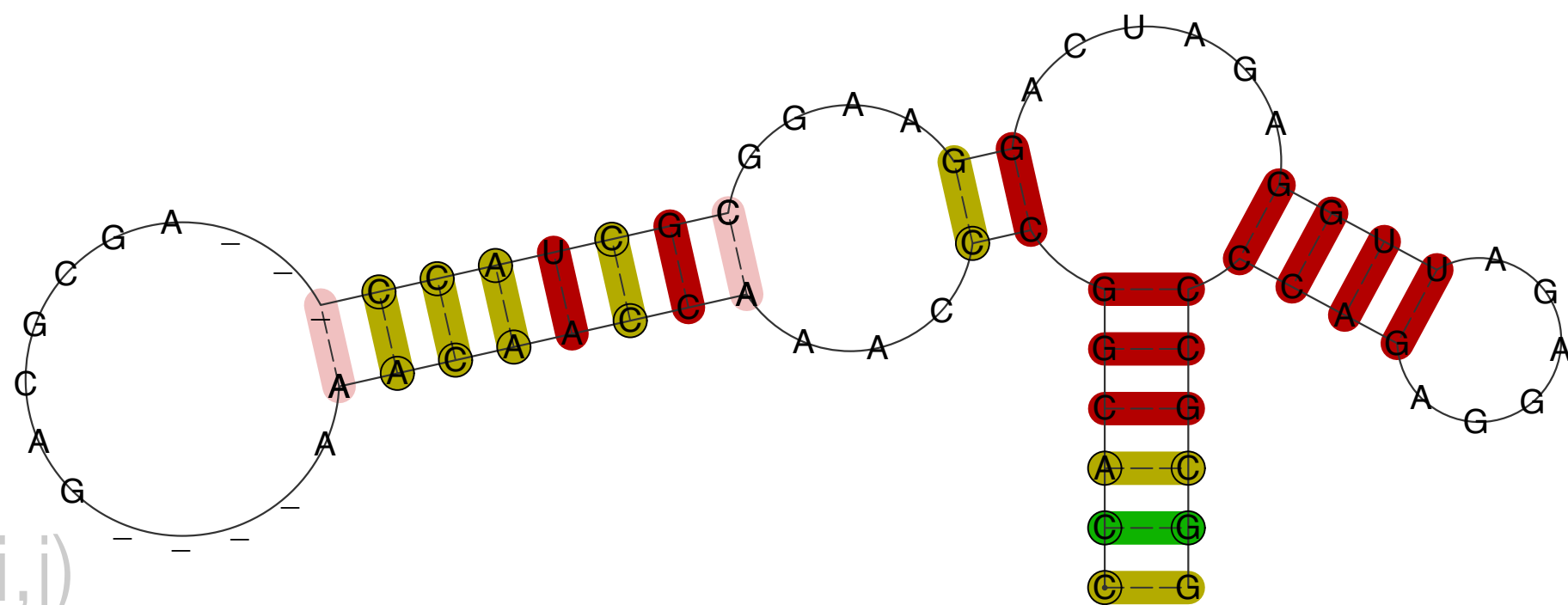
- High mutation rate in RNA viruses due to error-prone RdRP
- For base pair (i,j): GC/CG/AU/UA/GU/UG
- Consistent mutation: different standard combinations
- Compensatory mutation: both positions are mutated
- Presence of both strongly supports predicted base pair (i,j)



		types of pairs:					
		1	2	3	4	5	6
incompatible pairs in column	0						
	1						
	2						
allowed pairs:		A-U U-A		G-C C-G		G-U U-G	

RNA Covariation as Evolutionary Trait

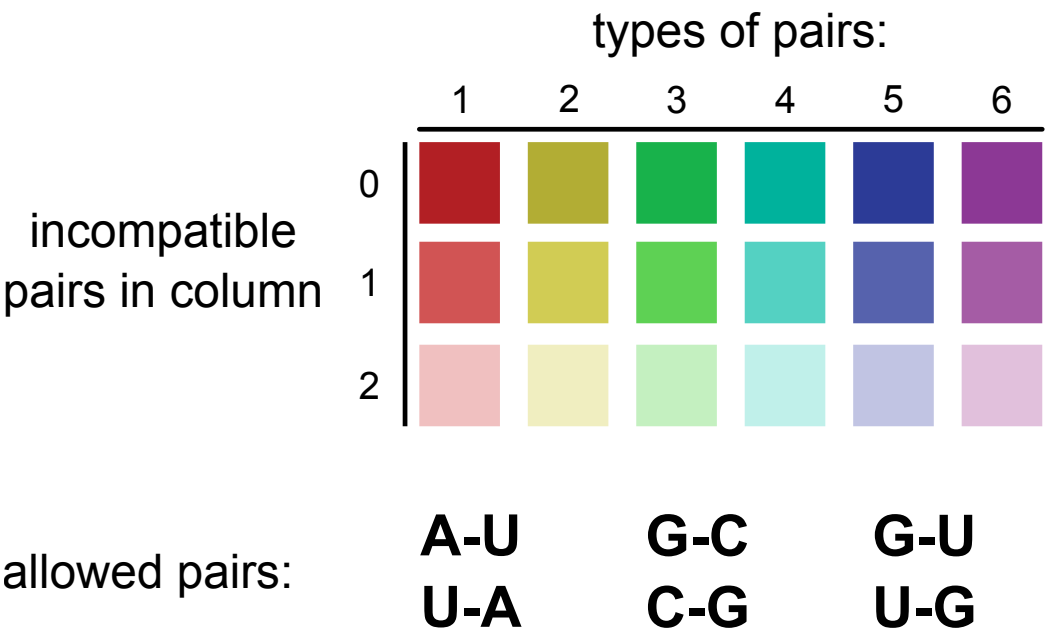
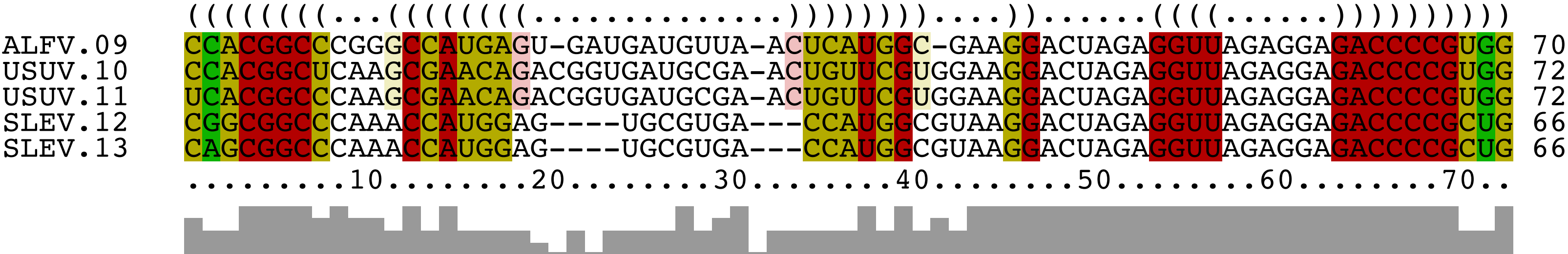
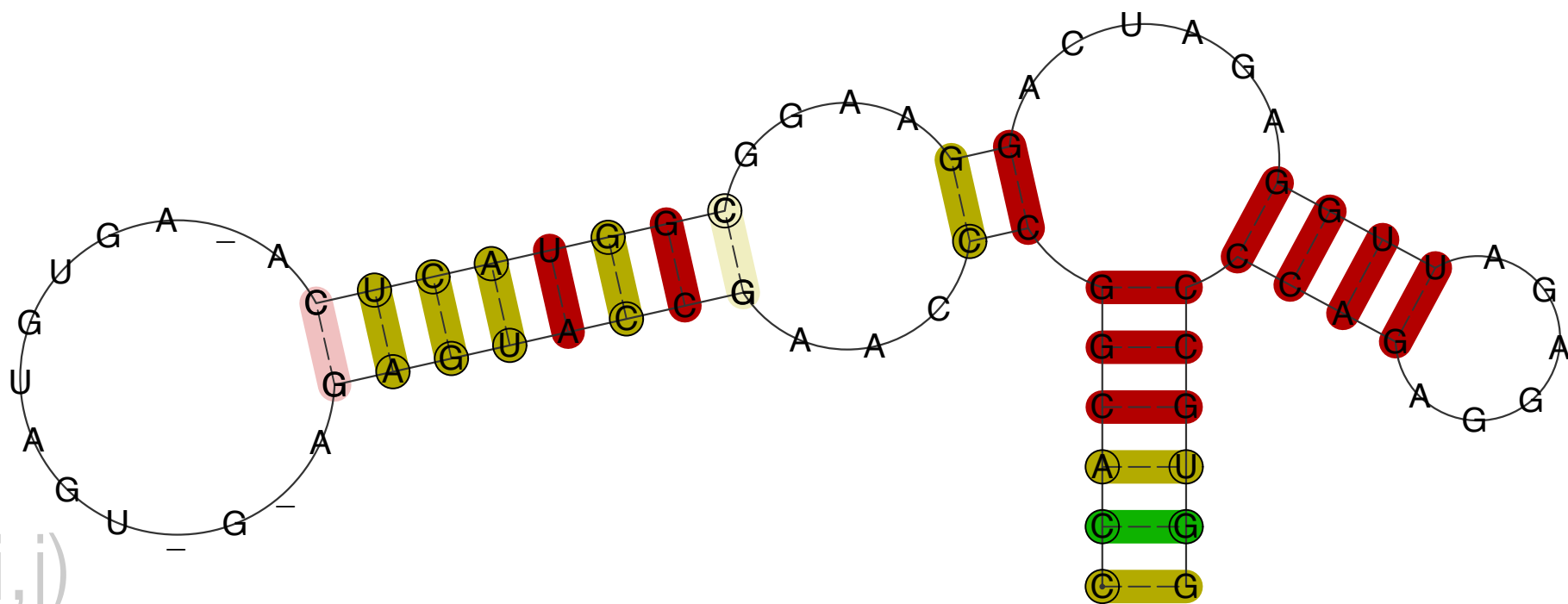
- High mutation rate in RNA viruses due to error-prone RdRP
- For base pair (i,j): GC/CG/AU/UA/GU/UG
- Consistent mutation: different standard combinations
- Compensatory mutation: both positions are mutated
- Presence of both strongly supports predicted base pair (i,j)



		types of pairs:					
		1	2	3	4	5	6
incompatible pairs in column	0						
	1						
	2						
allowed pairs:		A-U U-A		G-C C-G		G-U U-G	

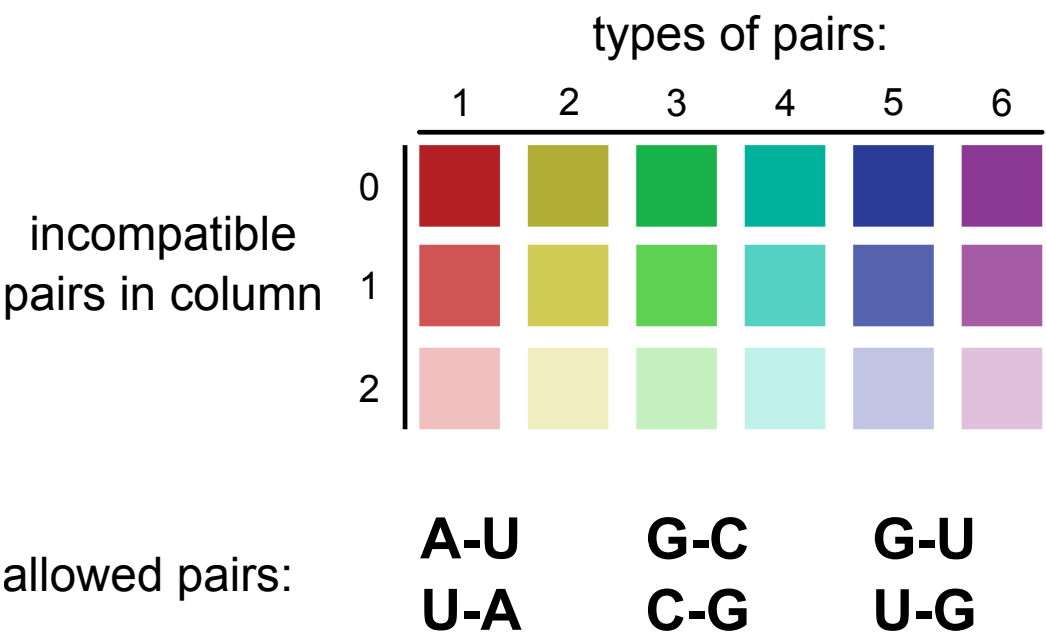
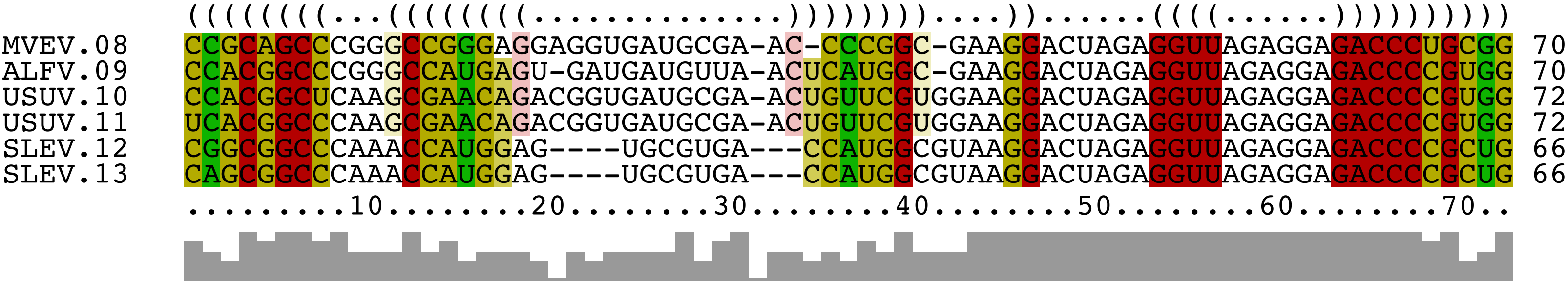
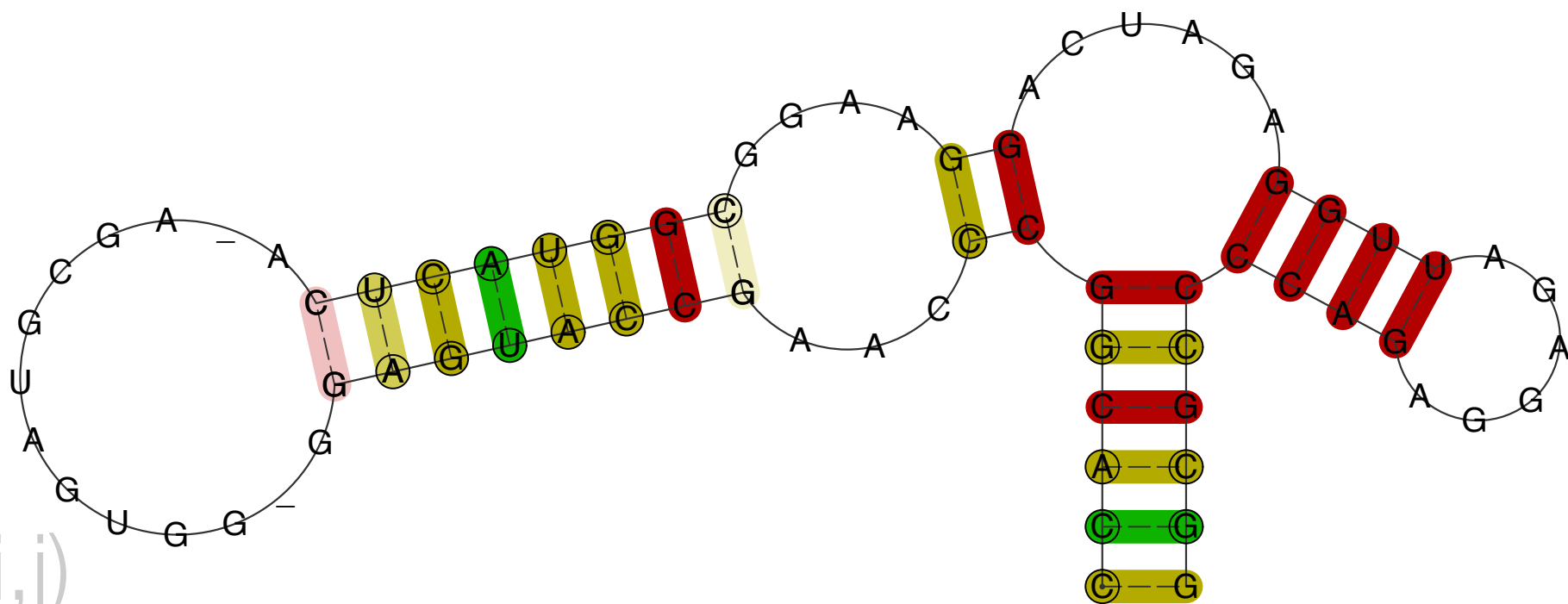
RNA Covariation as Evolutionary Trait

- High mutation rate in RNA viruses due to error-prone RdRP
- For base pair (i,j): GC/CG/AU/UA/GU/UG
- Consistent mutation: different standard combinations
- Compensatory mutation: both positions are mutated
- Presence of both strongly supports predicted base pair (i,j)

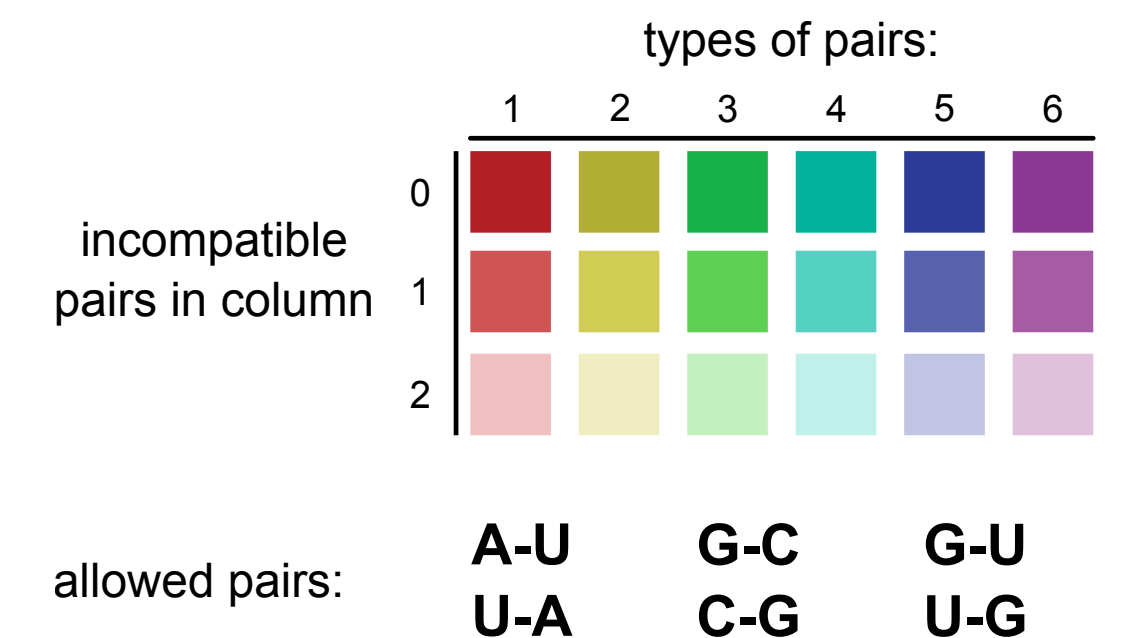
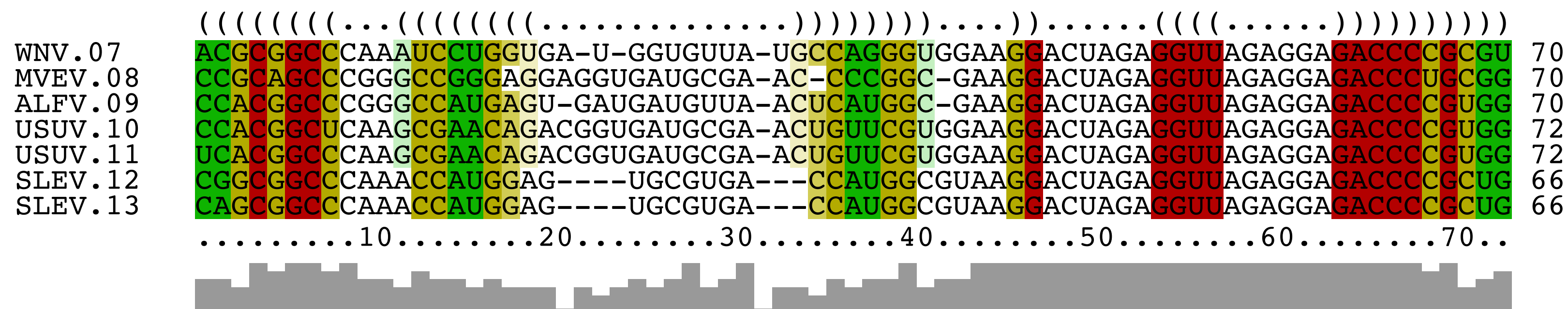
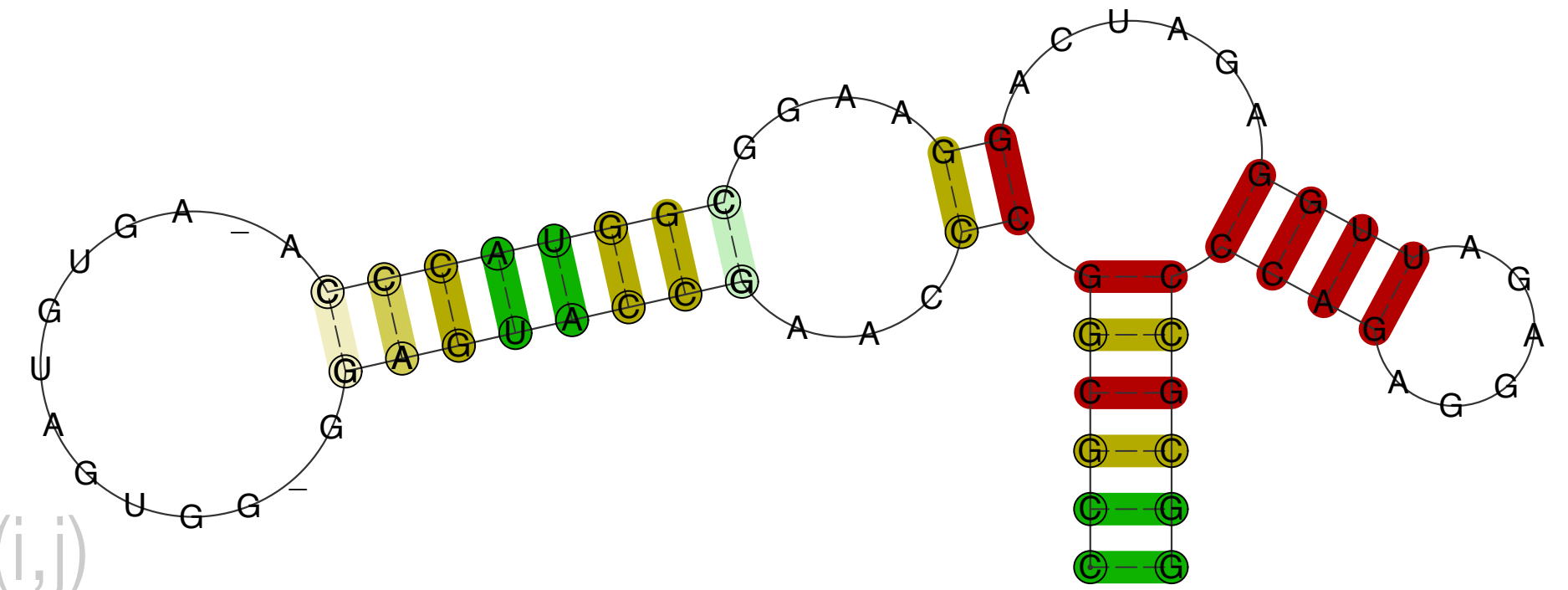


RNA Covariation as Evolutionary Trait

- High mutation rate in RNA viruses due to error-prone RdRP
- For base pair (i,j): GC/CG/AU/UA/GU/UG
- Consistent mutation: different standard combinations
- Compensatory mutation: both positions are mutated
- Presence of both strongly supports predicted base pair (i,j)

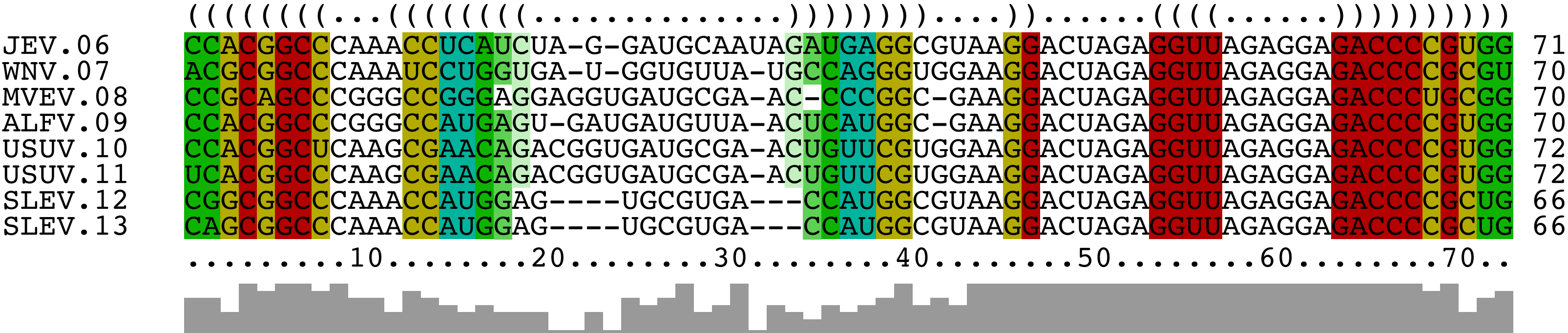
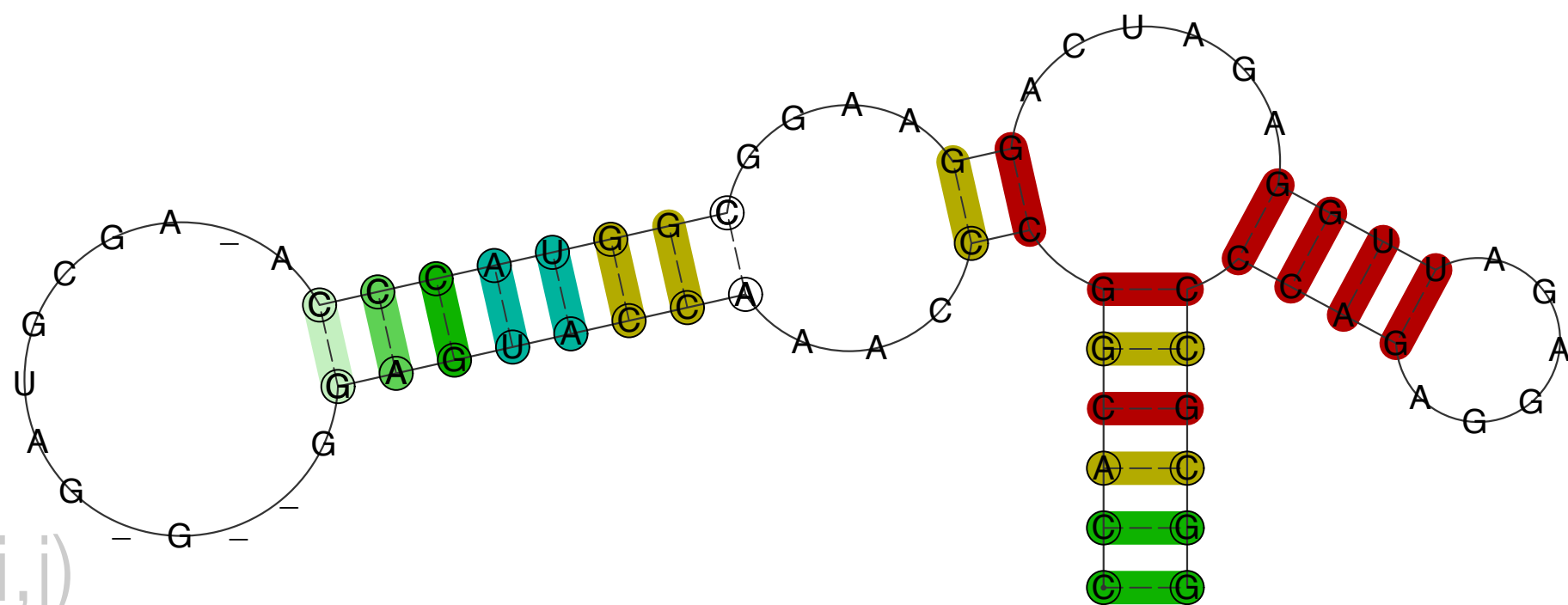


RNA Covariation as Evolutionary Trait



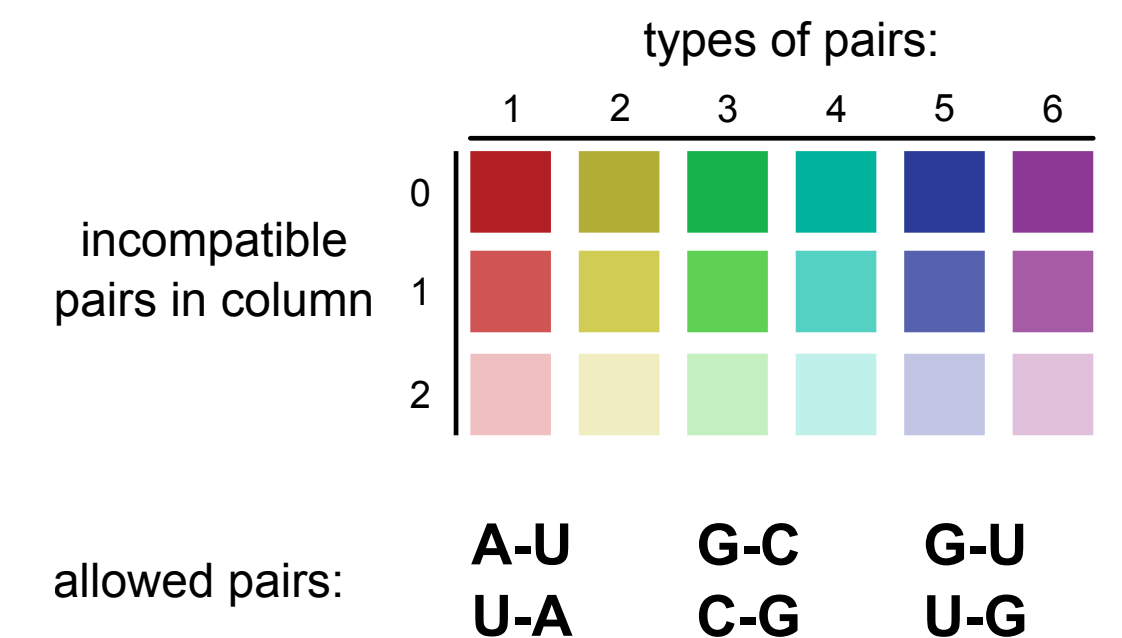
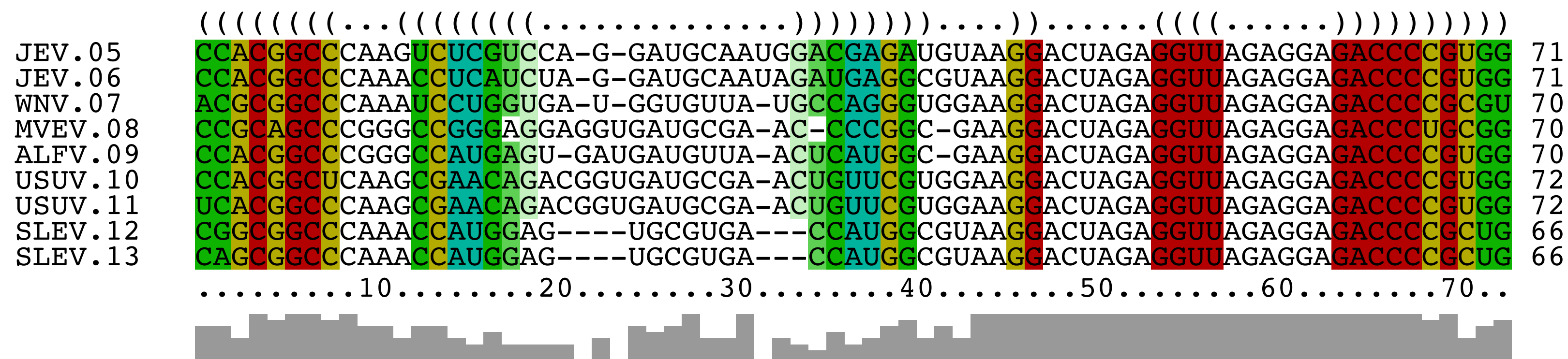
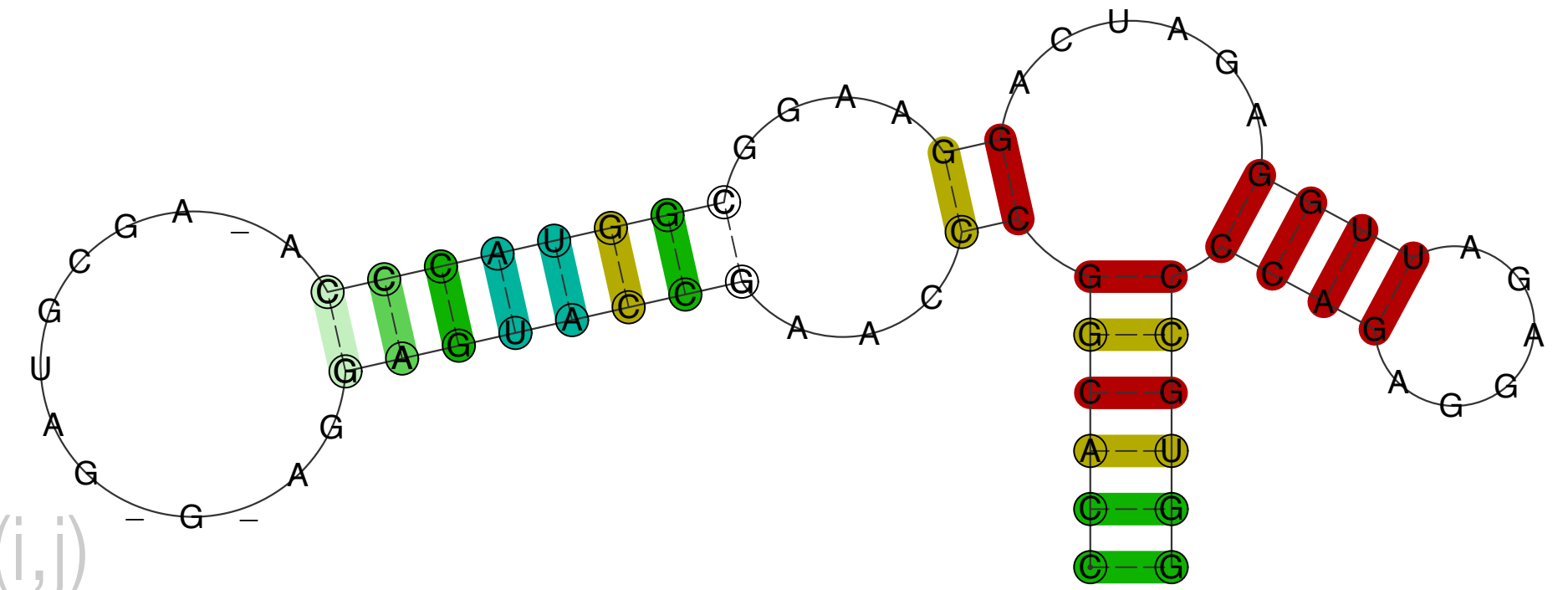
RNA Covariation as Evolutionary Trait

- High mutation rate in RNA viruses due to error-prone RdRP
- For base pair (i,j): GC/CG/AU/UA/GU/UG
- Consistent mutation: different standard combinations
- Compensatory mutation: both positions are mutated
- Presence of both strongly supports predicted base pair (i,j)



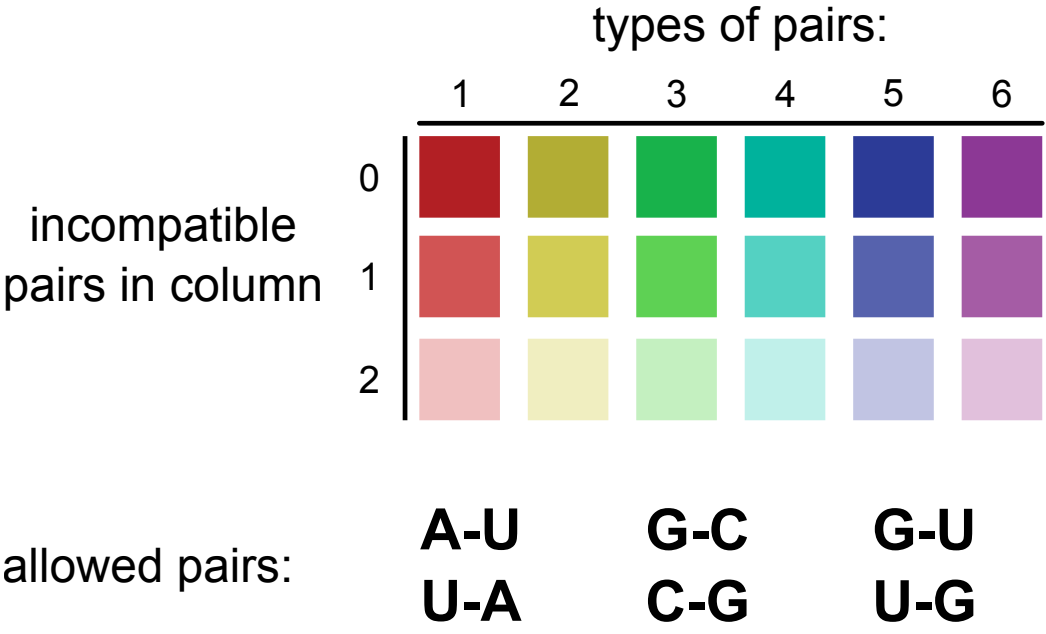
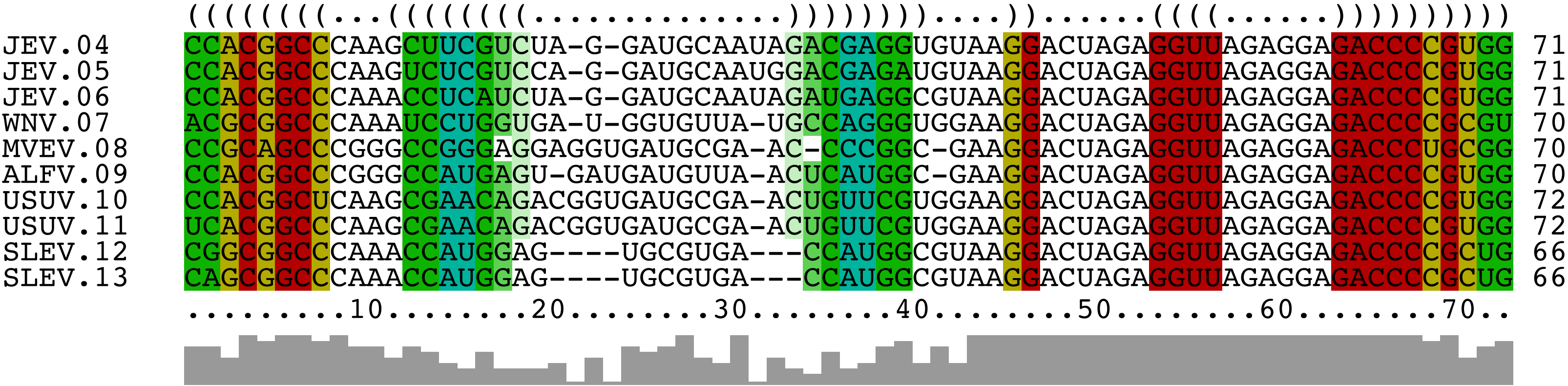
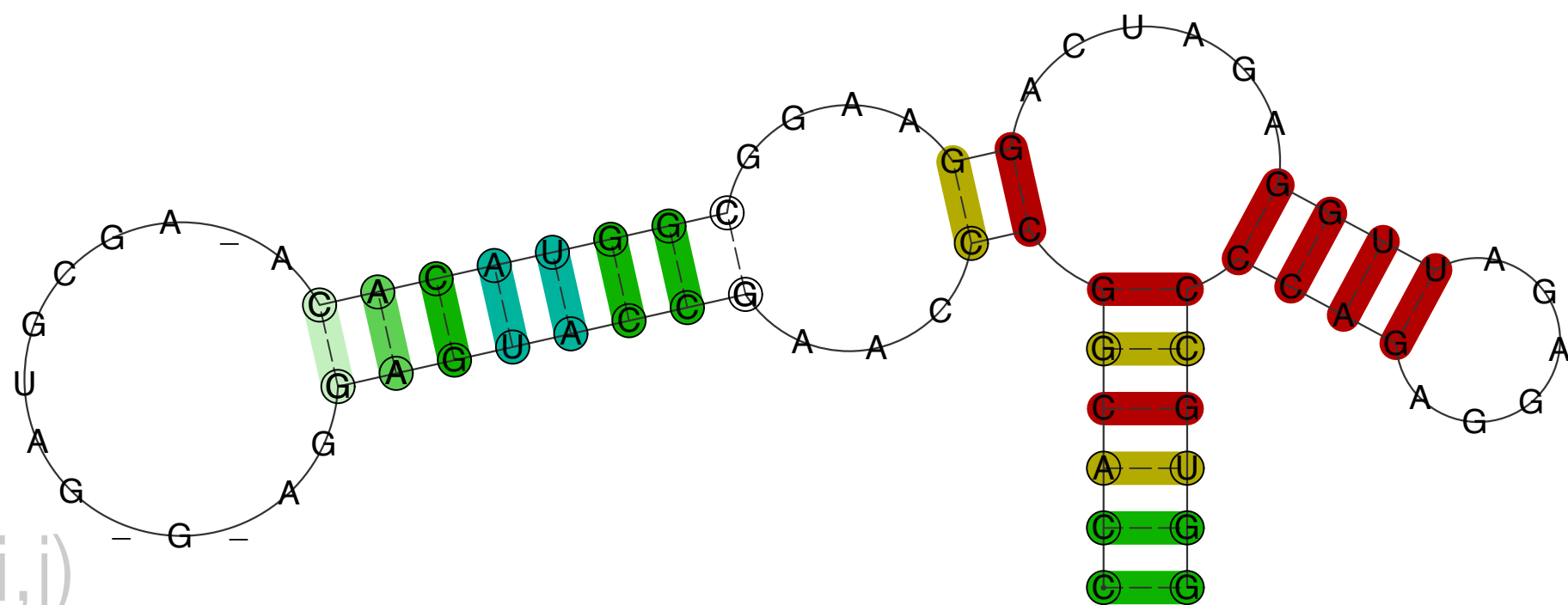
		types of pairs:					
		1	2	3	4	5	6
incompatible pairs in column	0	red	olive	green	teal	blue	purple
	1	red	olive	green	teal	blue	purple
	2	pink	yellow	light green	light blue	light purple	light purple
allowed pairs:		A-U U-A		G-C C-G		G-U U-G	

RNA Covariation as Evolutionary Trait

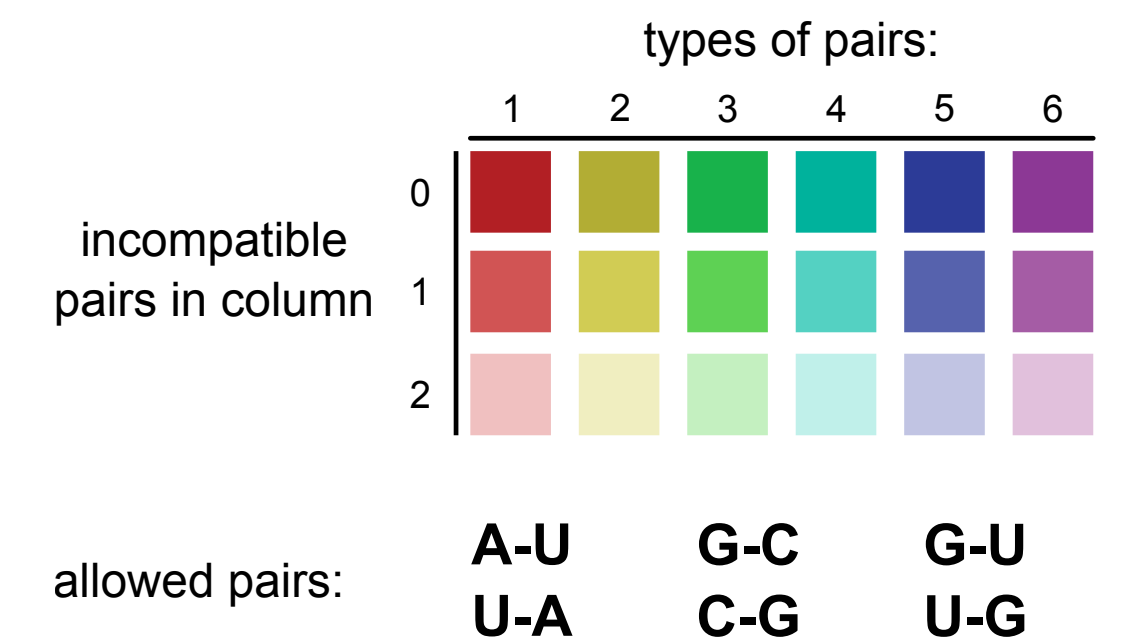
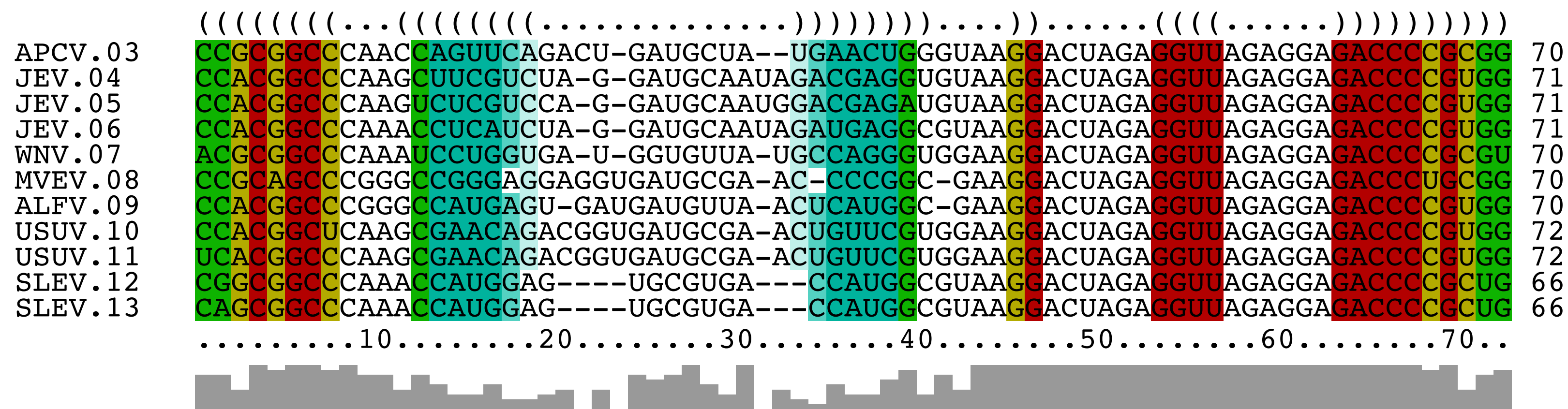
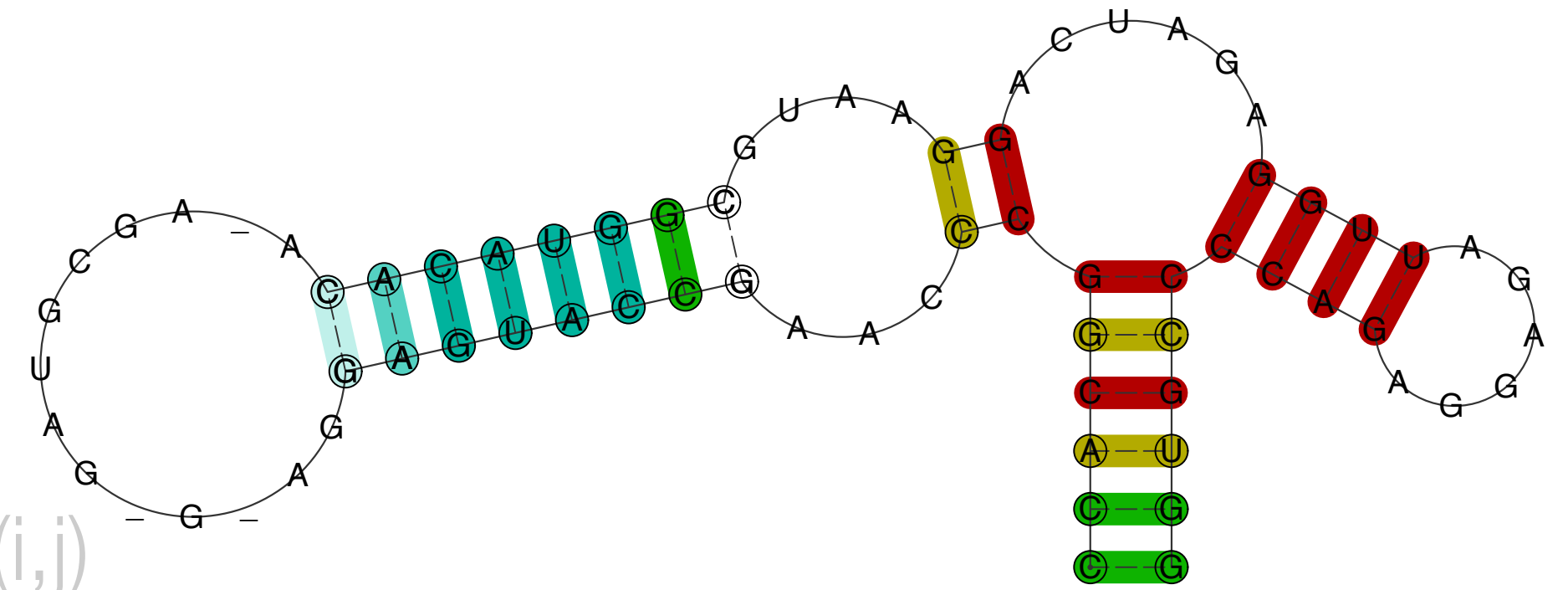


RNA Covariation as Evolutionary Trait

- High mutation rate in RNA viruses due to error-prone RdRP
- For base pair (i,j): GC/CG/AU/UA/GU/UG
- Consistent mutation: different standard combinations
- Compensatory mutation: both positions are mutated
- Presence of both strongly supports predicted base pair (i,j)

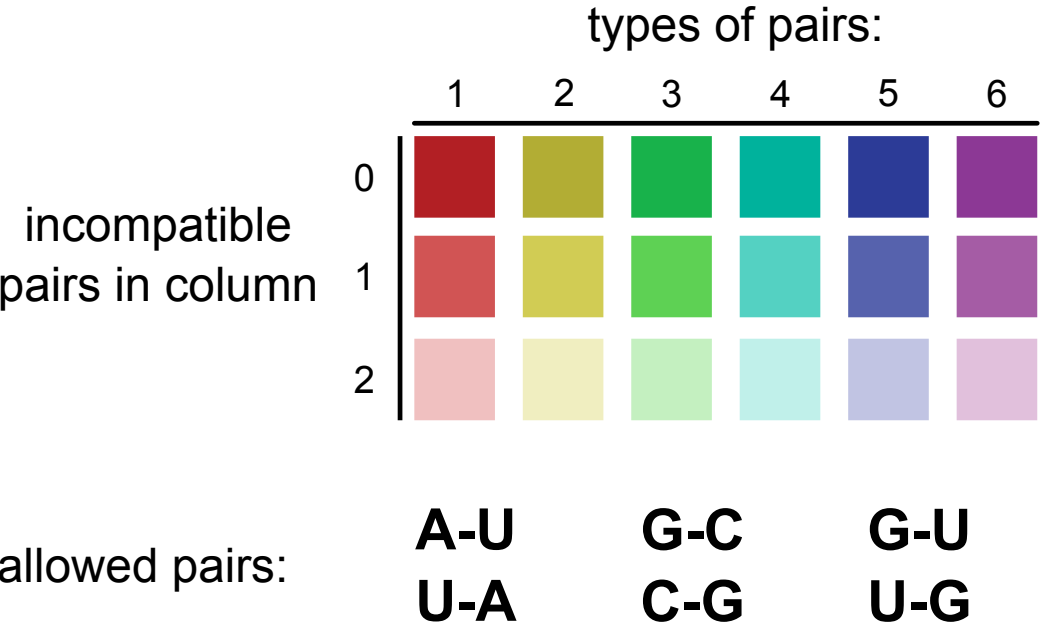
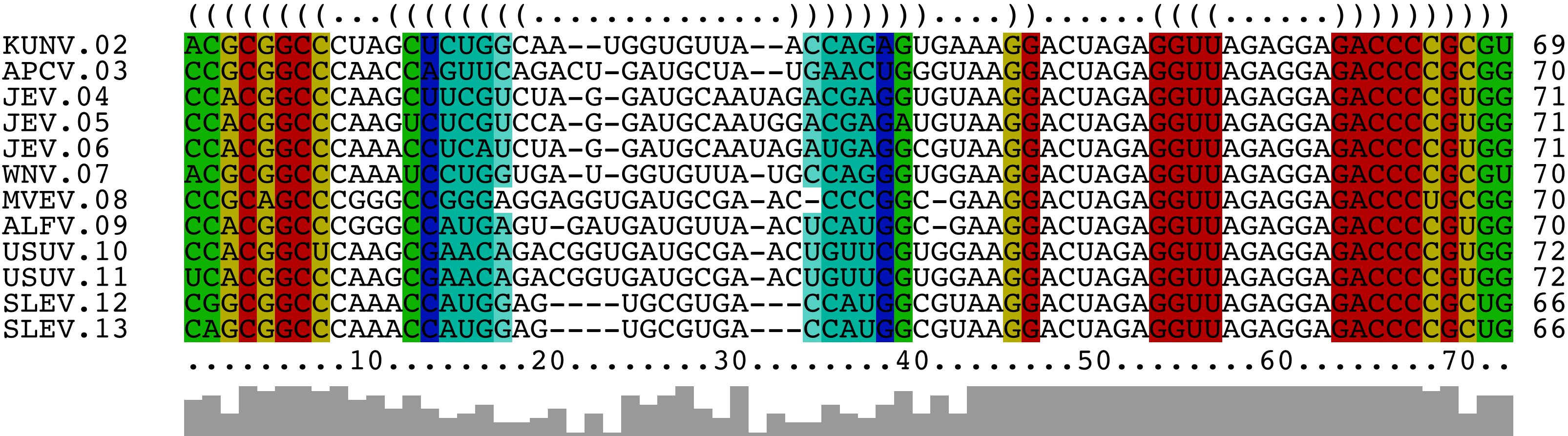
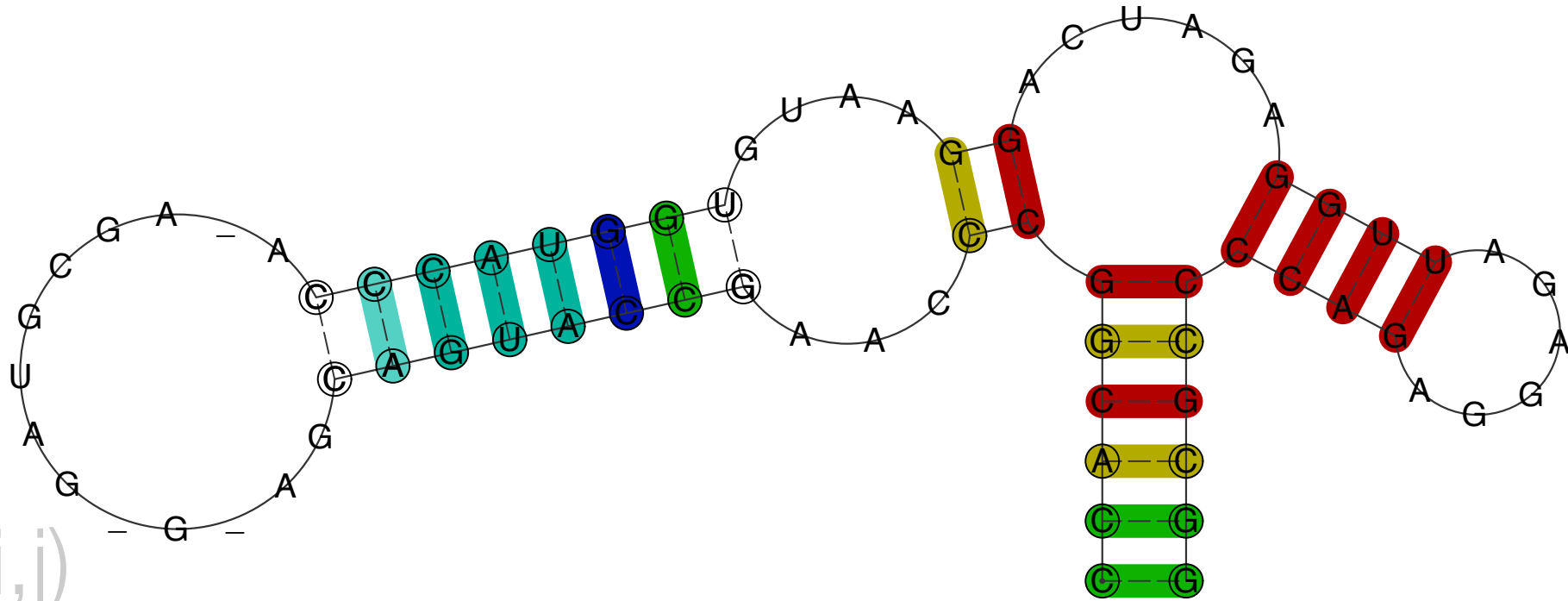


RNA Covariation as Evolutionary Trait



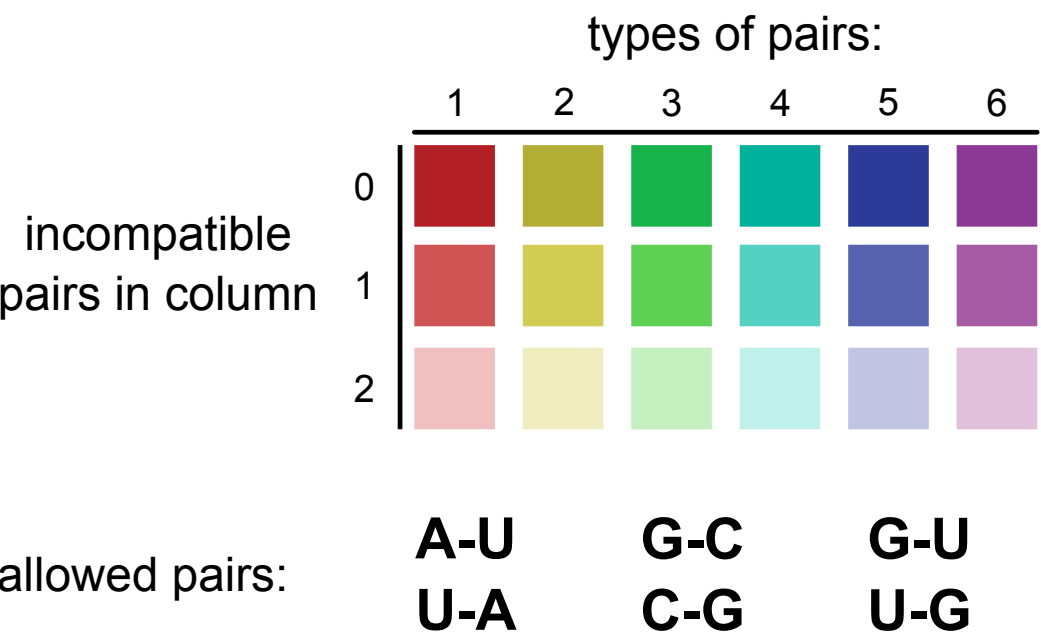
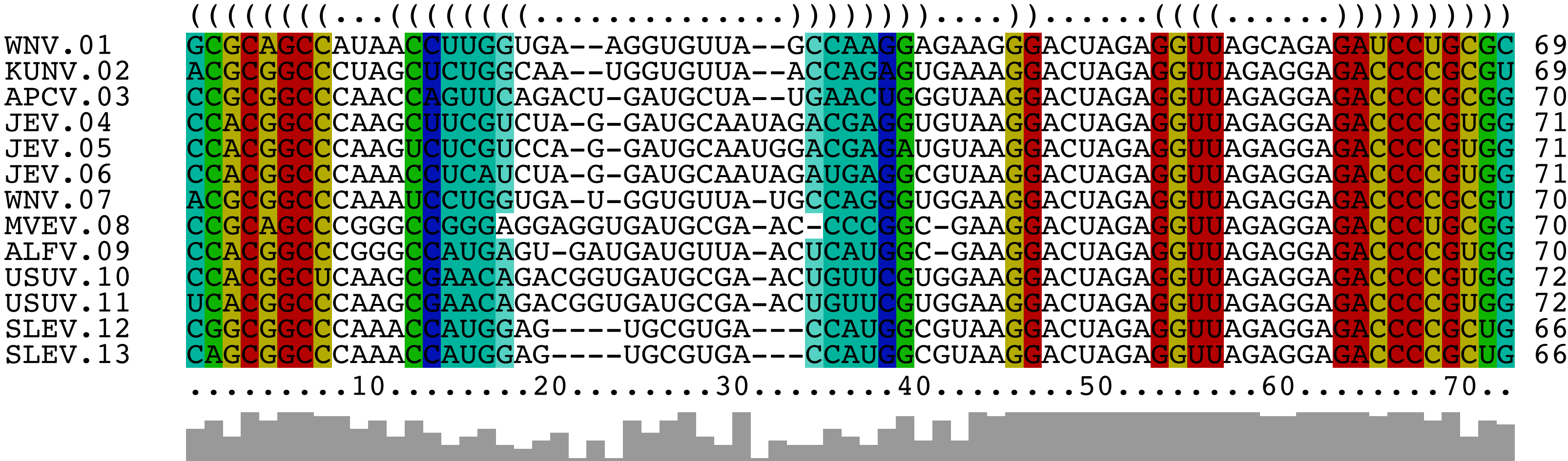
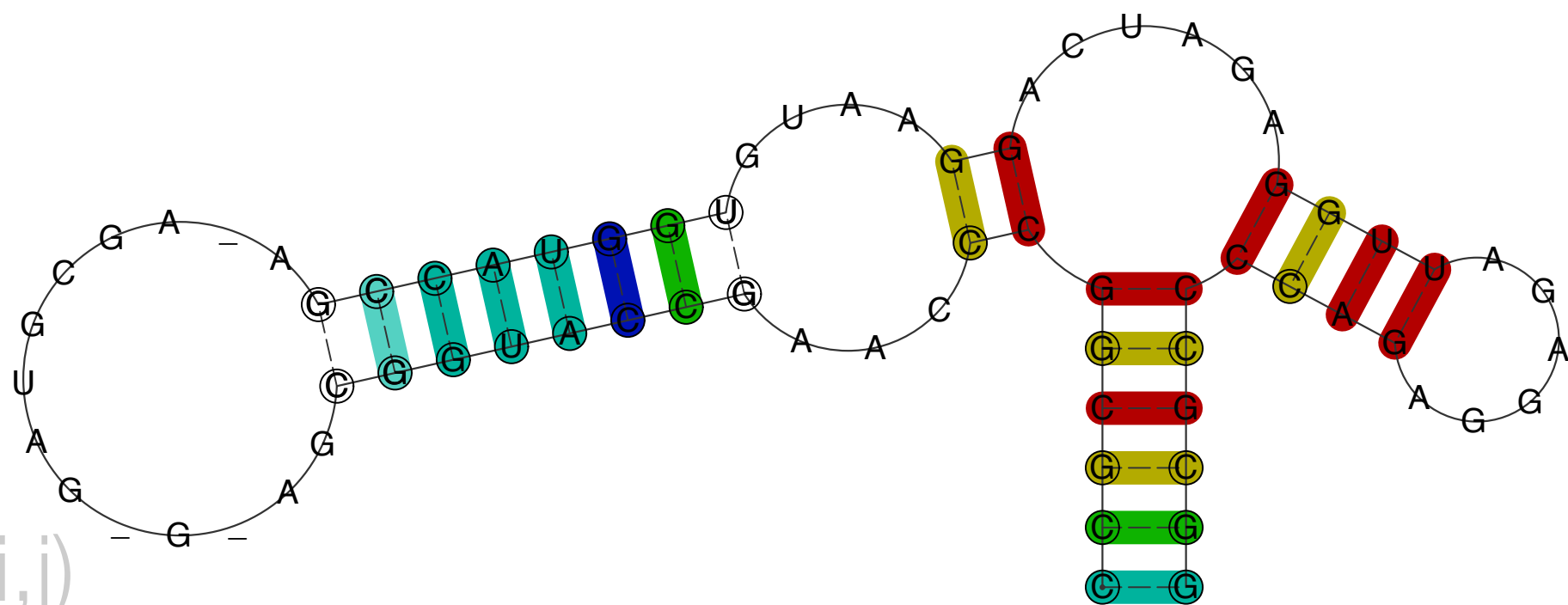
RNA Covariation as Evolutionary Trait

- High mutation rate in RNA viruses due to error-prone RdRP
- For base pair (i,j): GC/CG/AU/UA/GU/UG
- Consistent mutation: different standard combinations
- Compensatory mutation: both positions are mutated
- Presence of both strongly supports predicted base pair (i,j)



RNA Covariation as Evolutionary Trait

- High mutation rate in RNA viruses due to error-prone RdRP
- For base pair (i,j): GC/CG/AU/UA/GU/UG
- Consistent mutation: different standard combinations
- Compensatory mutation: both positions are mutated
- Presence of both strongly supports predicted base pair (i,j)

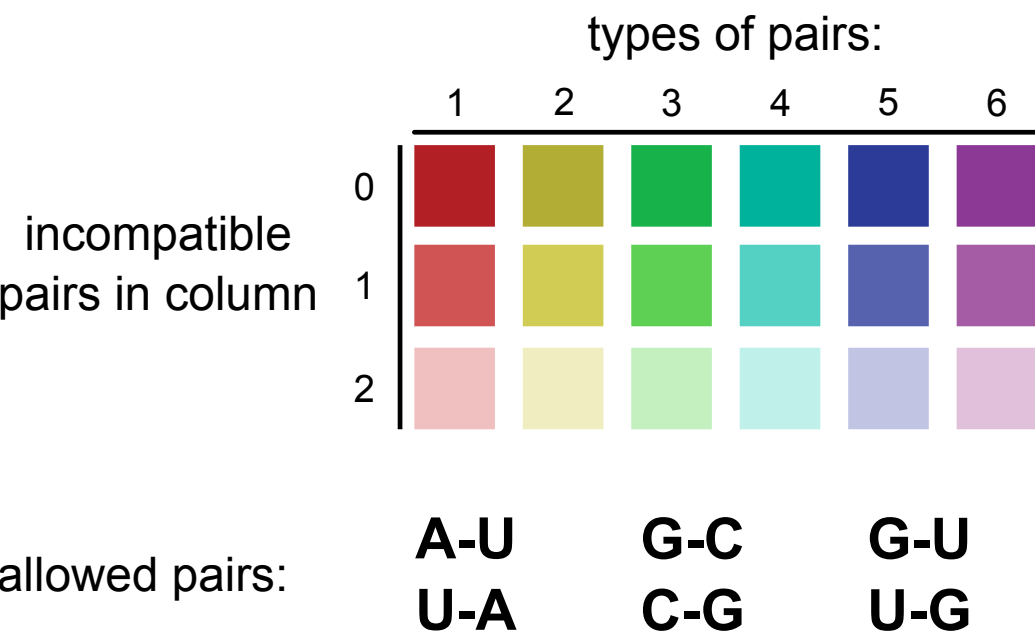
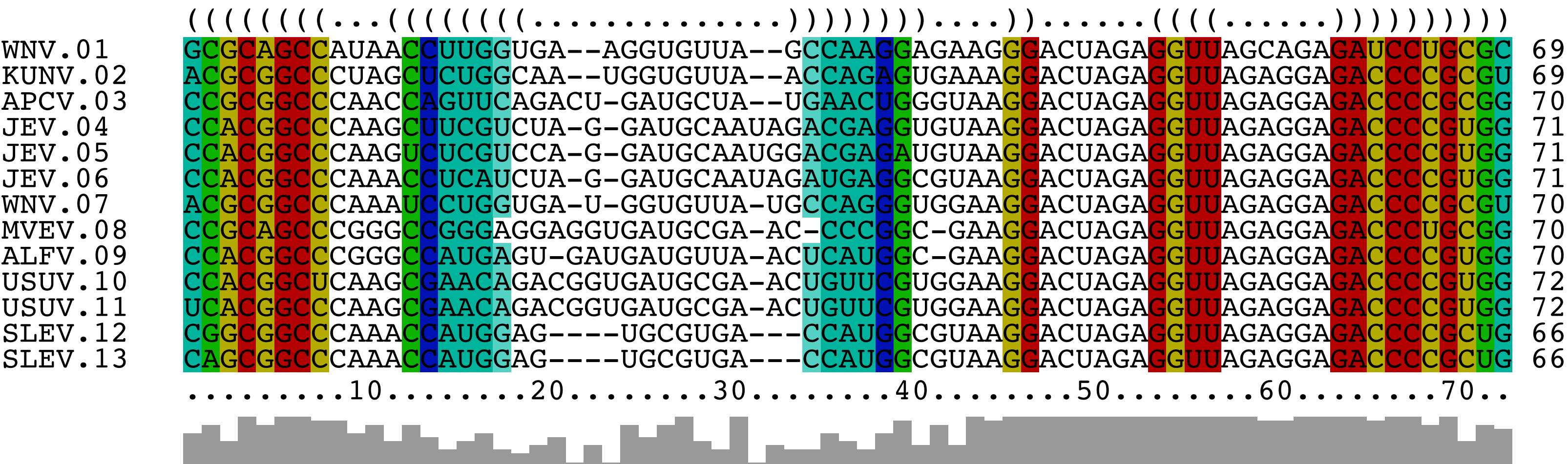
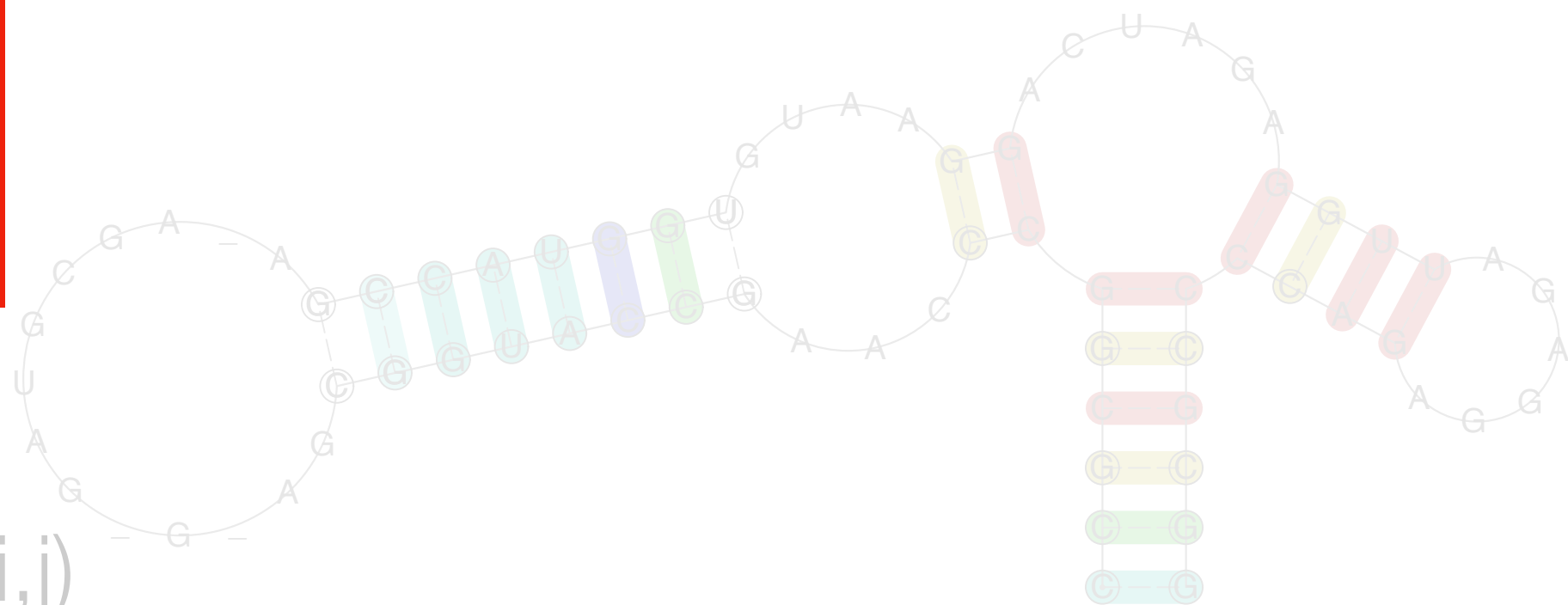


RNA Covariation as Evolutionary Trait

- High mutation rate in RNA viruses due to error-prone RdRP

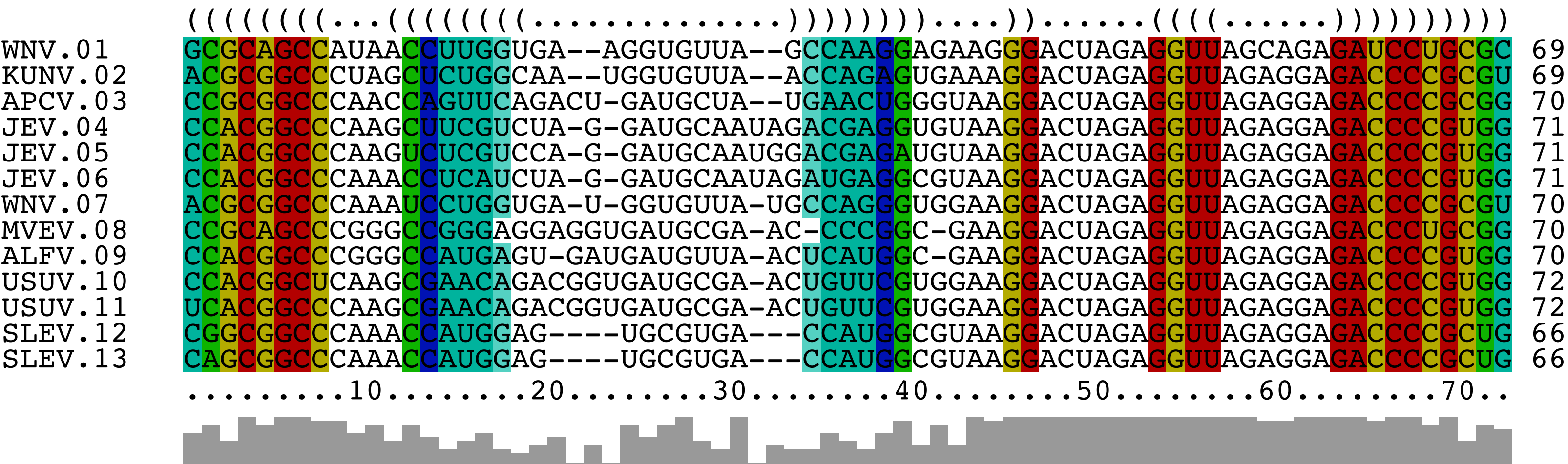
We use structural RNA alignments and Covariance Models to find conserved elements

- Compensatory mutation: both positions are mutated
- Presence of both strongly supports predicted base pair (i,j)



RNA Covariation as Evolutionary Trait

- High mutation rate in RNA viruses due to error-prone RdRP
- **We use structural RNA alignments and Covariance Models to find conserved elements**
- Compensatory mutation: both positions are mutated
- Presence of both strongly supports predicted base pair (i,j)



types of pairs:

	1	2	3	4	5	6
0	Red	Yellow	Green	Teal	Blue	Purple
1	Red	Yellow	Green	Teal	Blue	Purple
2	Pink	Light Yellow	Light Green	Light Teal	Light Blue	Light Purple

incompatible pairs in column

allowed pairs:

	A-U	G-C	G-U
U-A			
C-G			
U-G			

Part II:

Conserved RNA structures in flaviviruses

Flaviviruses as Global Health Threat

- Family *Flaviviridae* / genus *Flavivirus*; arthropod-borne (mainly mosquitoes & ticks)
- FV include (re-)emerging human pathogens like YFV, DENV, JEV, WNV, TBEV
- Clinical manifestations: headache, rash, (hemorrhagic) fever, meningitis, encephalitis
- FV neurotropism reported for YFV, WNV, TBEV, DENV, ZIKV
- ZIKV outbreak in the Americas 2015-2017
- Congenital neurotropism: high increase in microcephaly cases in newborns



Source: Wikipedia

Flaviviruses as Global Health Threat

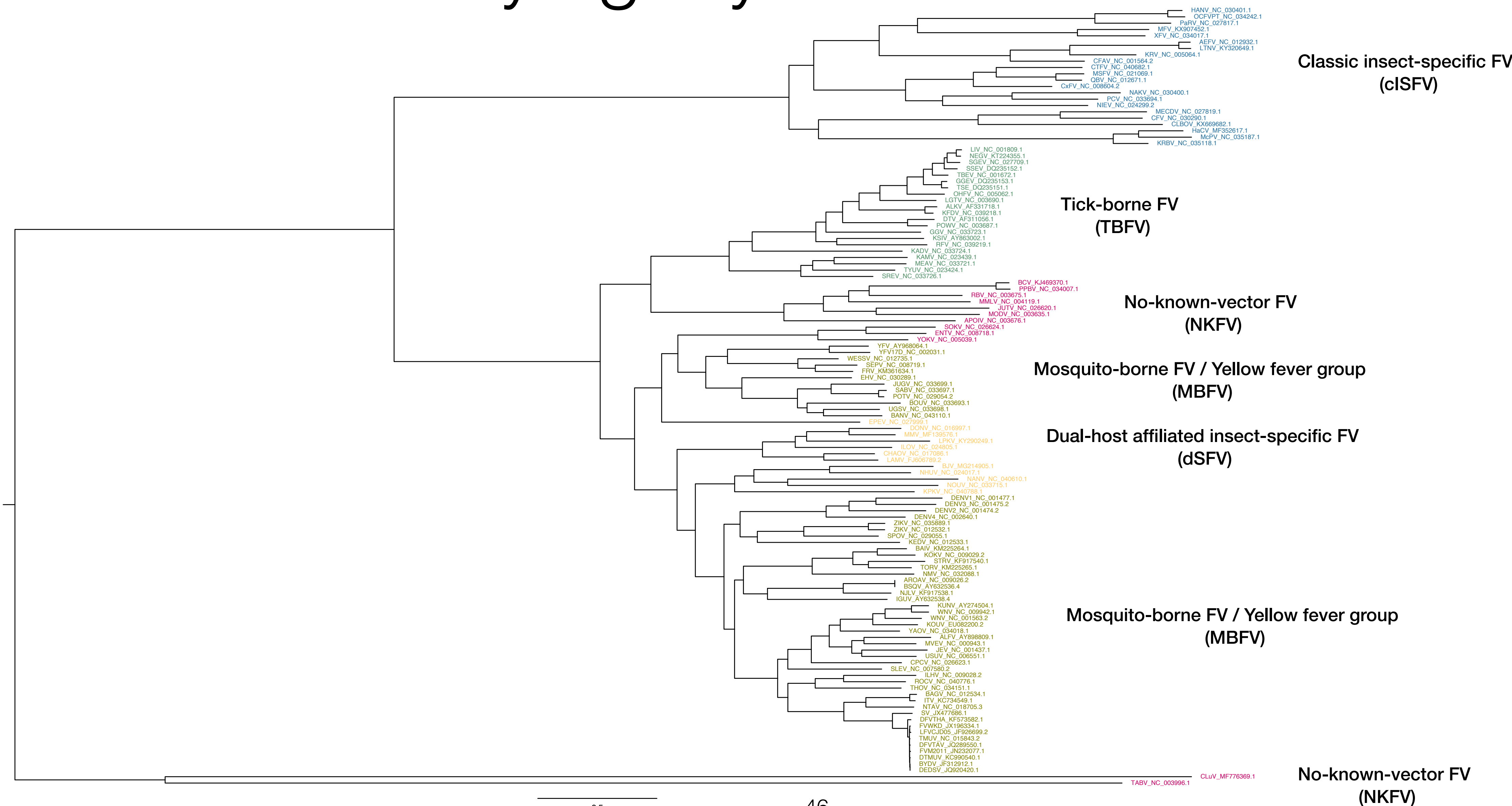
- Family *Flaviviridae* / genus *Flavivirus*; arthropod-borne (mainly mosquitoes & ticks)
- FV include (re-)emerging human pathogens like YFV, DENV, JEV, WNV, TBEV
- Clinical manifestations: headache, rash, (hemorrhagic) fever, meningitis, encephalitis
- FV neurotropism reported for YFV, WNV, TBEV, DENV, ZIKV
- ZIKV outbreak in the Americas 2015-2017
- Congenital neurotropism: high increase in microcephaly cases in newborns



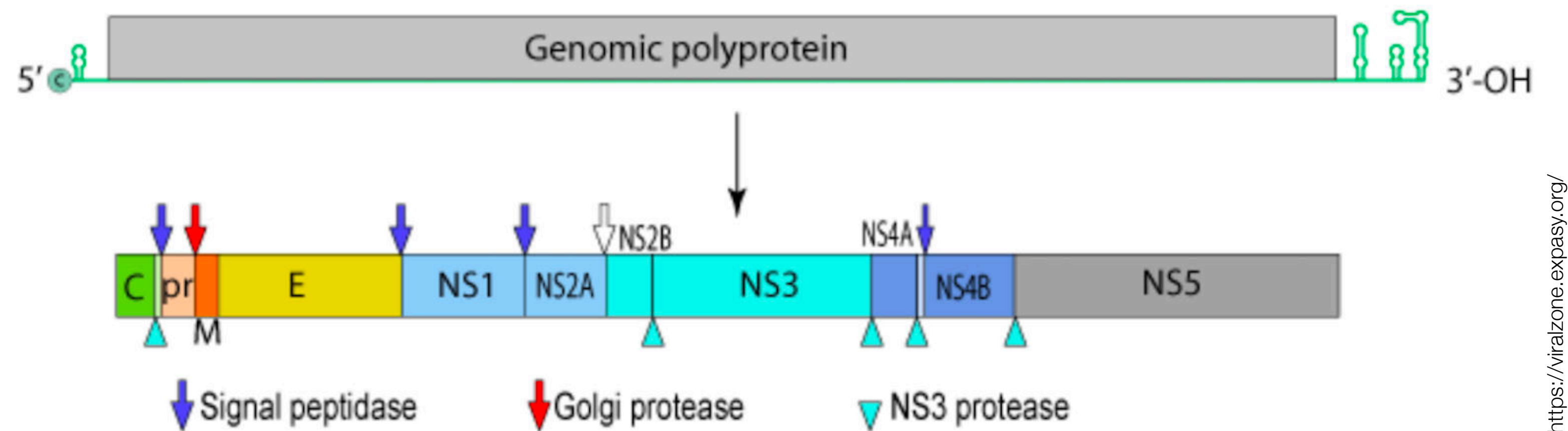
Source: Wikipedia

Vaccine only available for YFV

Flavivirus Phylogeny

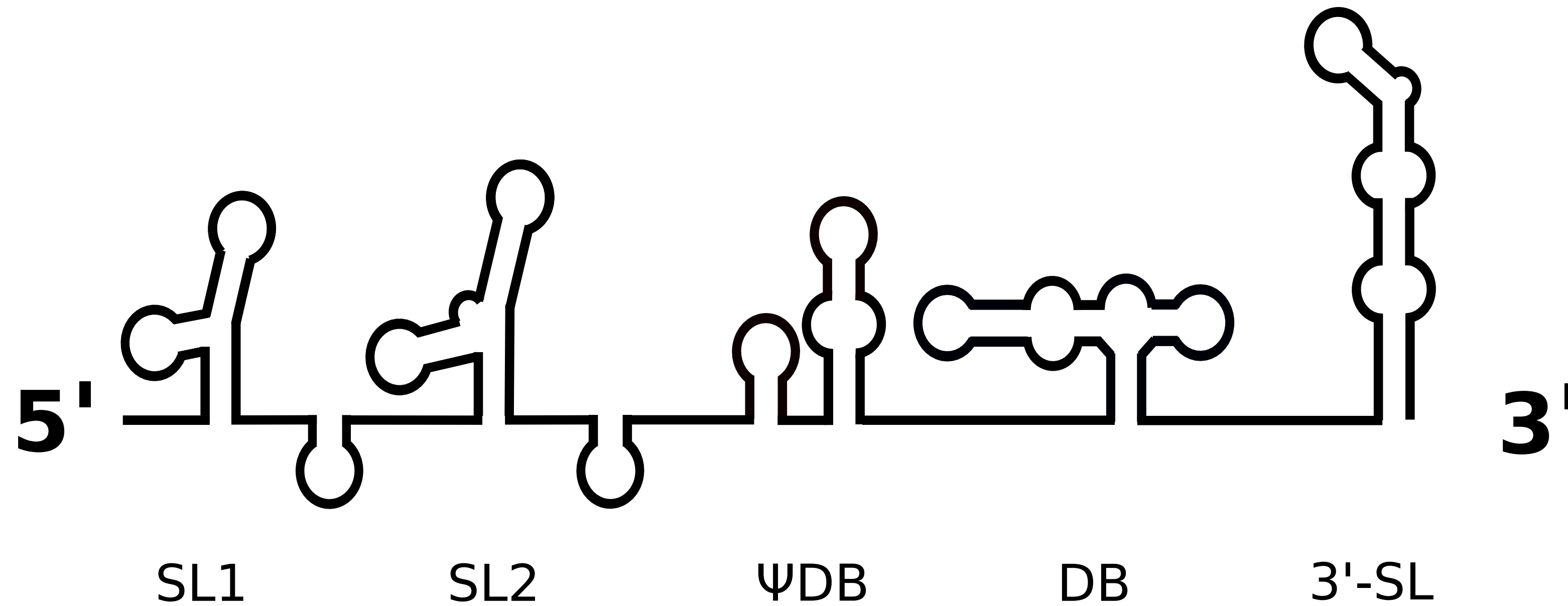


Flavivirus Genome Organization

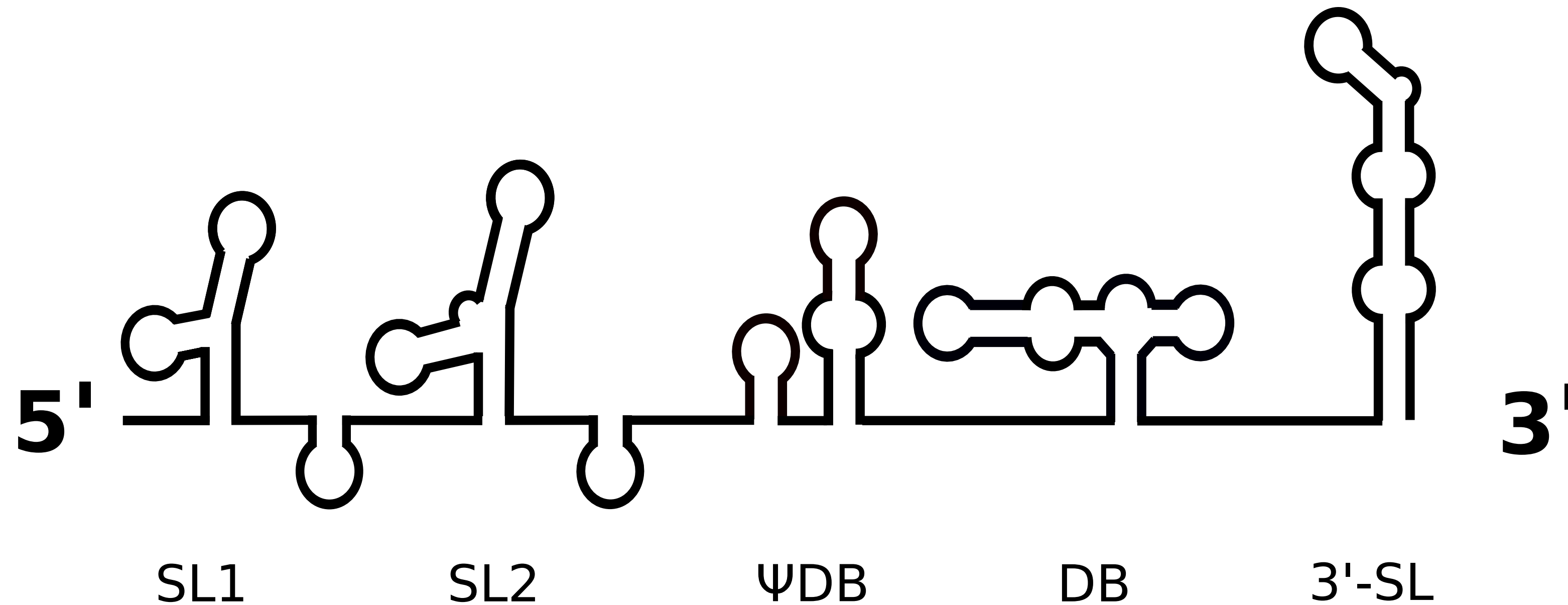


- Non-segmented, single-stranded, (+)-sense RNA genomes of 10-12kB length
- Capped, non-polyadenylated
- Encode a single ORF, flanked by highly structures 5'UTR and 3'UTR

Conserved RNAs in Viral 3'UTR: ZIKV

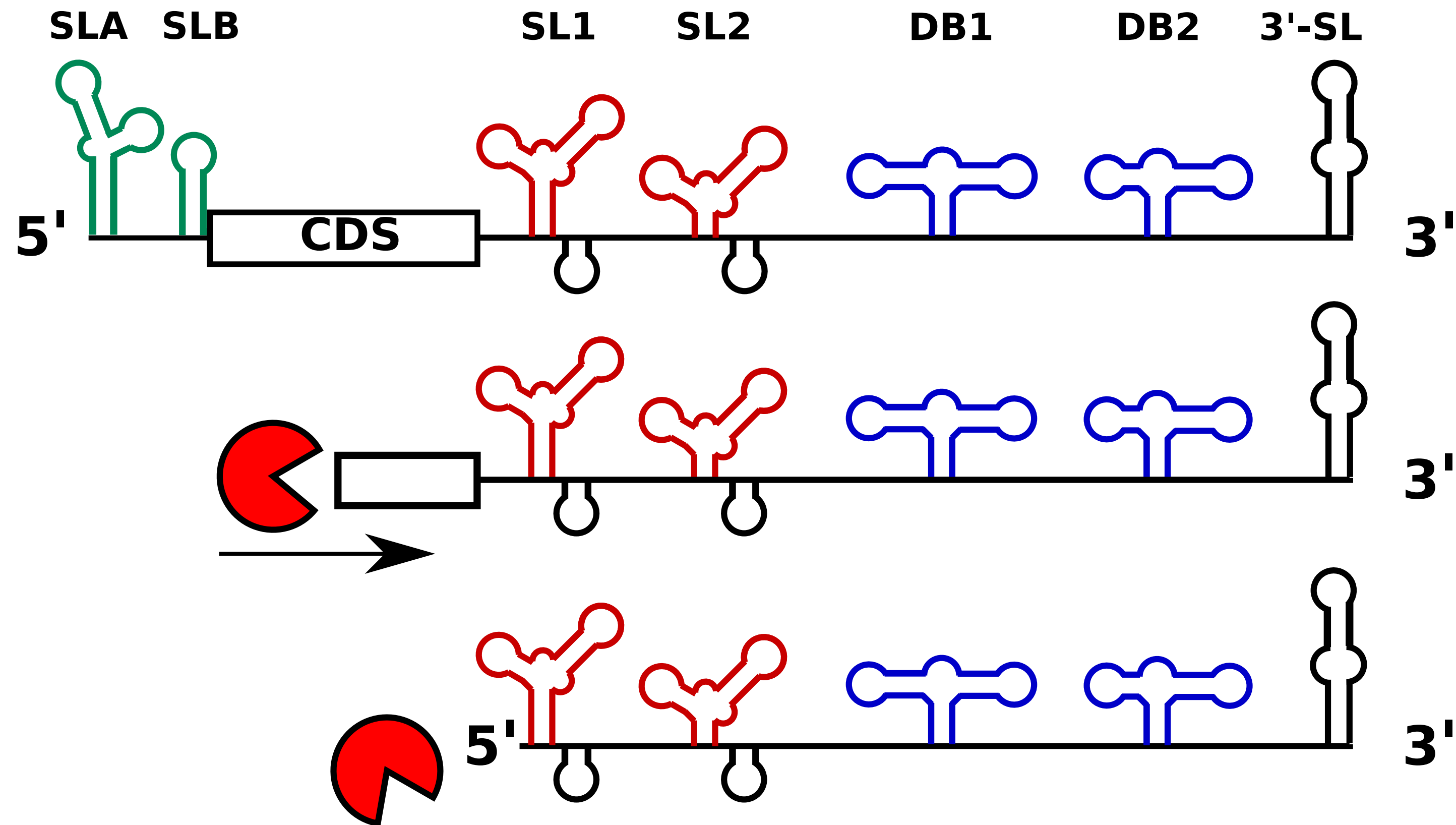


Conserved RNAs in Viral 3'UTR: ZIKV



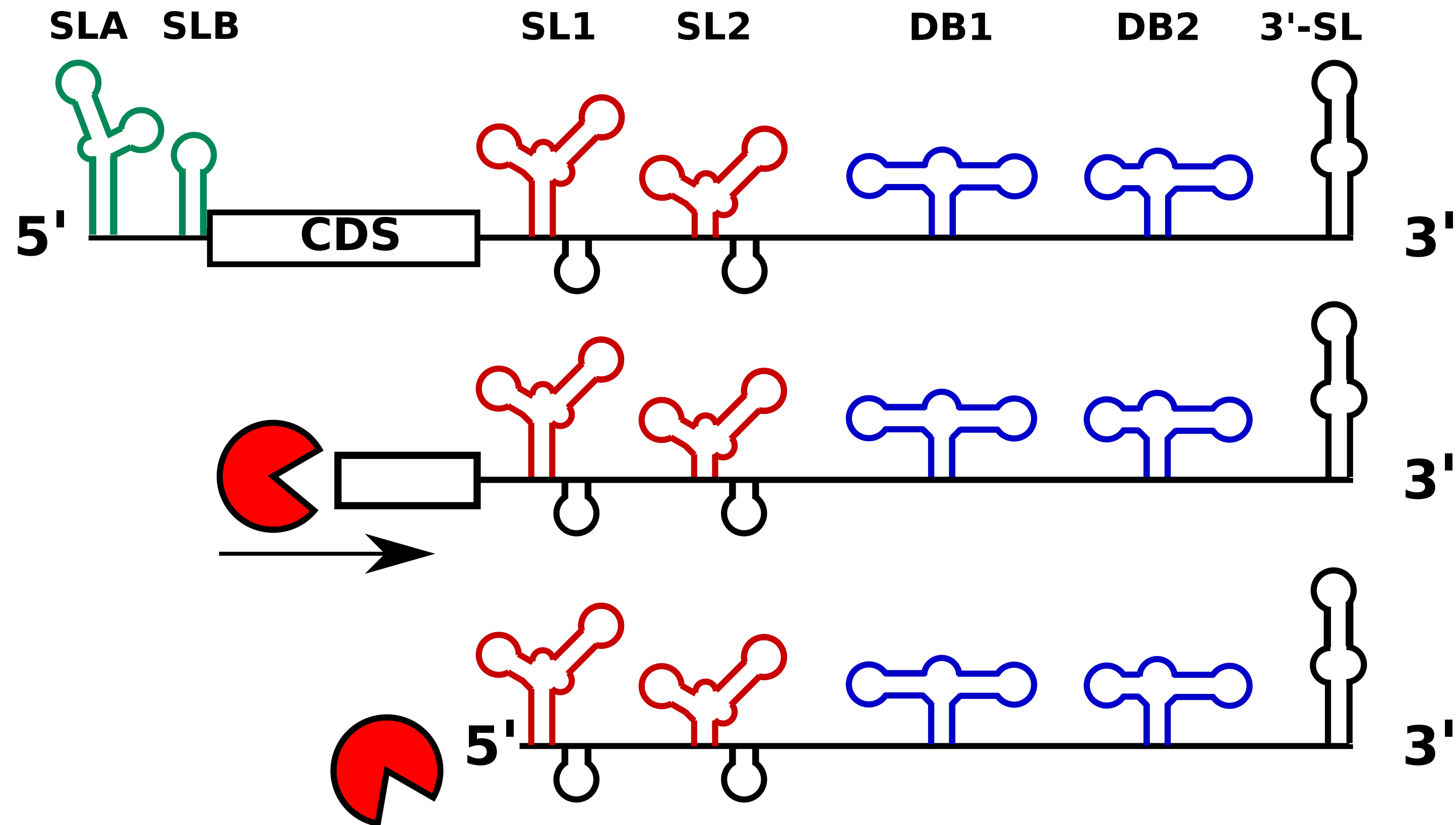
- Accumulation of short flavivirus RNA (sfRNA) upon infection
- Stable decay intermediates produced by partial exoribonuclease degradation
- Xrn1 is efficiently stalled at conserved xrRNA structures

Conserved RNAs in Viral 3'UTR



- Accumulation of short flavivirus RNA (sfRNA) upon infection
- Stable decay intermediates produced by partial exoribonuclease degradation
- Xrn1 is efficiently stalled at conserved xrRNA structures

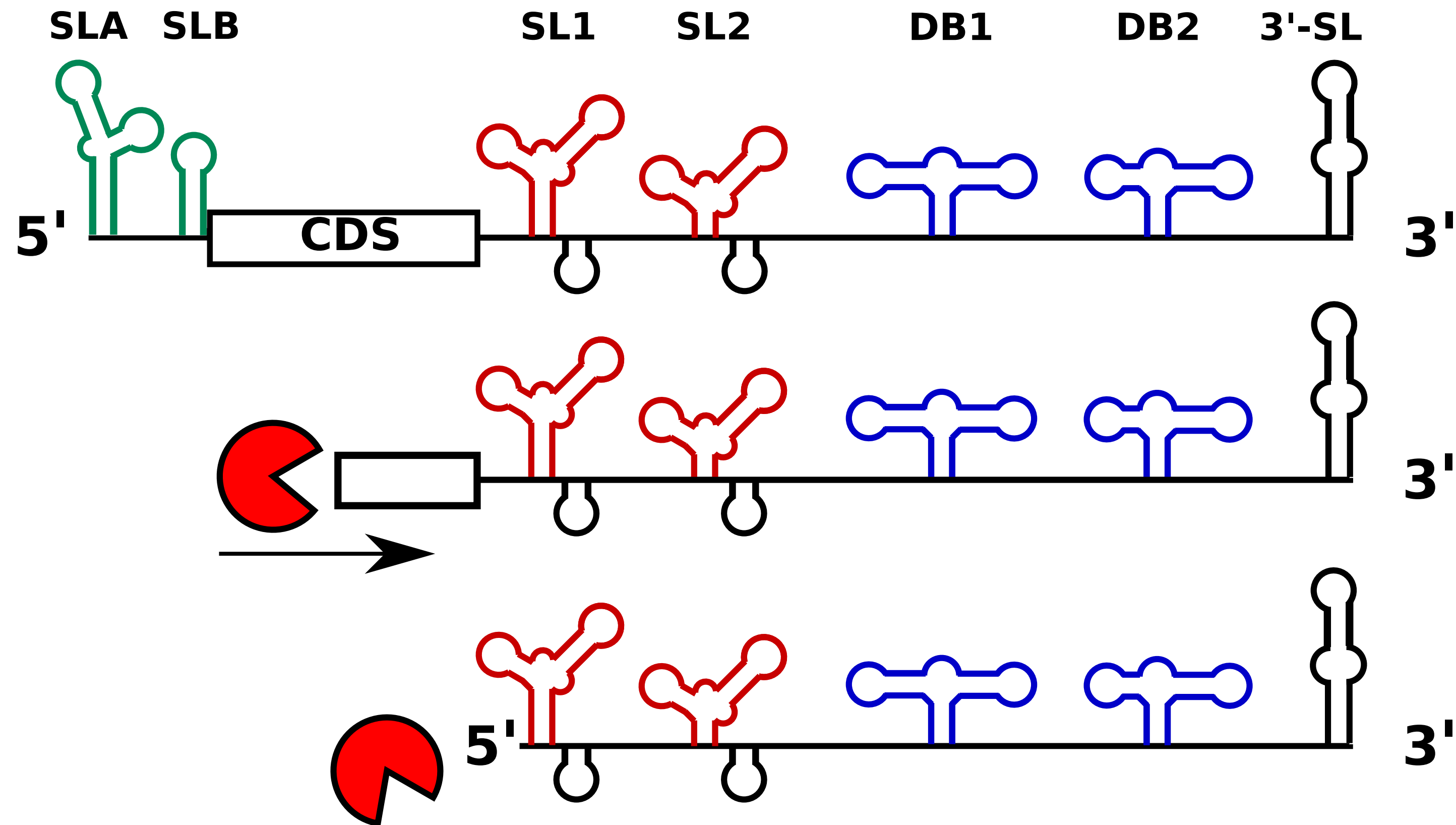
Conserved RNAs in Viral 3'UTR



**sfRNA mediates
pathogenicity !**

- Accumulation of short flavivirus RNA (sfRNA) upon infection
- Stable decay intermediates produced by partial exoribonuclease degradation
- Xrn1 is efficiently stalled at conserved xrRNA structures

Conserved RNAs in Viral 3'UTR

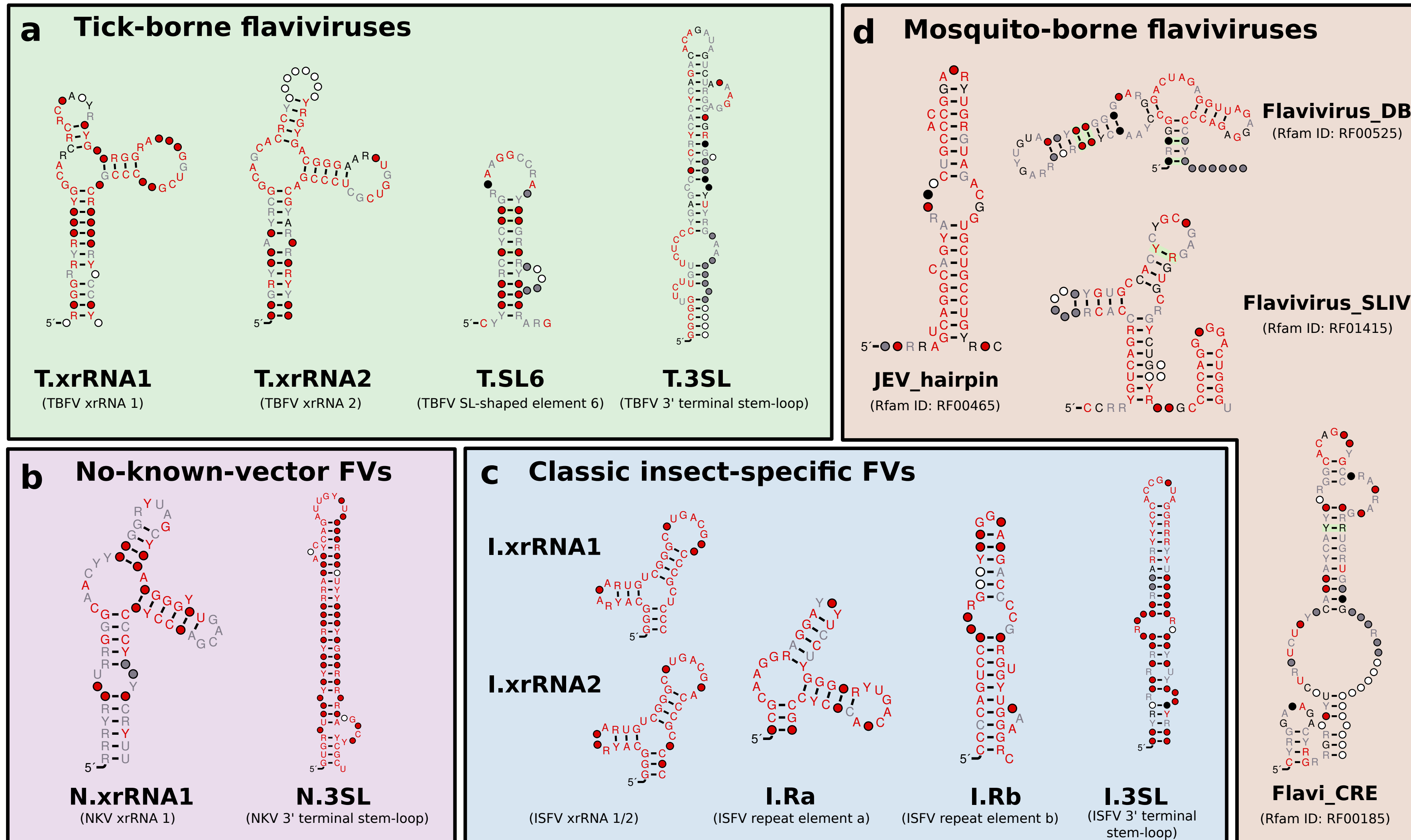


**sfRNA mediates
pathogenicity !**

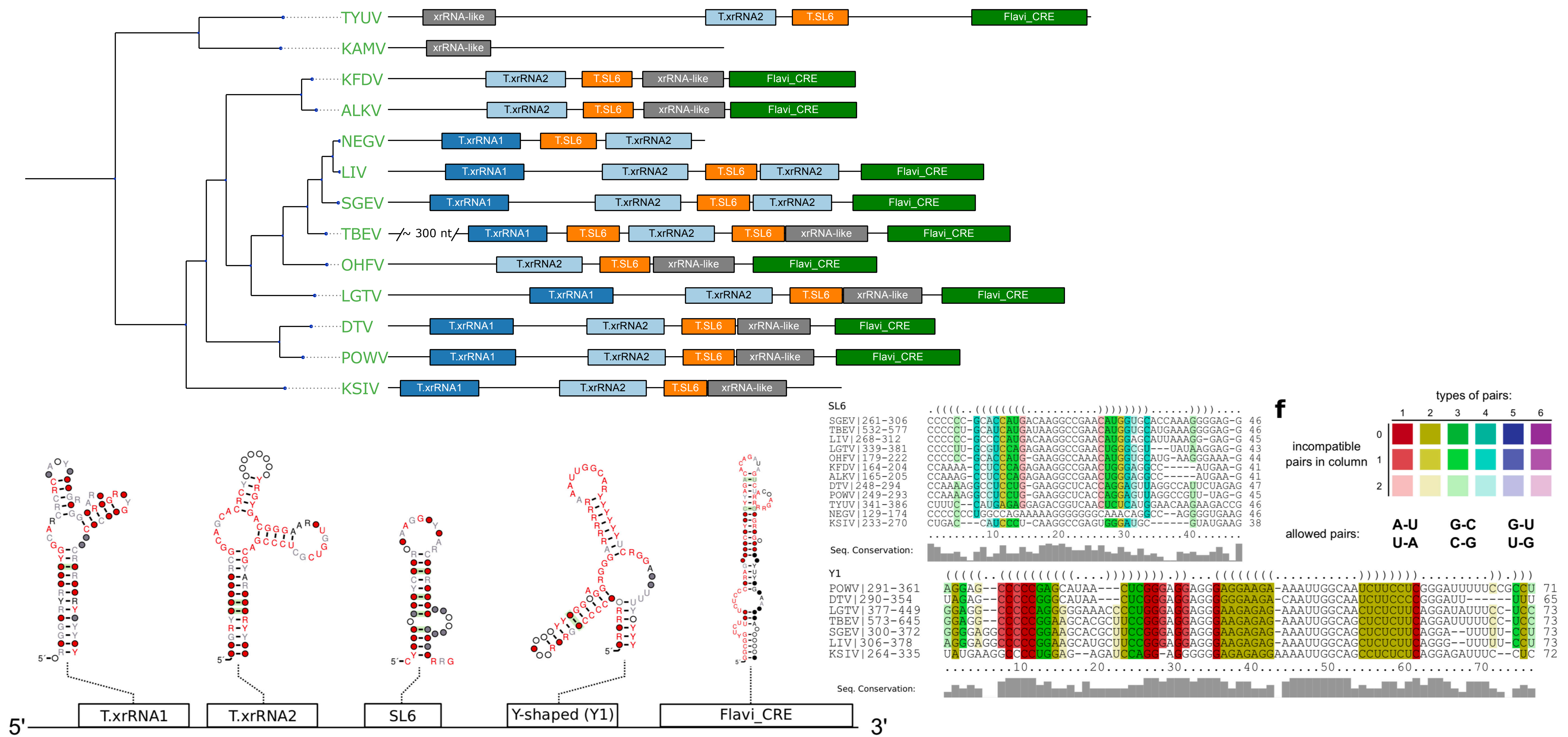
How to find them?

- Accumulation of short flavivirus RNA (sfRNA) upon infection
- Stable decay intermediates produced by partial exoribonuclease degradation
- Xrn1 is efficiently stalled at conserved xrRNA structures

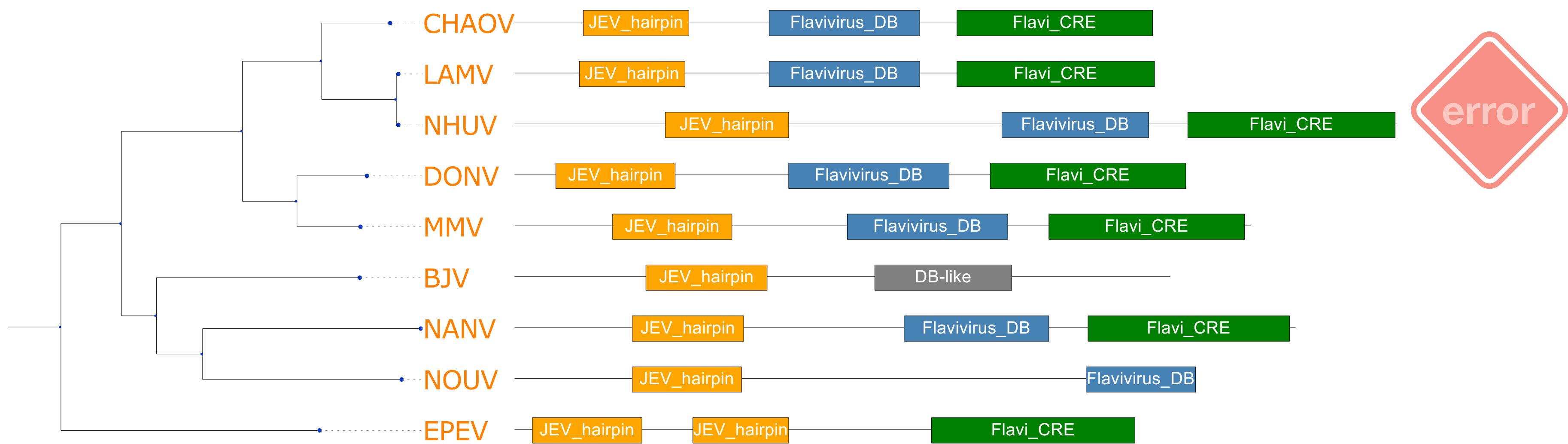
Consensus Structures & Covariance Models



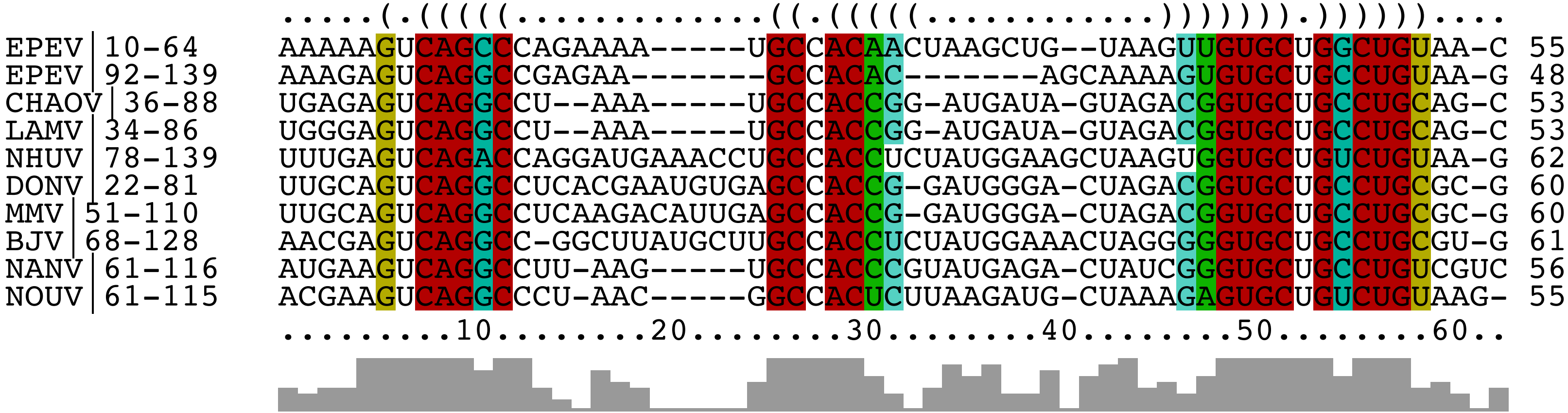
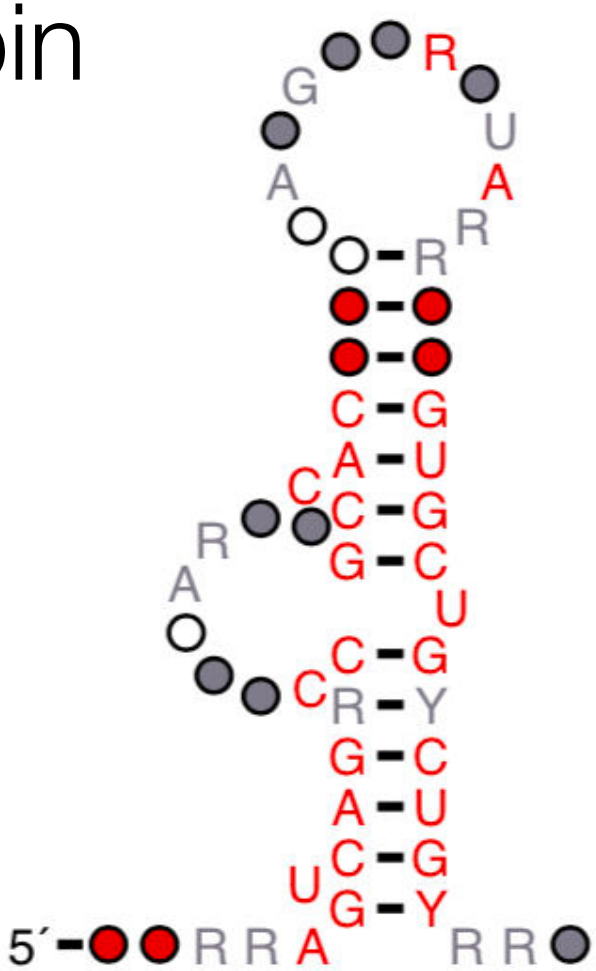
Tick-Borne Flaviviruses



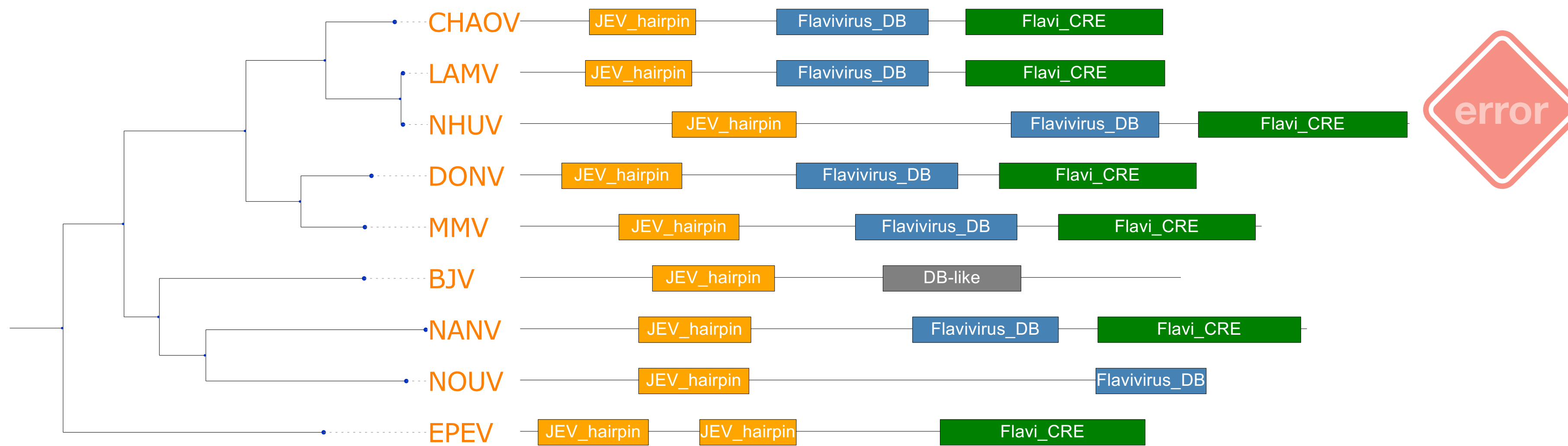
Dual-Host Affiliated Insect-Specific FVs



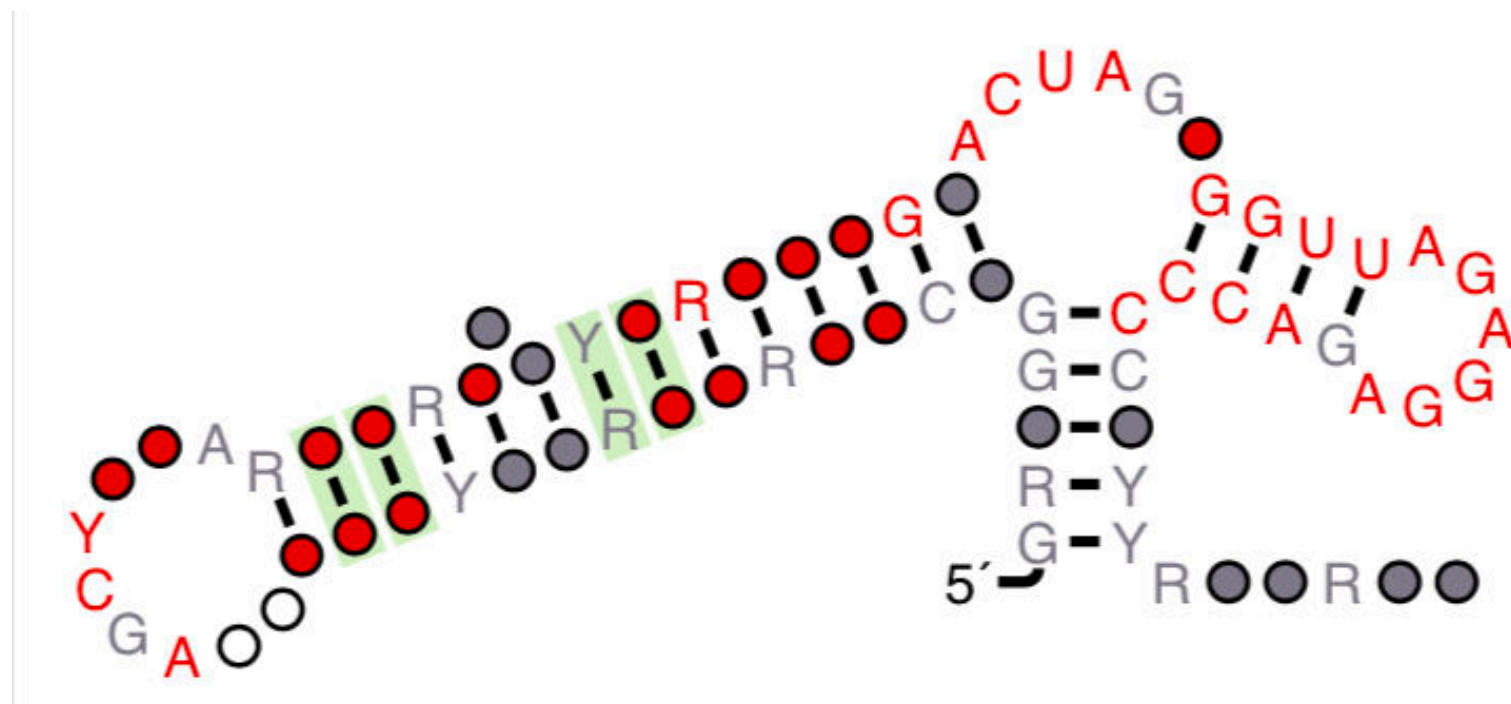
JEV_hairpin



Dual-Host Affiliated Insect-Specific FVs



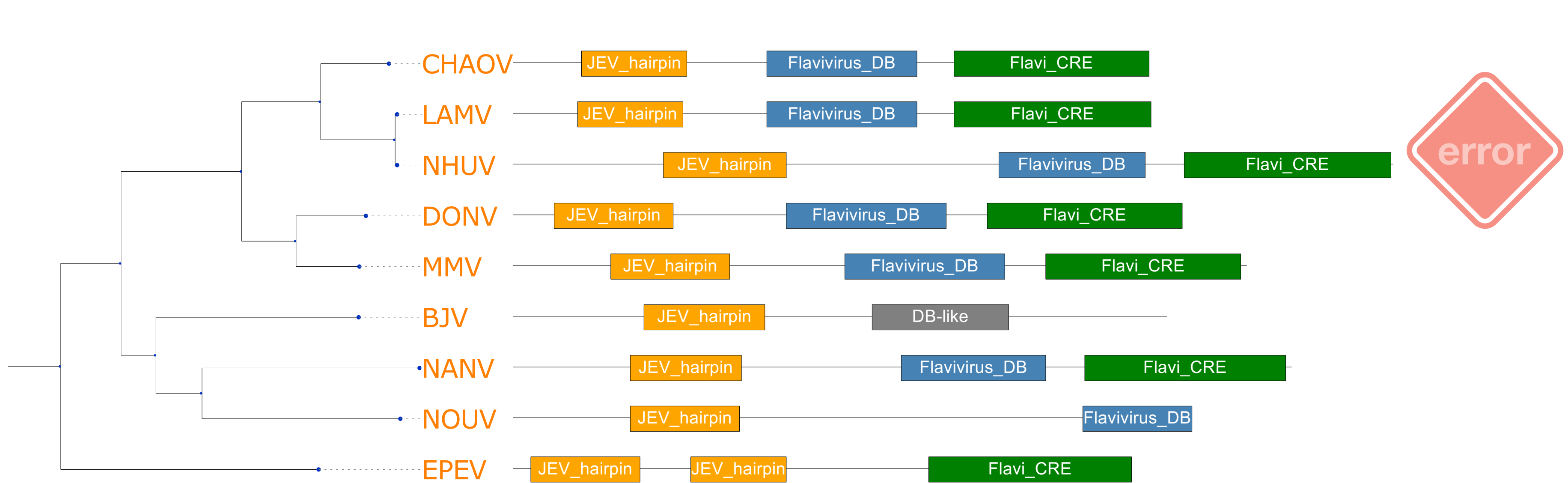
Flavivirus_DB



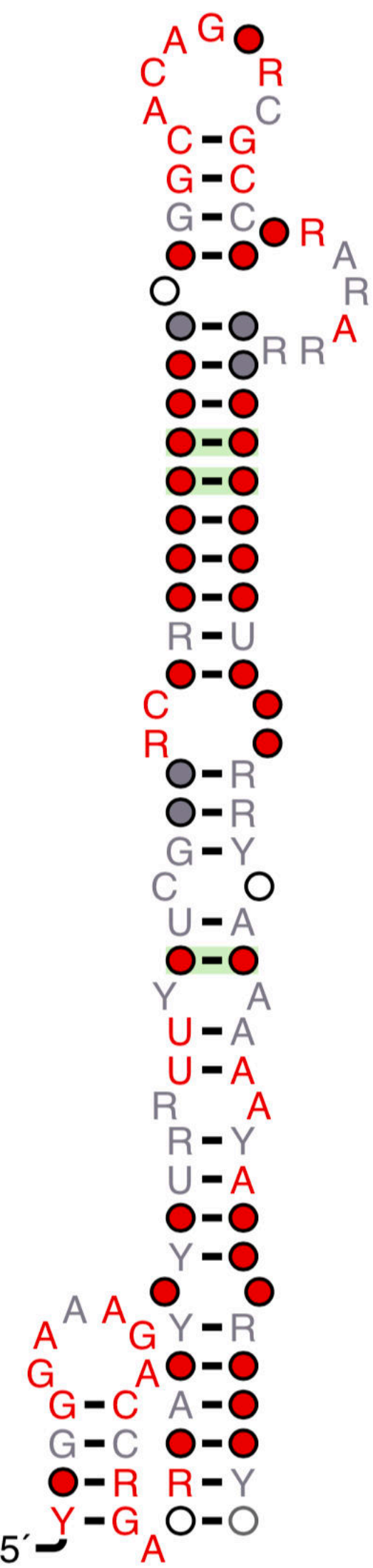
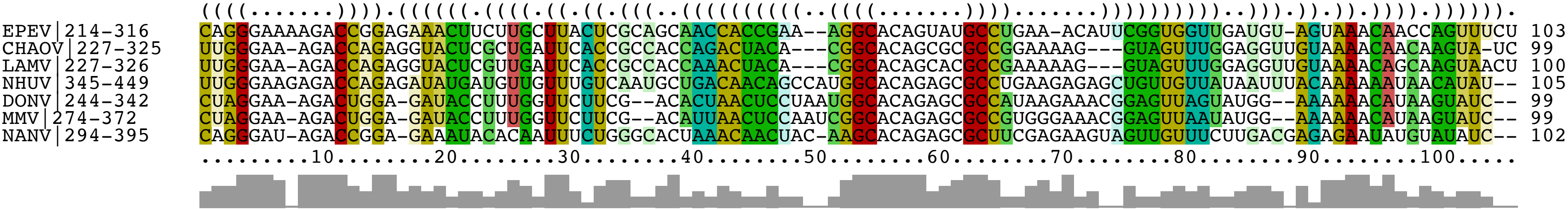
CHAOV	131-206
LAMV	131-206
NHUV	250-323
DONV	141-221
MMV	171-251
NANV	200-272
NOUV	293-347

[illegible]

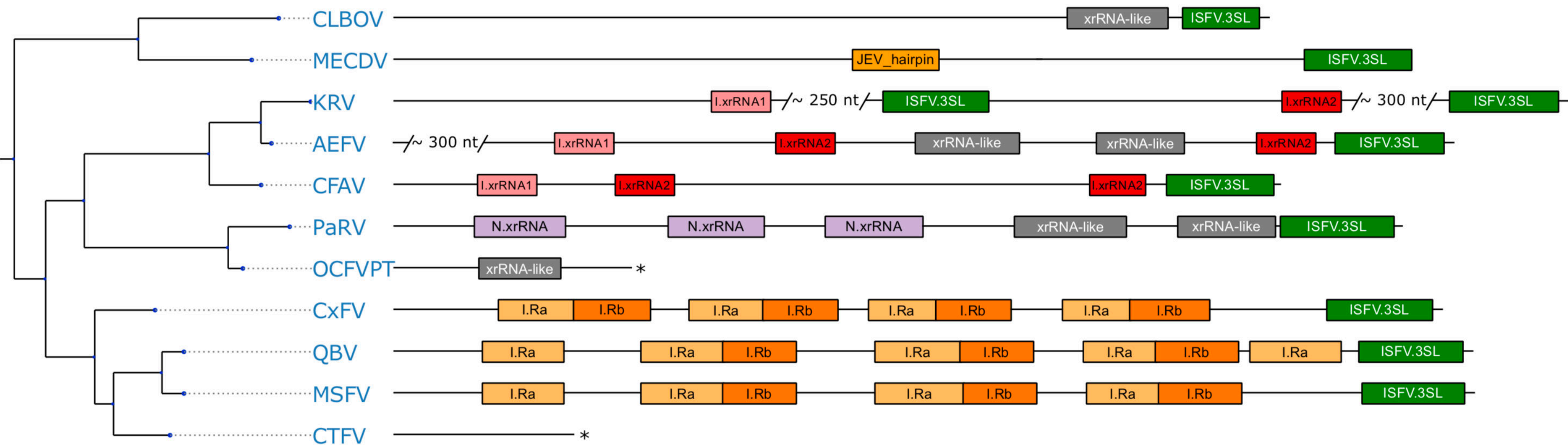
Dual-Host Affiliated Insect-Specific FVs



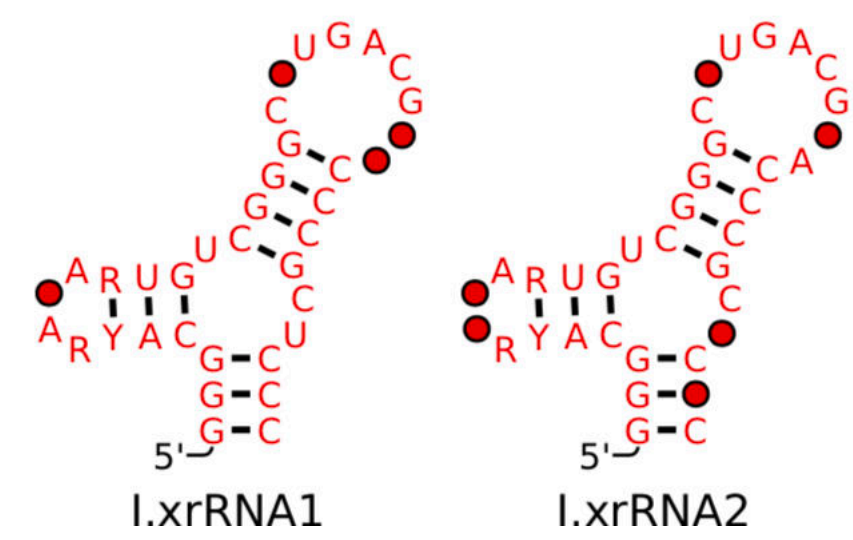
Flavi_CRE



Classic Insect-Specific FVs



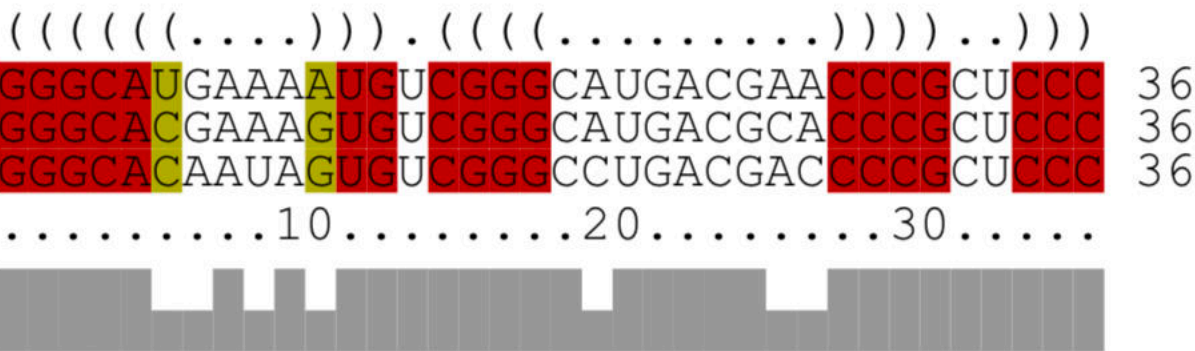
KRV / AEFV / CFAV



I.xrRNA1

CFAV| 53-88
KRV| 199-234
AEFV| 382-417

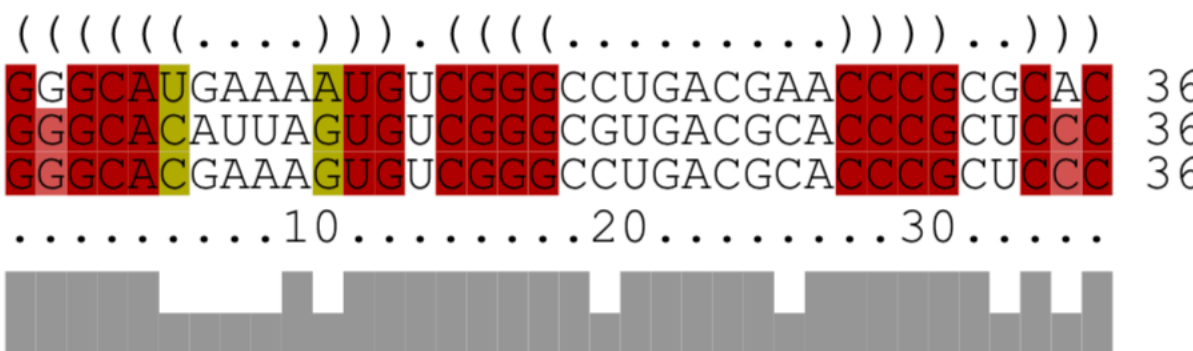
Seq. conservation



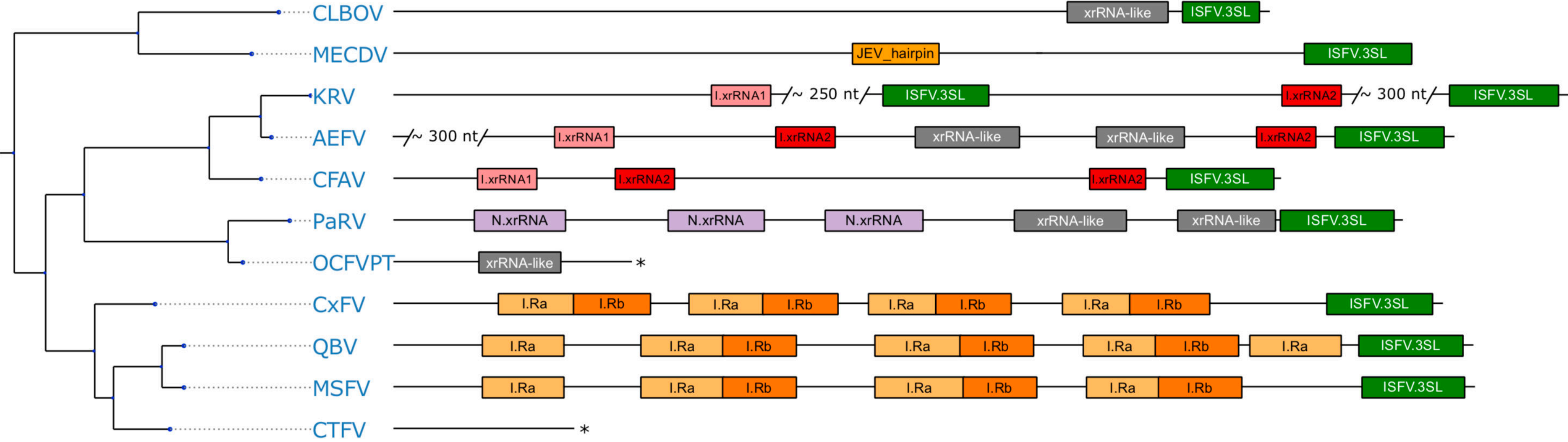
I.xrRNA2

AEFV| 520-555
CFAV| 139-174
KRV| 795-830

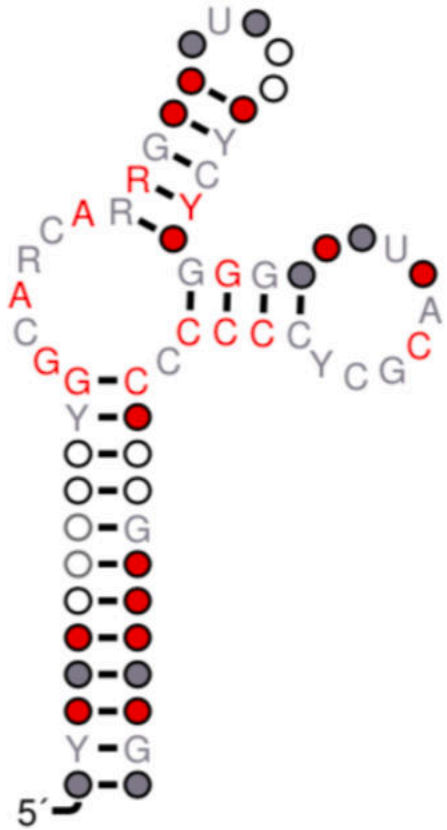
Seq. conservation



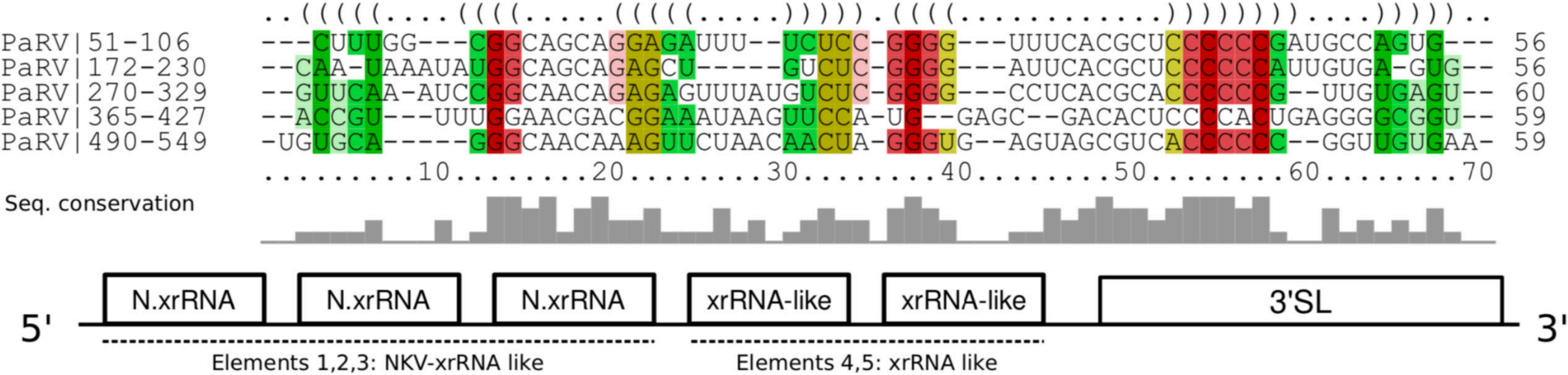
Classic Insect-Specific FVs



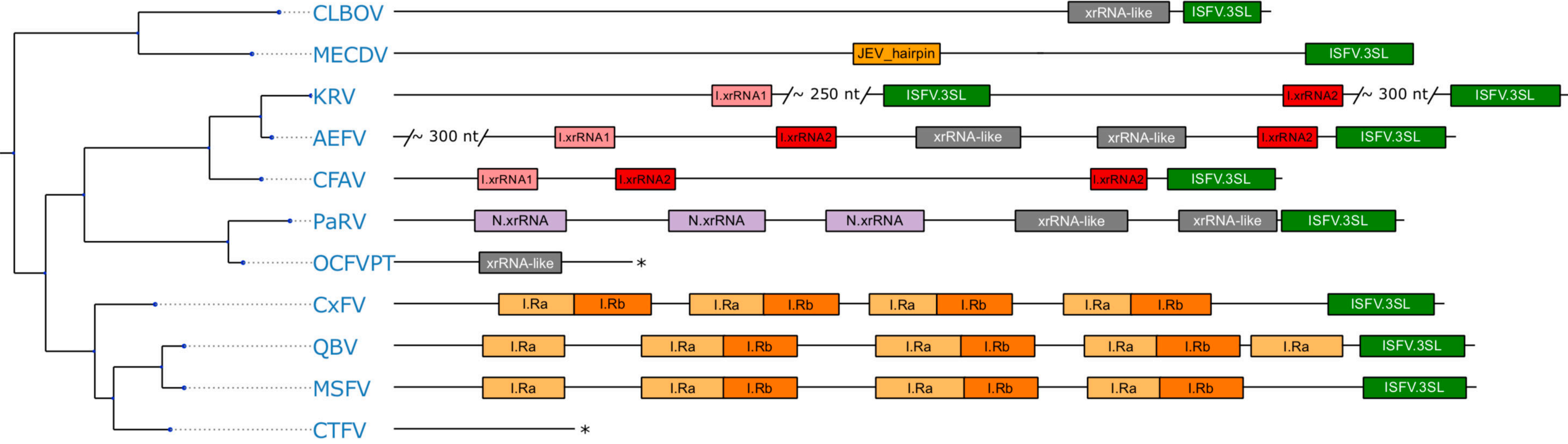
PaRV



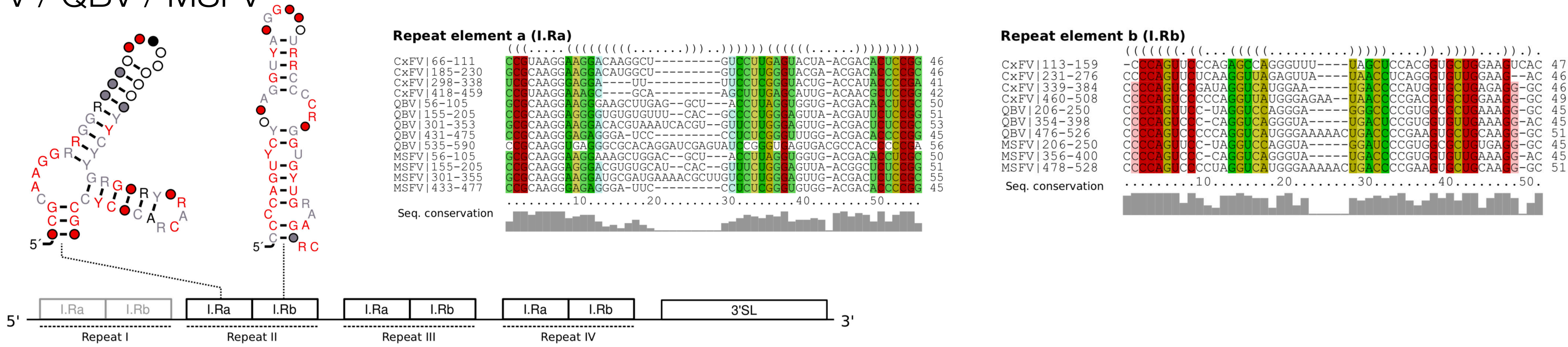
PaRV xrRNA-like element



Classic Insect-Specific FVs



CxFV / QBV / MSFV

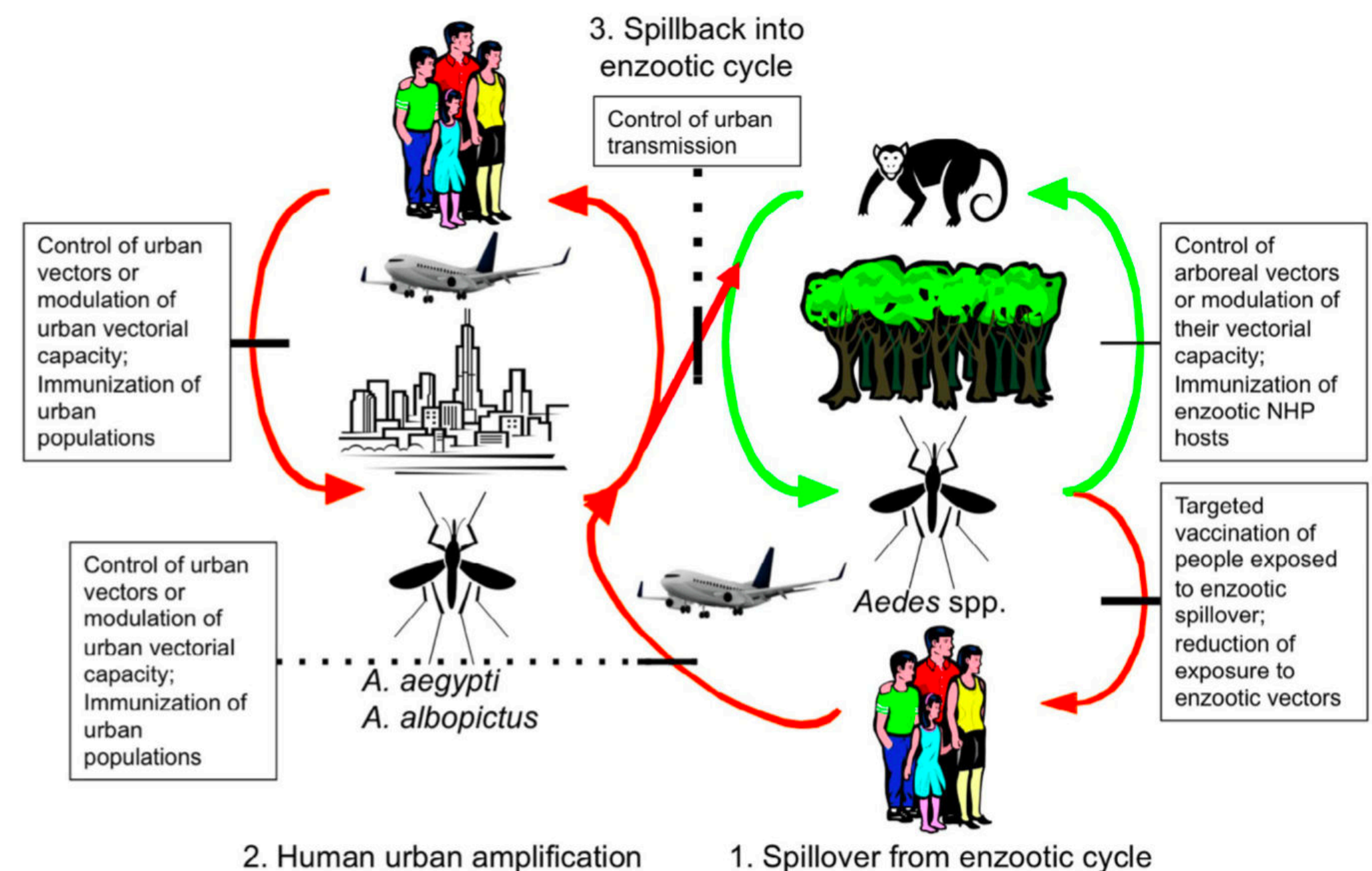


Part II:

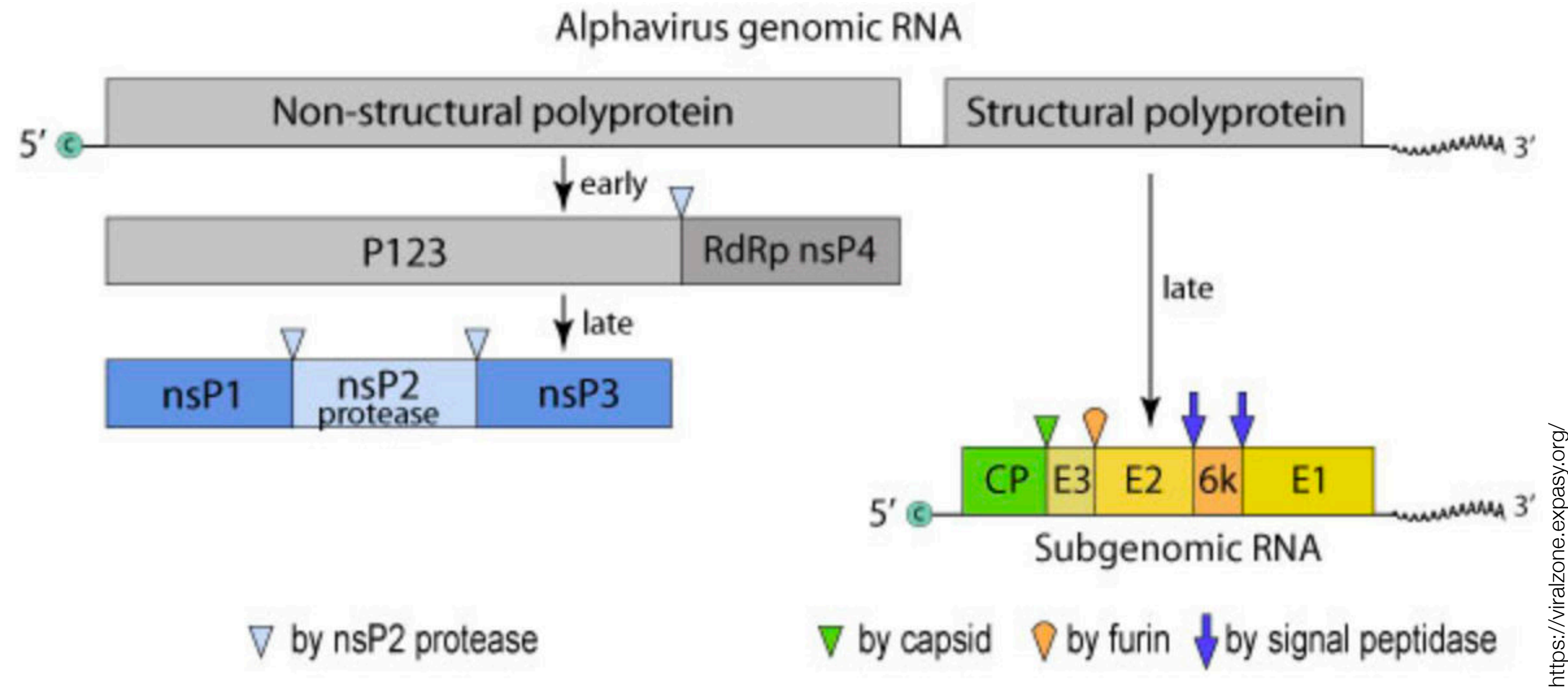
Conserved RNA structures in alphaviruses

Chikungunya Virus (CHIKV)

- Family *Togaviridae* / genus *Alphavirus*; mosquito-borne (*Aedes* spp.)
- Single-stranded (+) sense RNA virus
- Chikungunya fever: febrile illness, arthralgia, rash, rarely causes hemorrhagic complications
- Enzootic in tropical and subtropical regions of Africa
- First outbreak described 1952 in Tanzania
- No vaccine available

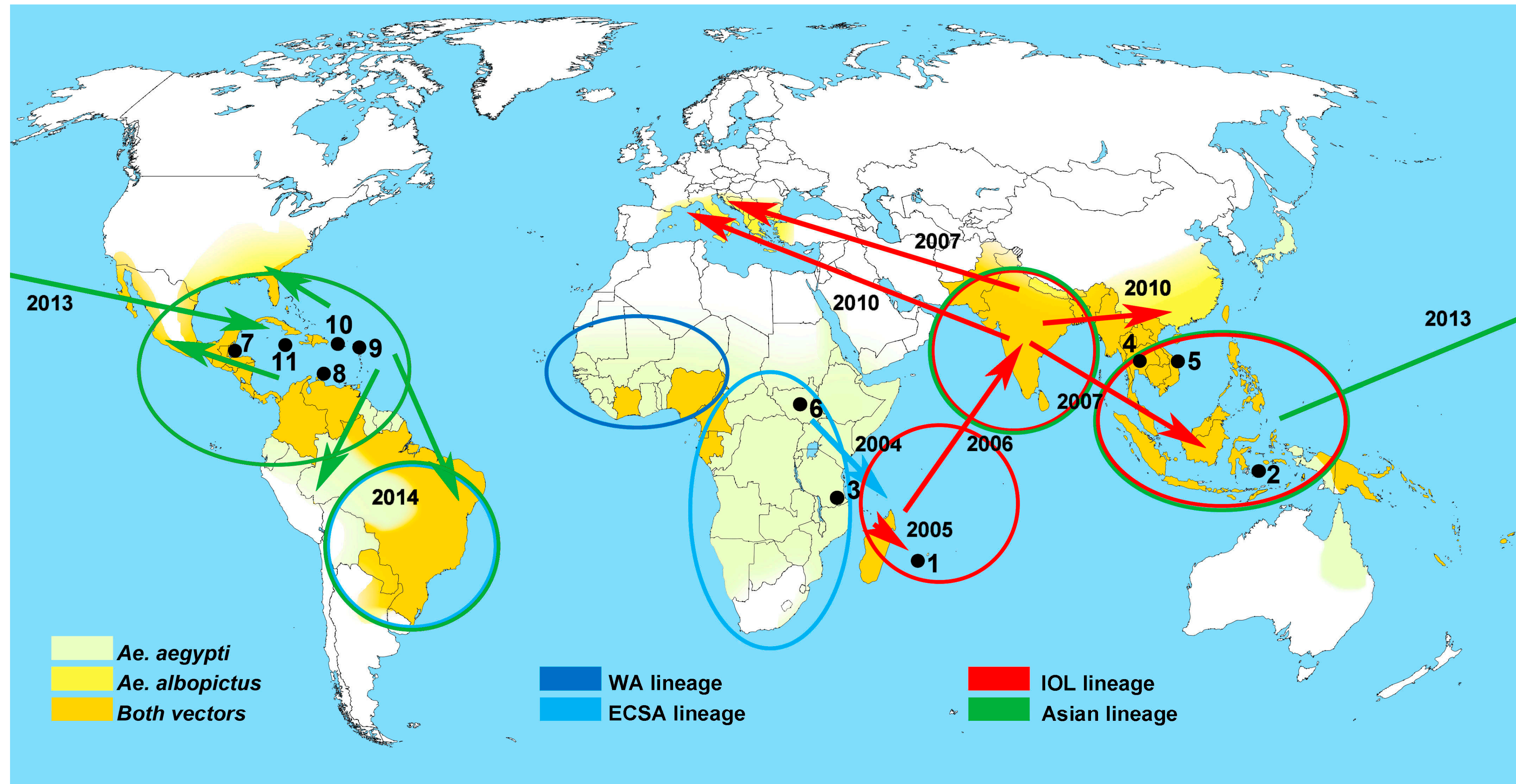


Alphavirus Genome Organization

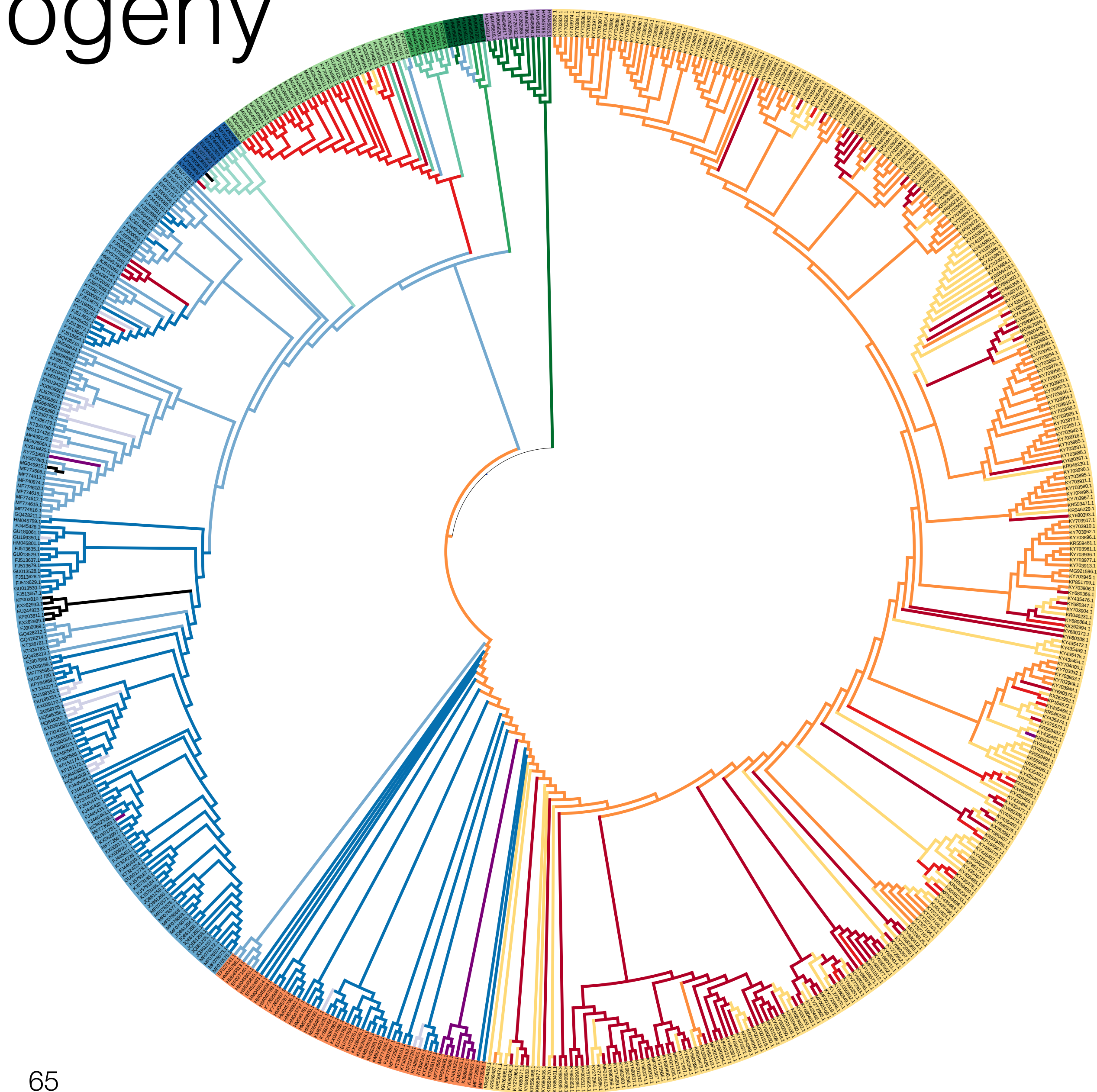


- Non-segmented, single-stranded, (+)-sense RNA genomes of 11-12kB length
- Capped and polyadenylated
- AV genomes appear to host cells as mRNA for immediate translation upon entry into the cytoplasm

CHIKV Epidemic Spread



Updated CHIKV Phylogeny



- 590 CHIKV genomes
- iq-tree / SH-aLRT

CHIKV 3'UTR



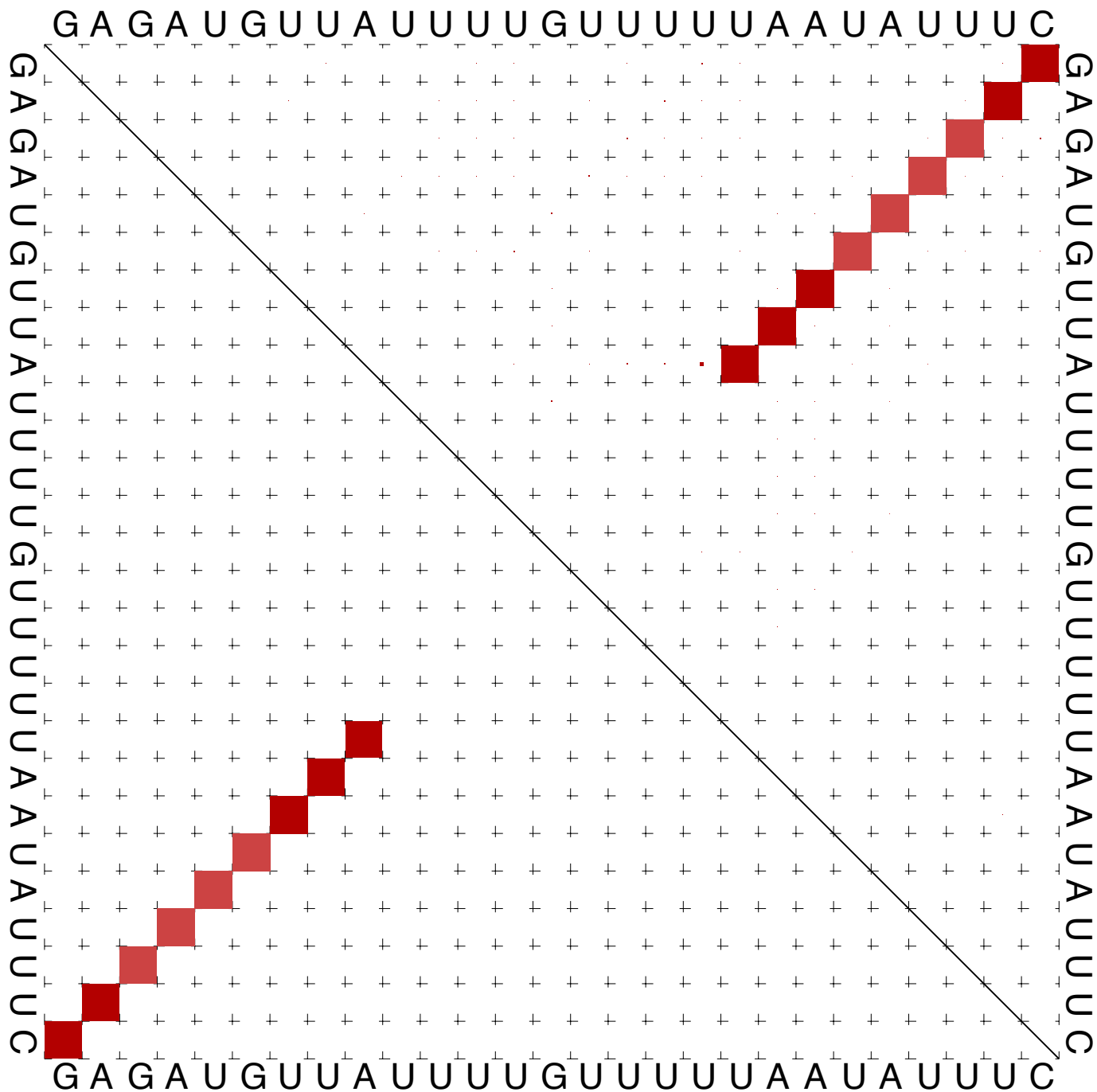
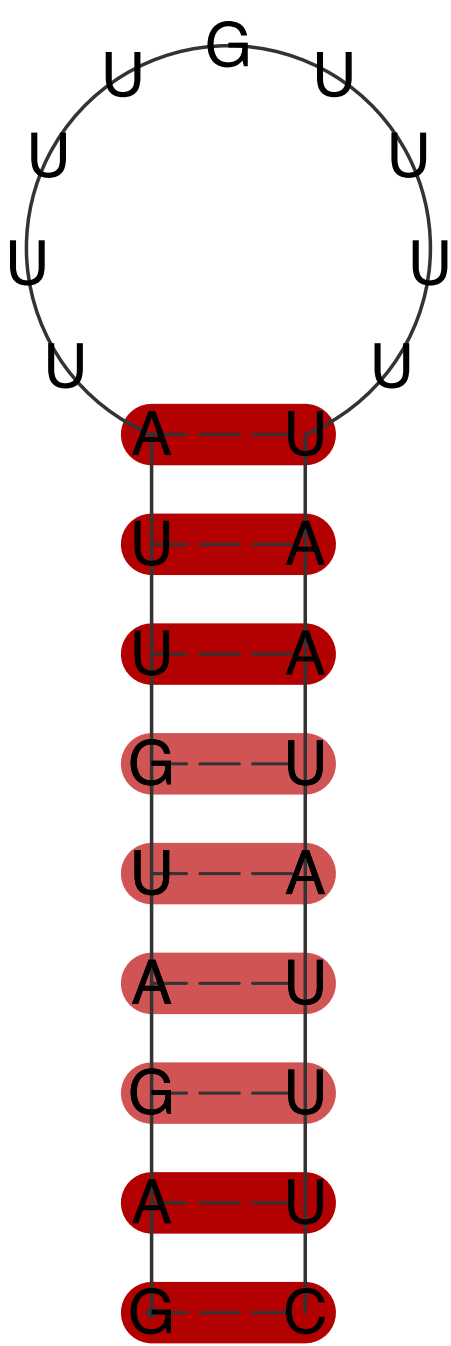
- 110 full length CHIKV 3'UTRs
- Variable length (510nt — 930nt)
- Structural alignments + Covariance models
- Thermodynamic modelling based on ViennaRNA Package

CHIKV 3'UTR: Conserved RNA Structures

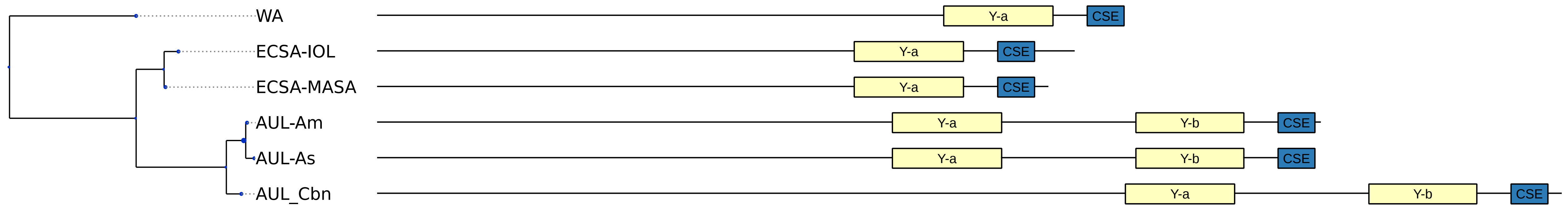


Conserved Sequence Element (CSE)

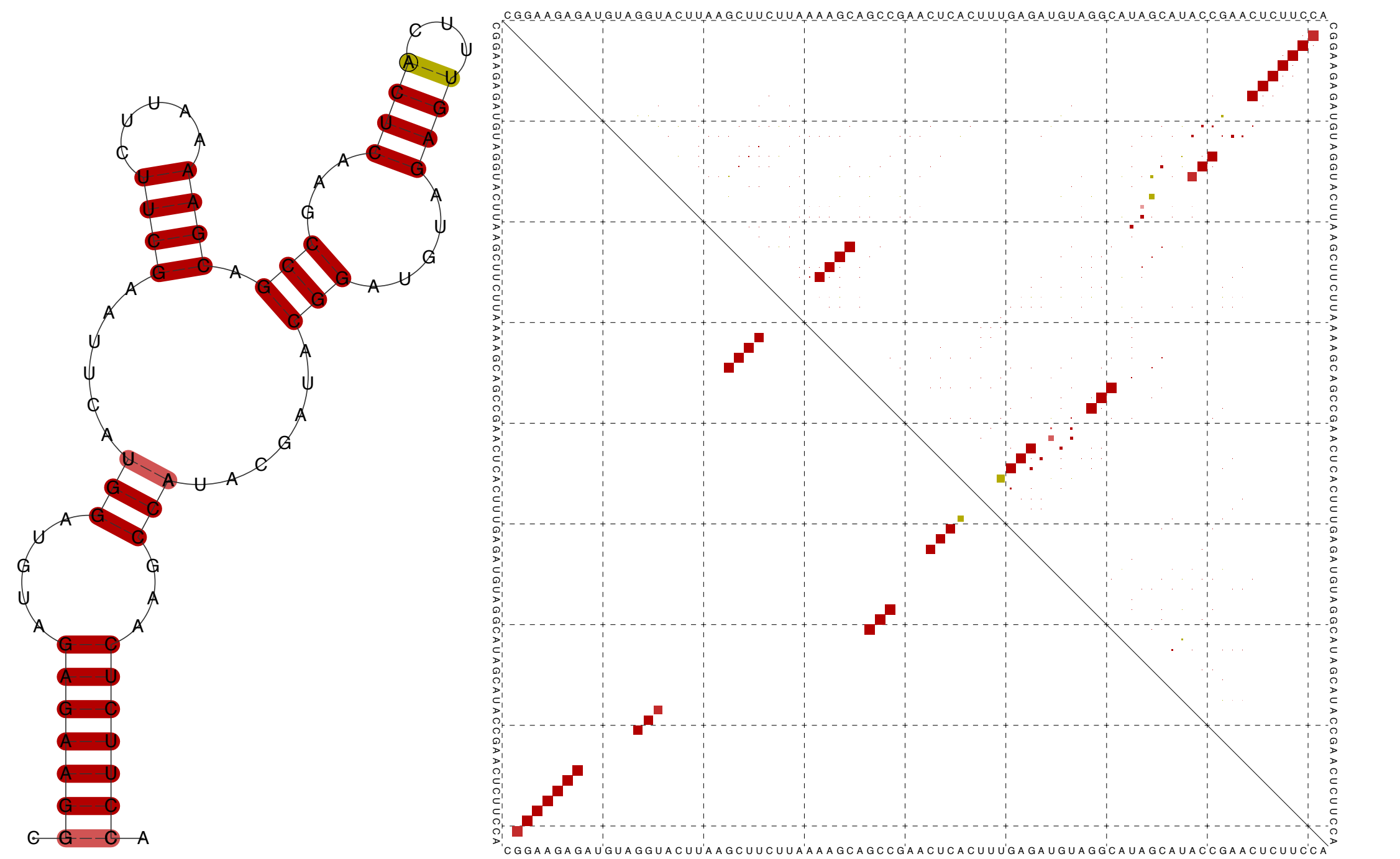
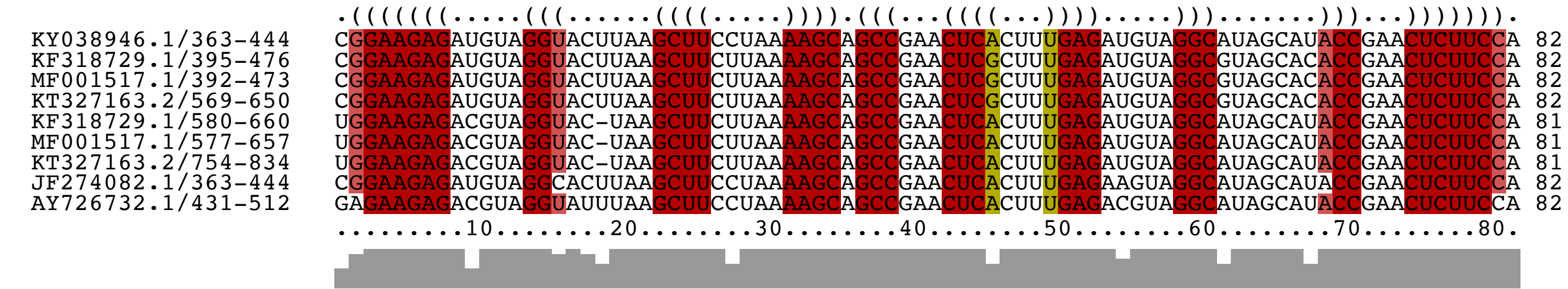
	((((((((.....))))))))))	
KF318729.1/688-714	GAGAUGUUAUUUUGUUUUUAAUAUUUC	27
KT327163.2/862-888	GAGAUGUUAUUUUGUUUUUAAUAUUUC	27
JF274082.1/472-498	GAGAUGUUAUUUUGUUUUUAAUAUUUC	27
KY038946.1/472-498	GAGAUGUUAUUUUGUUUUUAAUAUUUC	27
AY726732.1/540-566	GAGAUGUUAUUUUGUUUUUAAUAUUUC	27
MF001517.1/685-707	GAGAUGUUAUUUUGUUUUUAA----UC	23
	10.....20.....	



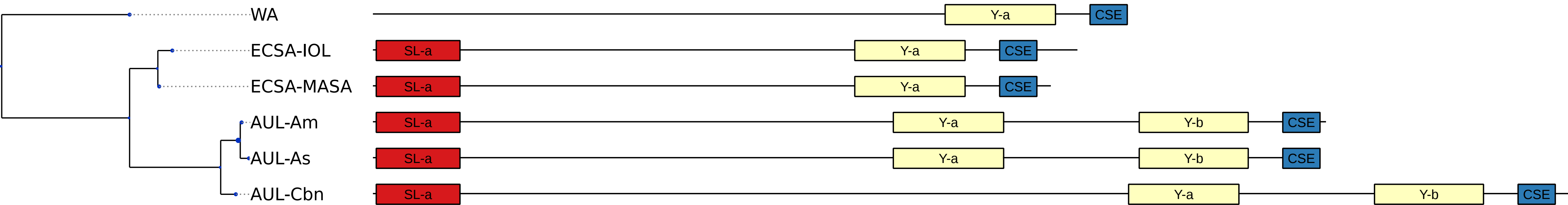
CHIKV 3'UTR: Conserved RNA Structures



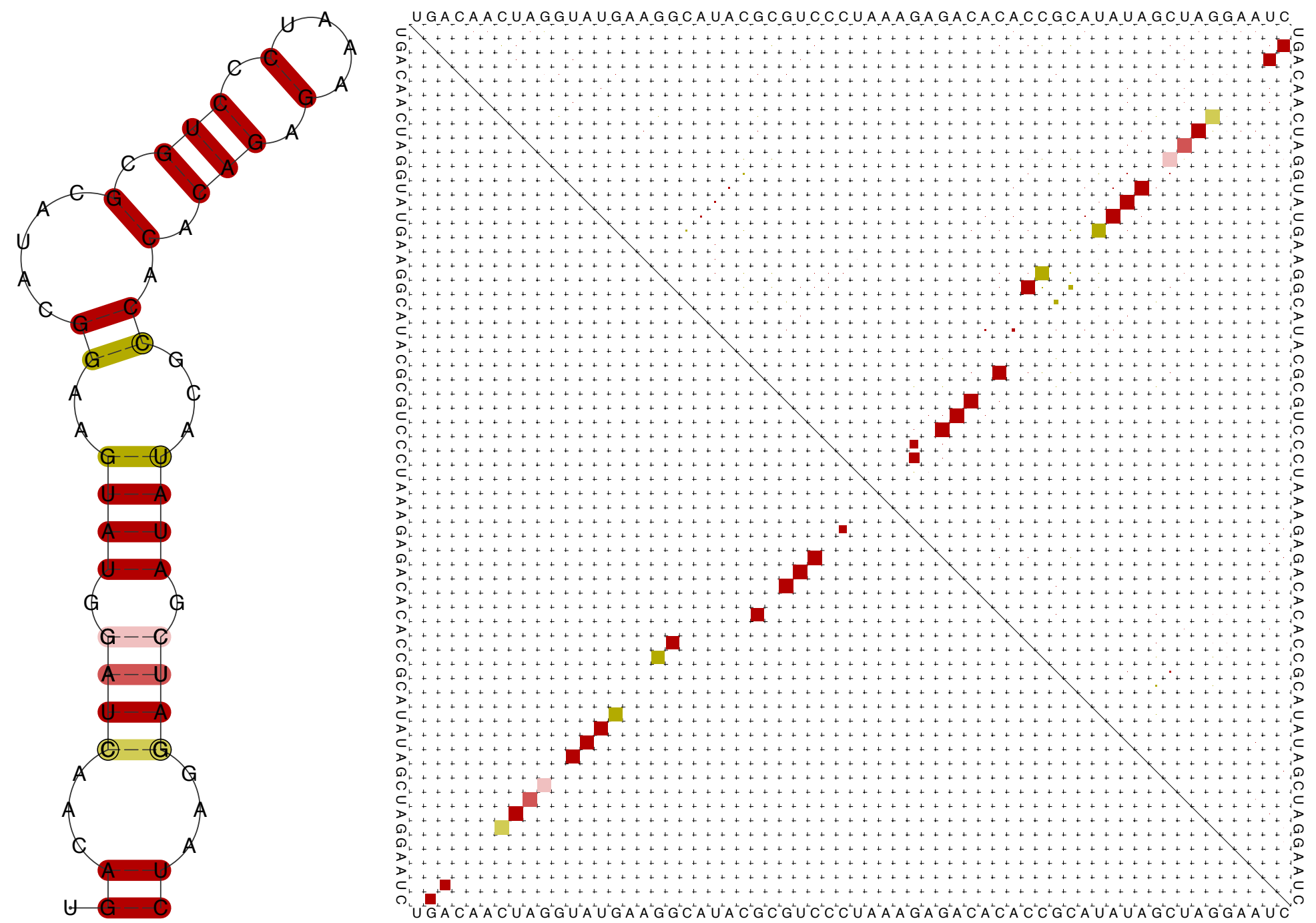
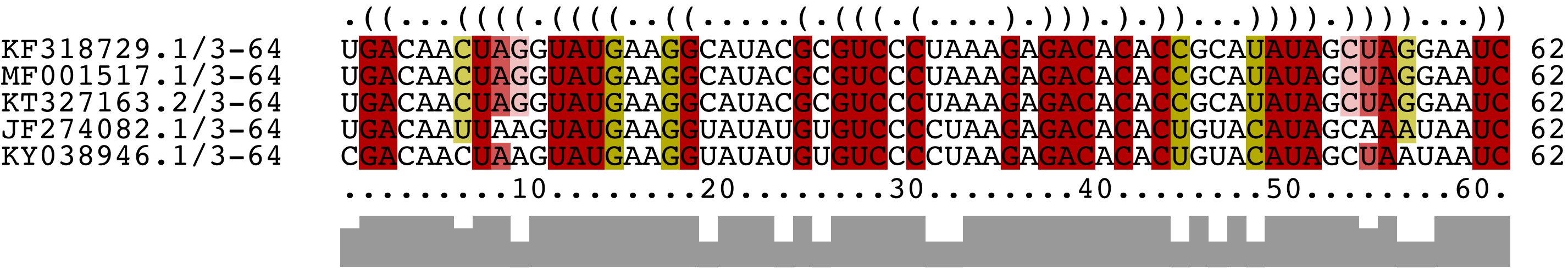
Y-shaped Element (SL-Y)



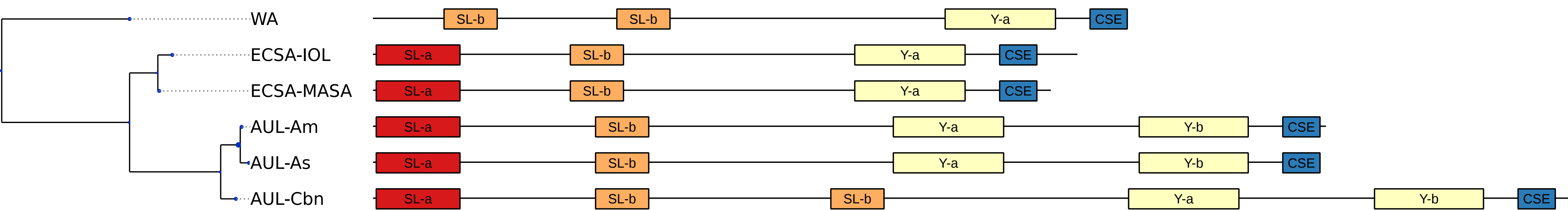
CHIKV 3'UTR: Conserved RNA Structures



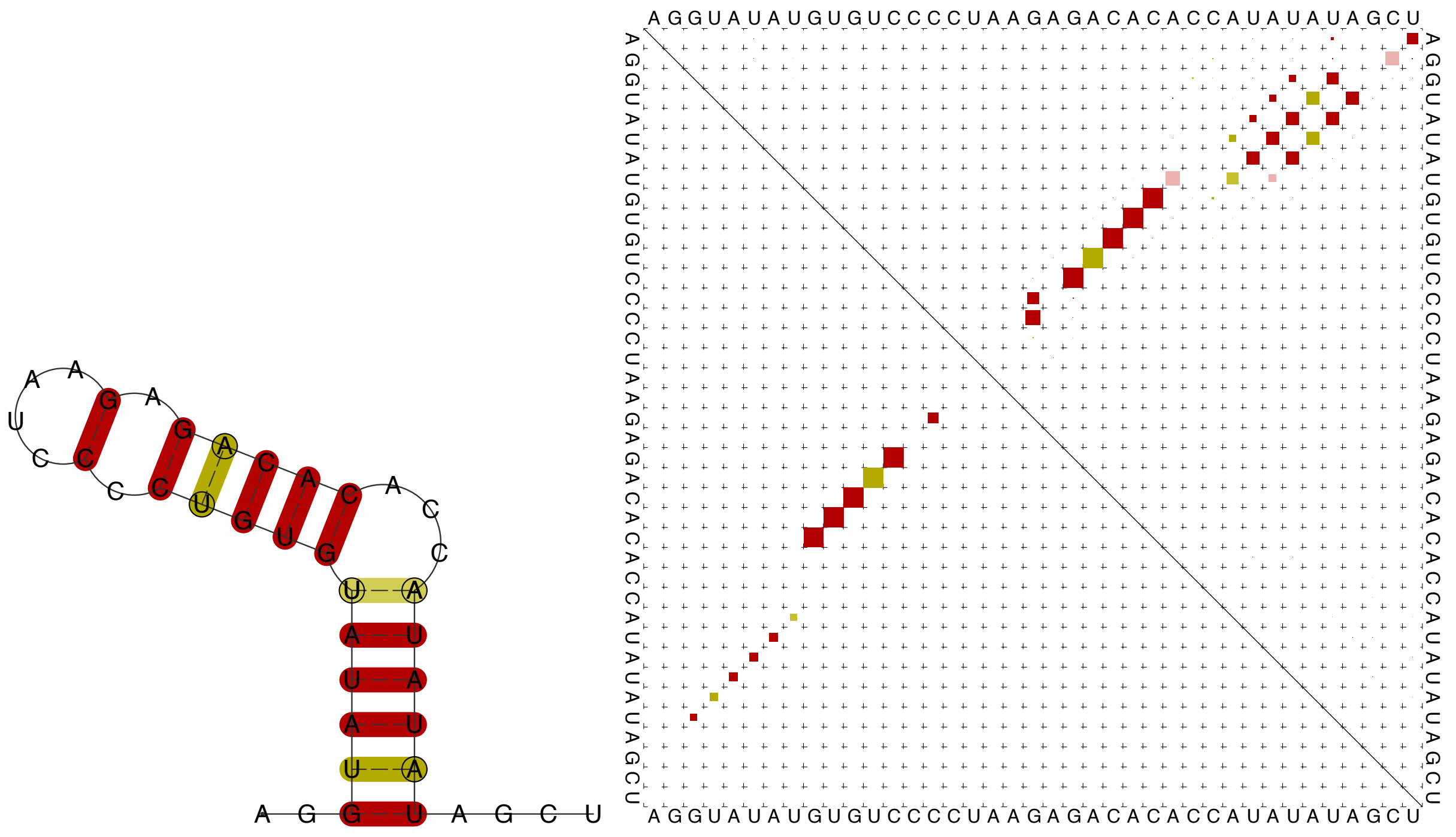
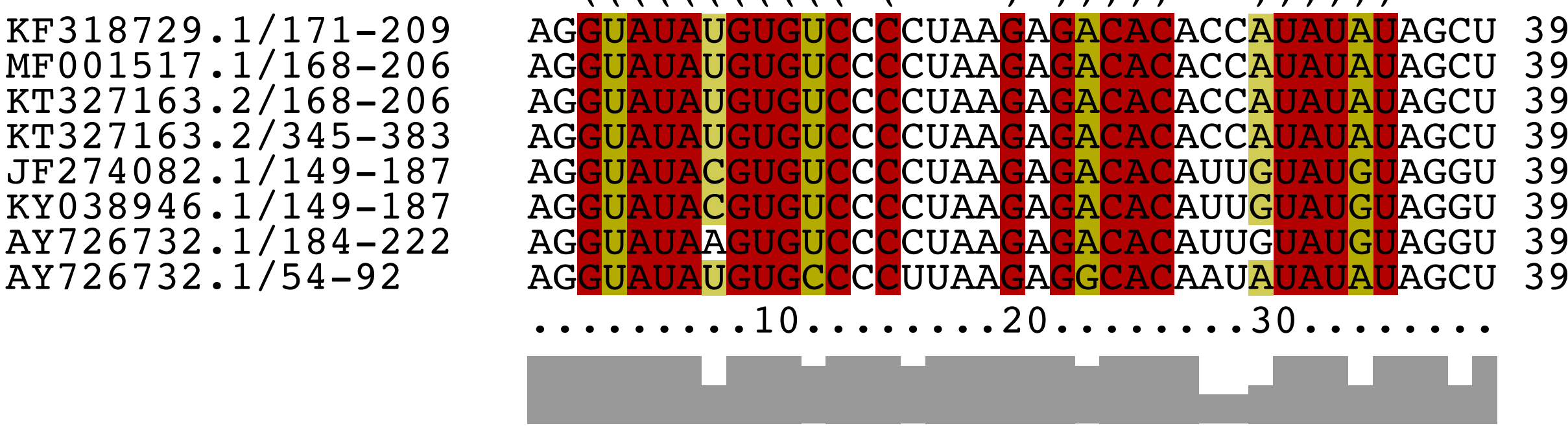
Stem-loop a (SL-a)



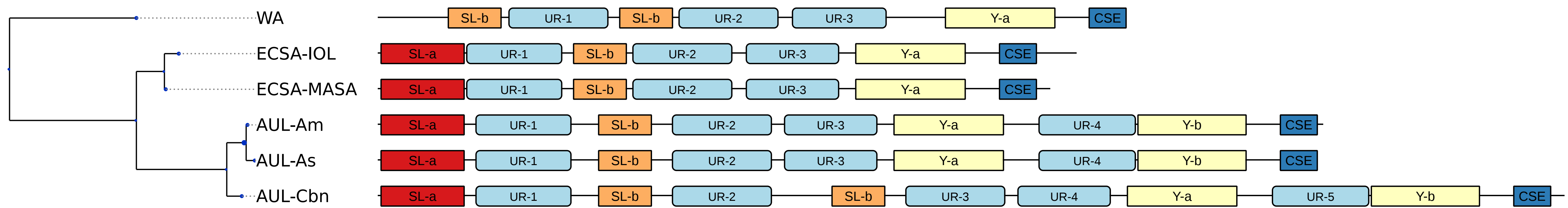
CHIKV 3'UTR: Conserved RNA Structures



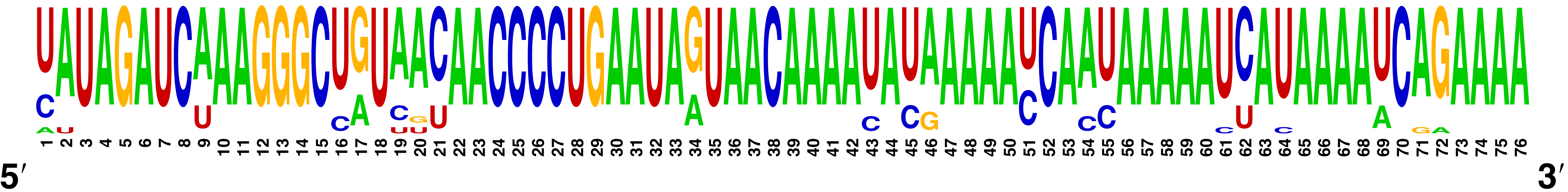
Stem-loop b (SL-b)



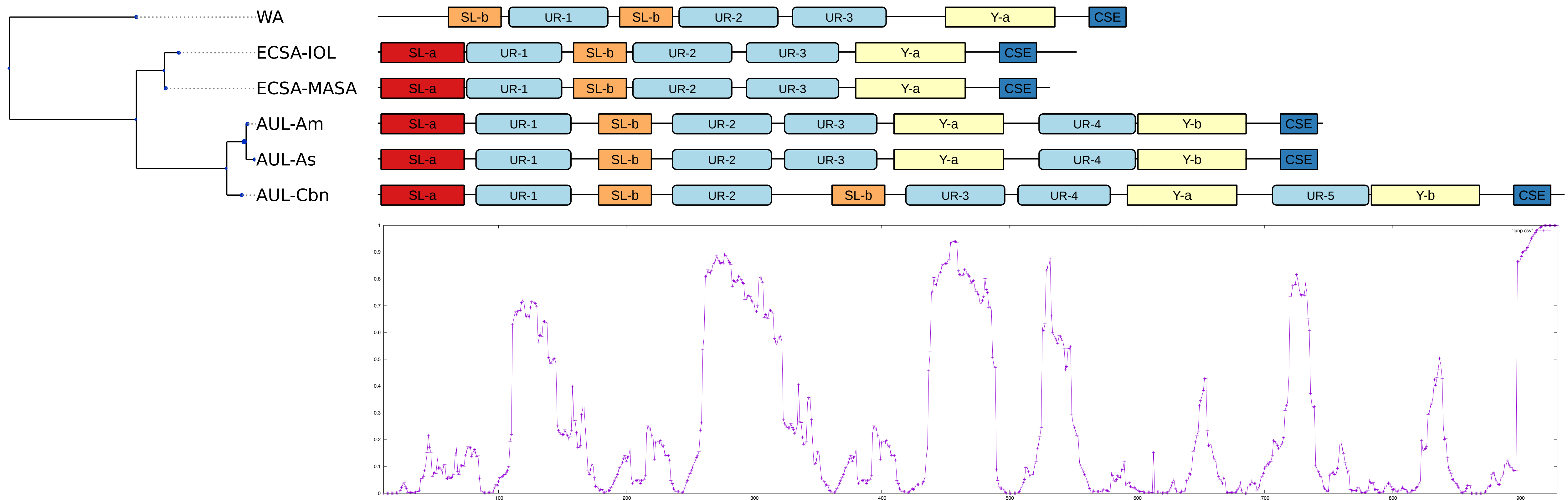
CHIKV 3'UTR: Conserved RNA Elements



Unstructured repeat (UR)



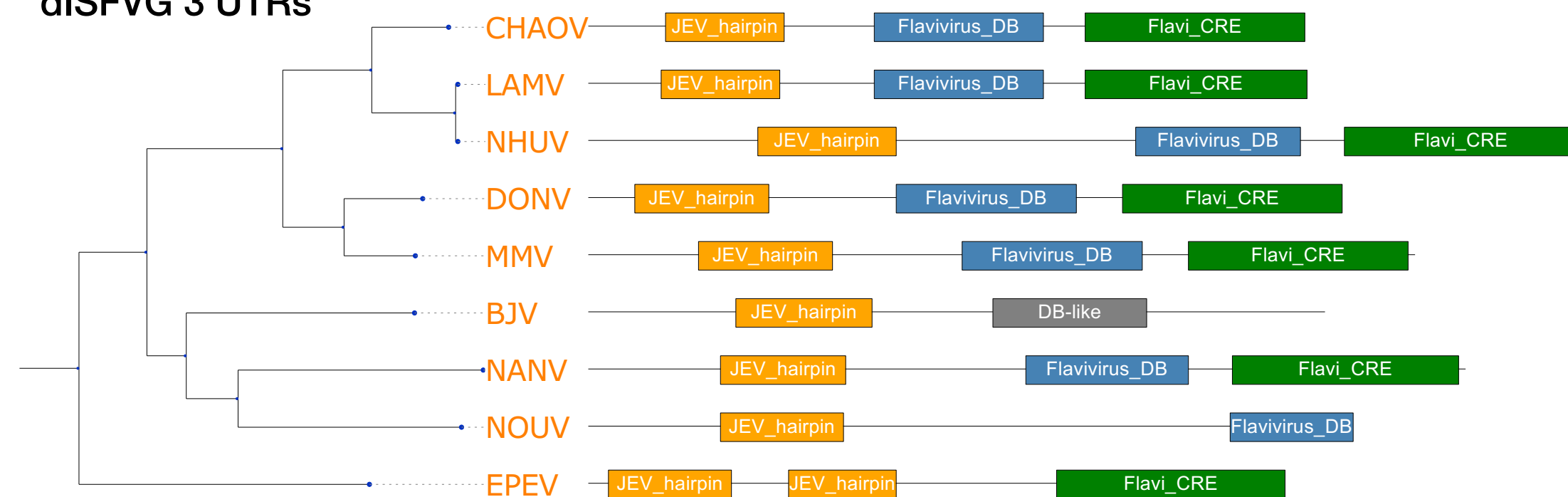
CHIKV-Cbn Accessibility



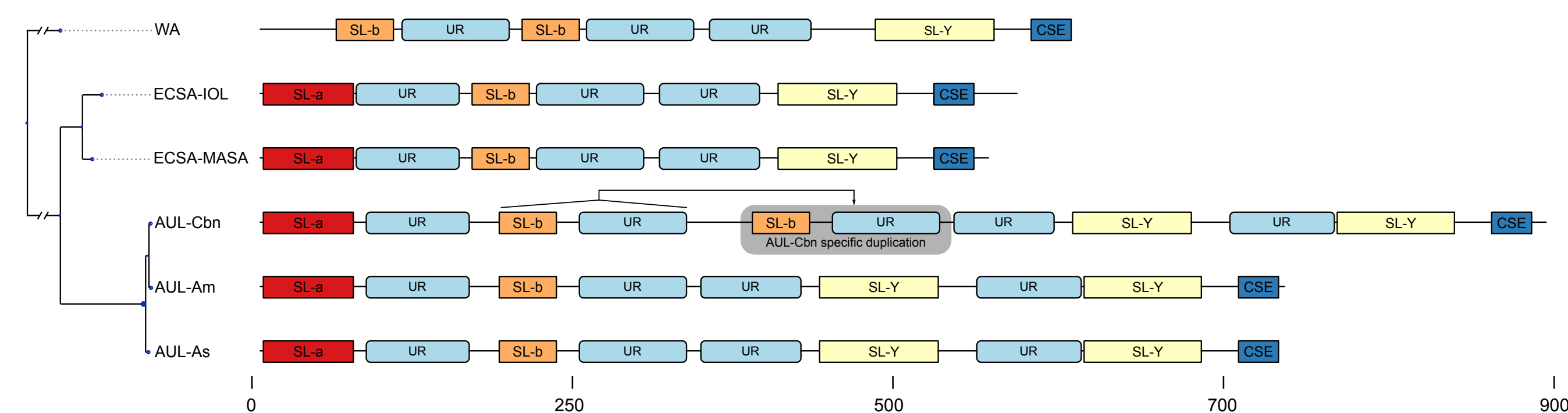
Summary

- Functional RNA structures are evolutionarily conserved
- Structural alignments and covariance models
- Varied architecture of conserved RNAs in flavivirus UTRs
- Lineage-specific non-coding RNA architecture in Alphaviruses
- Tools freely available through ViennaRNA Package

dISFVG 3'UTRs



CHIKV 3'UTRs



Acknowledgements

TBI Vienna

Roman Ochsenreiter
Ronny Lorenz
Ivo L. Hofacker

Univ. San Diego

Adriano de Bernardi Schneider

Functional RNA Structures in the 3' UTR of Tick-Borne, Insect-Specific and No-Known-Vector Flaviviruses. *Viruses* 11.3 (2019): 298. <https://doi.org/10.3390/v11030298>

Updated phylogeny of Chikungunya virus suggests lineage-specific RNA architecture. *Viruses* 11.9 (2019): 798. <https://doi.org/10.3390/v11090798>

Musashi binding elements in Zika and related Flavivirus 3' UTRs: A comparative study in silico. *Scientific reports* 9.1 (2019): 6911. <https://doi.org/10.1038/s41598-019-43390-5>

Twitter: [@mtwolfinger](https://twitter.com/mtwolfinger)